

| **SECTORAL BREAKDOWN - PART 1**

# OIL & GAS INDUSTRY

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# The Three Great Themes That Underlie **The Story of Oil**

- The first is the rise and development of capitalism and modern business. Oil is the world's biggest and most pervasive business, the greatest of the great industries that arose in the last decades of the nineteenth century. Standard Oil, which thoroughly dominated the American petroleum industry by the end of that century, was among the world's very first and largest multinational enterprises.
- The expansion of the business in the twentieth century—encompassing everything from wildcat drillers, smooth-talking promoters, and domineering entrepreneurs to great corporate bureaucracies and state-owned companies—embodies the twentieth-century evolution of business, of corporate strategy, of technological change and market development, and indeed of both national and international economies.
- Throughout the history of oil, deals have been done and momentous decisions have been made—among men, companies, and nations—sometimes with great calculation and sometimes almost by accident. No other business so starkly and extremely defines the meaning of risk and reward—and the profound impact of chance and fate.





# The Three Great Themes That Underlie The Story of Oil

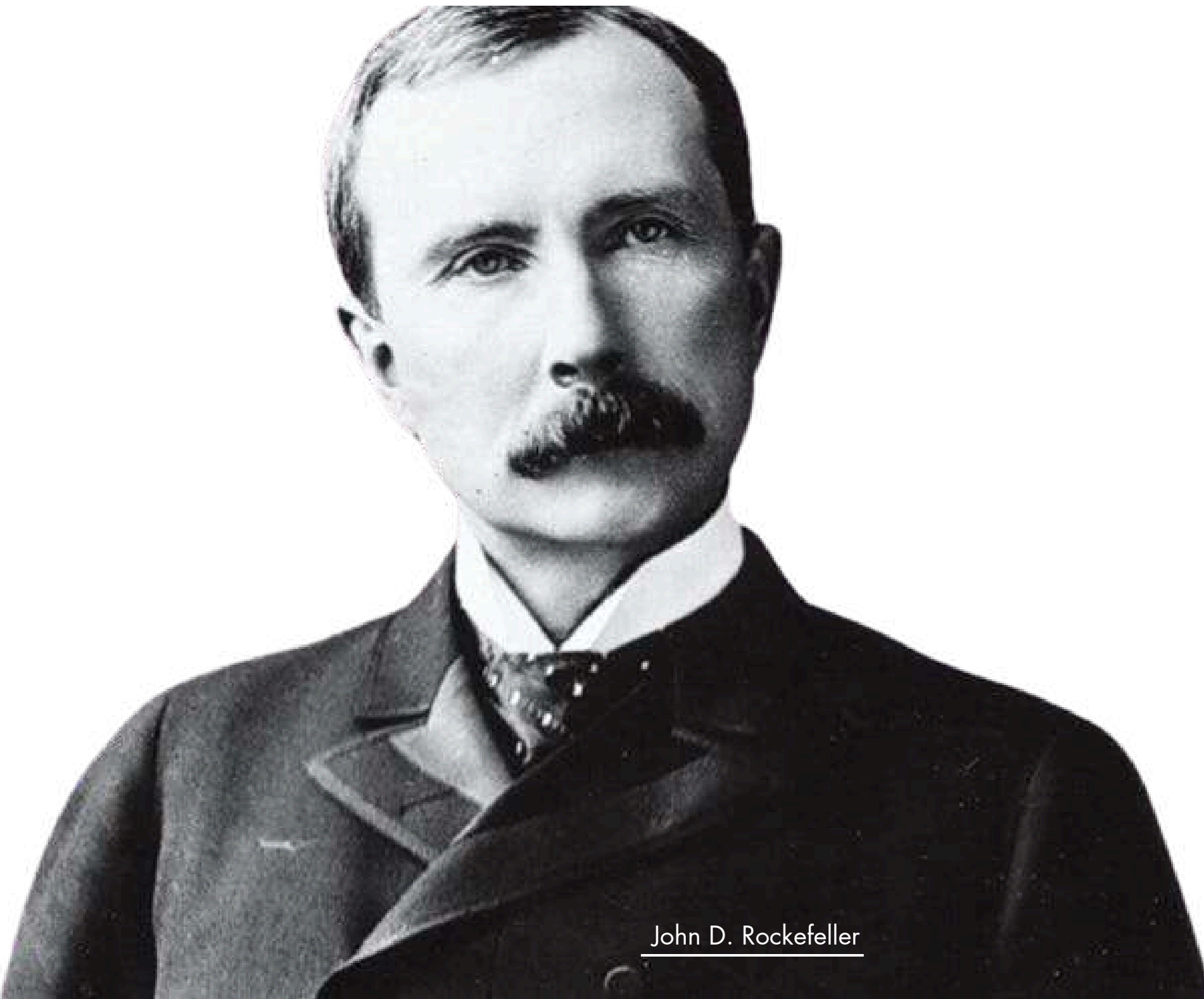
- The second theme is that of oil as a commodity intimately intertwined with national strategies and global politics and power. The battlefields of World War I established the importance of petroleum as an element of national power when the internal combustion machine overtook the horse and the coal-powered locomotive. Petroleum was central to the course and outcome of World War II in both the Far East and Europe.



- The Japanese attacked Pearl Harbor to protect their flank as they grabbed for the petroleum resources of the East Indies. Among Hitler's most important strategic objectives in the invasion of the Soviet Union was the capture of the oil fields in the Caucasus. But America's predominance in oil proved decisive, and by the end of the war German and Japanese fuel tanks were empty.
- In the Cold War years, the battle for control of oil between international companies and developing countries was a major part of the great drama of decolonization and emergent nationalism. The Suez Crisis of 1956, which truly marked the end of the road for the old European imperial powers, was as much about oil as about anything else. "Oil power" loomed very large in the 1970s, catapulting states heretofore peripheral to international politics into positions of great wealth and influence, and creating a deep crisis of confidence in the industrial nations that had based their economic growth upon oil. And oil was at the heart of the first post-Cold War crisis of the 1990s—Iraq's invasion of Kuwait. Yet oil has also proved that it can be fool's gold.
- The Shah of Iran was granted his most fervent wish, oil wealth, and it destroyed him. Oil built up Mexico's economy, only to undermine it. The Soviet Union—the world's second-largest exporter—squandered its enormous oil earnings in the 1970s and 1980s in a military buildup and a series of useless and, in some cases, disastrous international adventures. And the United States, once the world's largest producer and still its largest consumer, must import half of its oil supply, weakening its overall strategic position and adding greatly to an already burdensome trade deficit—a precarious position for a great power. With the end of the Cold War, a new world order is taking shape. Economic competition, regional struggles, and ethnic rivalries may replace ideology as the focus of international—and national—conflict, aided and abetted by the proliferation of modern weaponry. But whatever the evolution of this new international order, oil will remain the strategic commodity, critical to national strategies and international politics.



# The Three Great Themes That Underlie **The Story of Oil**



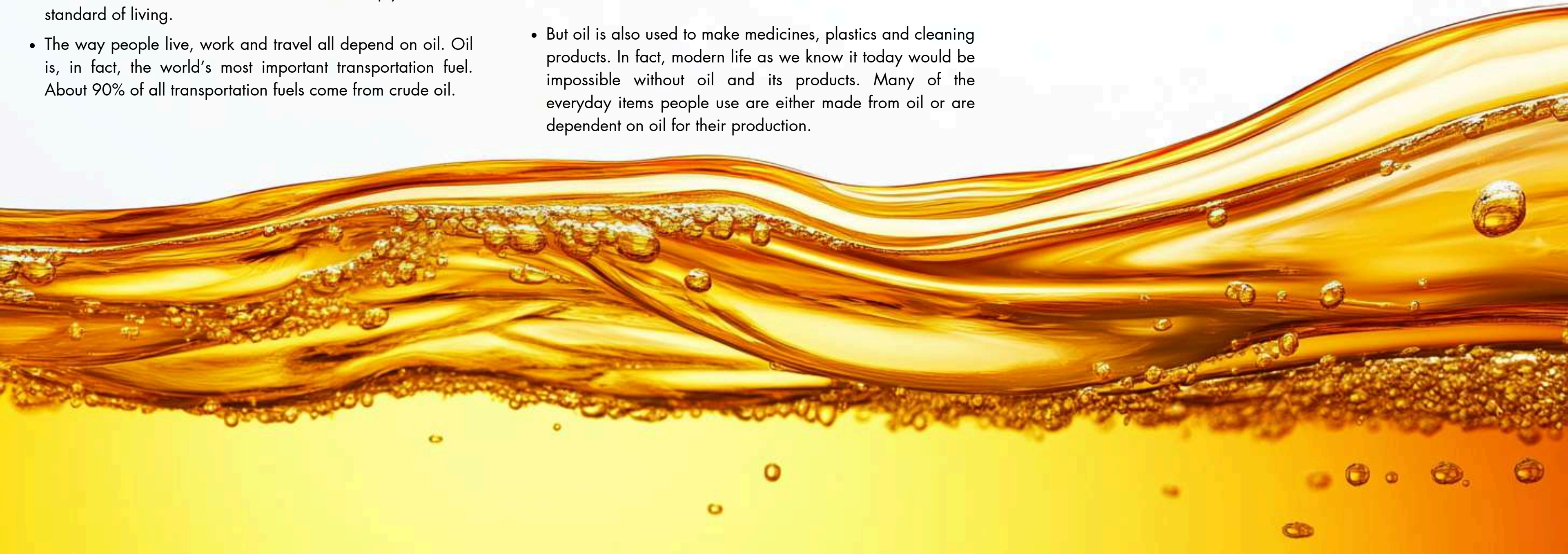
John D. Rockefeller

- A third theme in the history of oil illuminates how ours has become a "Hydrocarbon Society" and we, in the language of anthropologists, "Hydrocarbon Man." In its first decades, the oil business provided an industrializing world with a product called by the made-up name of "kerosene" and known as the "new light," which pushed back the night and extended the working day. At the end of the nineteenth century, John D. Rockefeller had become the richest man in the United States, mostly from the sale of kerosene.
- Gasoline was then only an almost useless by-product, which sometimes managed to be sold for as much as two cents a gallon, and, when it could not be sold at all, was run out into rivers at night. But just as the invention of the incandescent light bulb seemed to signal the obsolescence of the oil industry, a new era opened with the development of the internal combustion engine powered by gasoline. The oil industry had a new market, and a new civilization was born.
- In the twentieth century, oil, supplemented by natural gas, toppled King Coal from his throne as the power source for the industrial world. Oil also became the basis of the great postwar suburbanization movement that transformed both the contemporary landscape and our modern way of life.
- Today, we are so dependent on oil, and oil is so embedded in our daily doings, that we hardly stop to comprehend its pervasive significance. It is oil that makes possible where we live, how we live, how we commute to work, how we travel—even where we conduct our courtships. It is the lifeblood of suburban communities. Oil (and natural gas) are the essential components in the fertilizer on which world agriculture depends; oil makes it possible to transport food to the totally non-self-sufficient megacities of the world. Oil also provides the plastics and chemicals that are the bricks and mortar of contemporary civilization, a civilization that would collapse if the world's oil wells suddenly went dry.



# Why is Oil Important?

- Crude oil is a central part of modern life and the world's most important energy resource. We rely on it in many ways for the food we eat, the clothes we wear and the electronics we use at home and in the workplace. Without oil, we would not be able to continue to enjoy the same standard of living.
- The way people live, work and travel all depend on oil. Oil is, in fact, the world's most important transportation fuel. About 90% of all transportation fuels come from crude oil.
- It is also the raw material from which other important products are made, which have improved the quality of our lives over the past century and a half. It has been used to keep people's houses warm during the winter for centuries.
- But oil is also used to make medicines, plastics and cleaning products. In fact, modern life as we know it today would be impossible without oil and its products. Many of the everyday items people use are either made from oil or are dependent on oil for their production.
- Some of these products include heating oil, jet fuel, kerosene, dentures, diapers, fertilizers, lipstick, shampoo, deodorant, shaving cream, crayons, musical devices, cameras, computers, glue, contact lenses, toothpaste, synthetic fibres, tires, artificial hearts and even aspirin.





# Oil in everyday life

Whether as a fuel or a feedstock: oil is an integral part of your daily life.

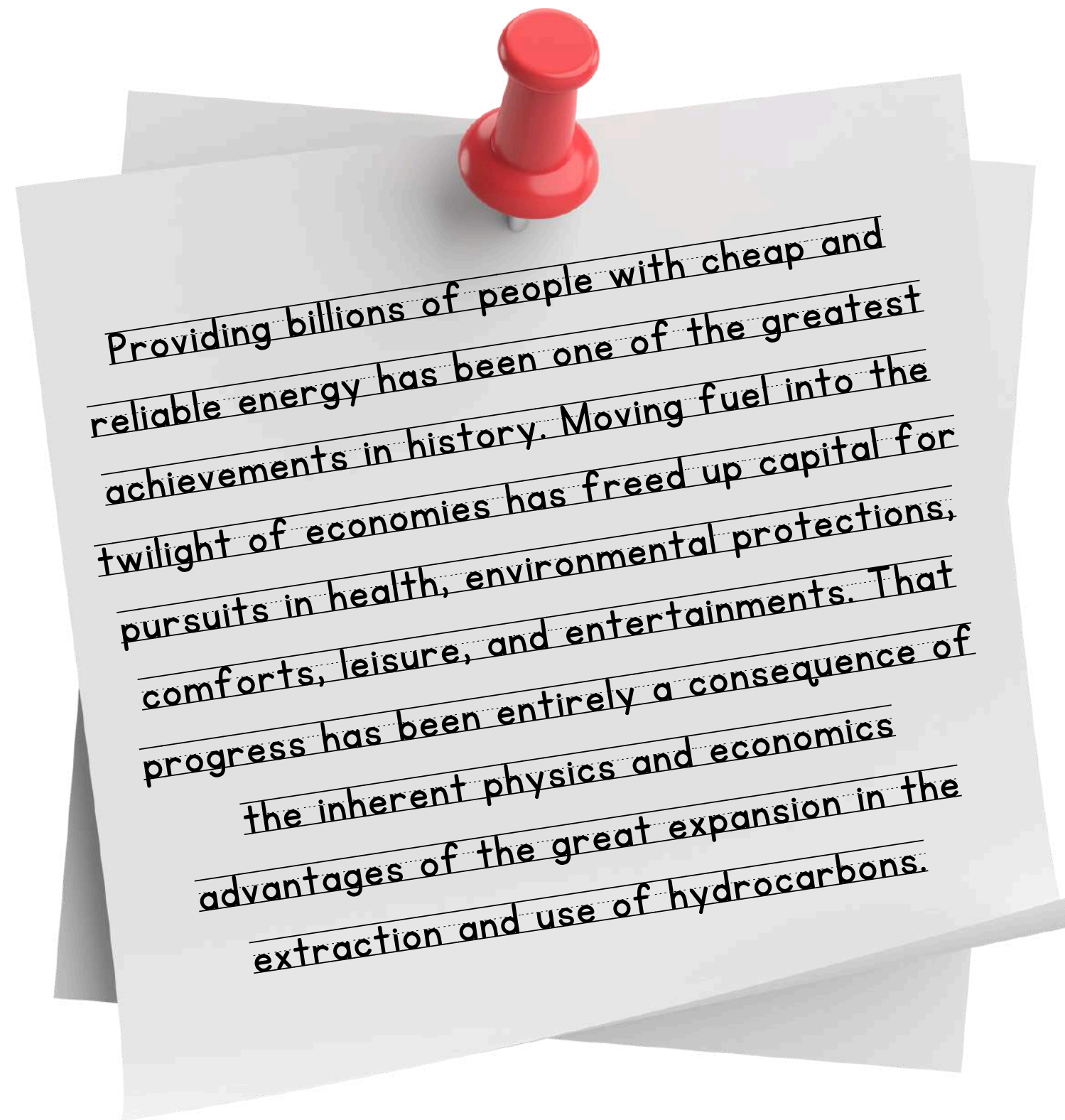


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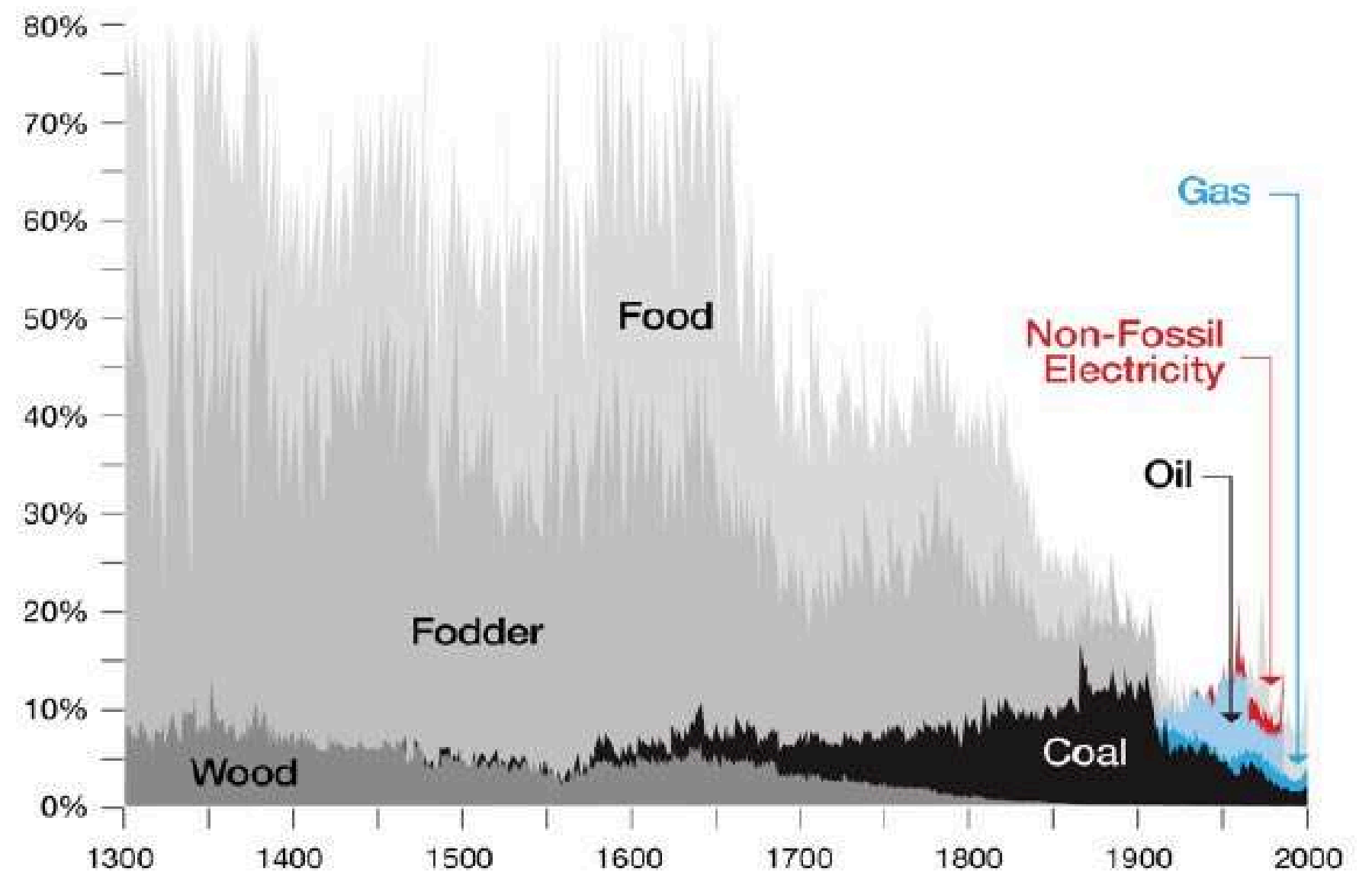
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Topic: Oil & Gas Industry



# GDP Spent on Energy



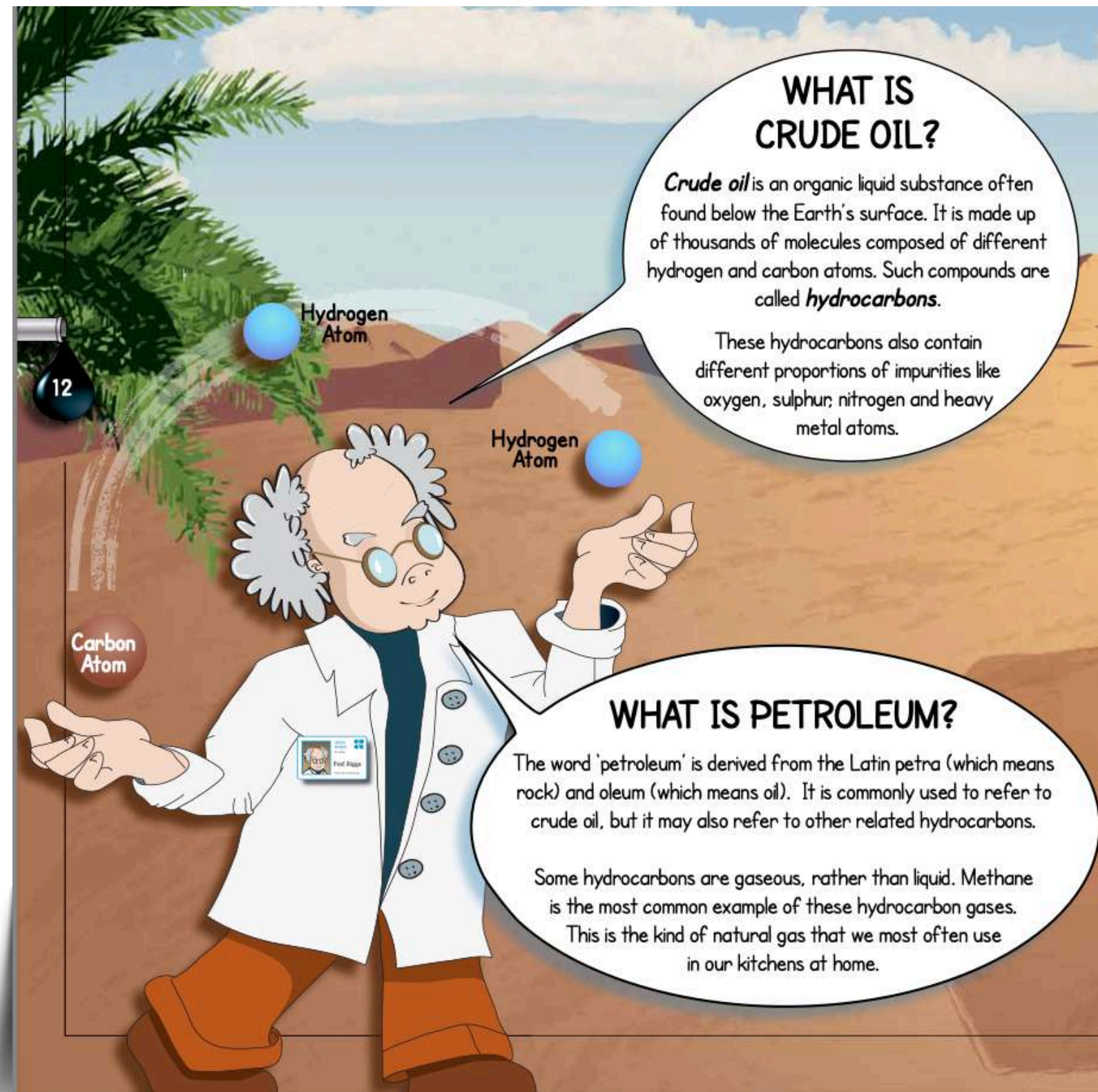
Share GDP Spent on Energy



Source: John W. Day et al., "The Energy Pillars of Society: Perverse Interactions of Human Resource Use, the Economy, and Environmental Degradation," *BioPhysical Economics and Resource Quality* 3, no. 2 (February 2018)



# Oil Basics



A cartoon scientist with white hair and glasses, wearing a white lab coat and orange boots, stands in a desert landscape. He is pointing towards a speech bubble. In the background, there are palm trees and a small building. A speech bubble from a tree on the left says '12'. Two blue spheres are labeled 'Hydrogen Atom' and a red sphere is labeled 'Carbon Atom'.

### WHAT IS CRUDE OIL?

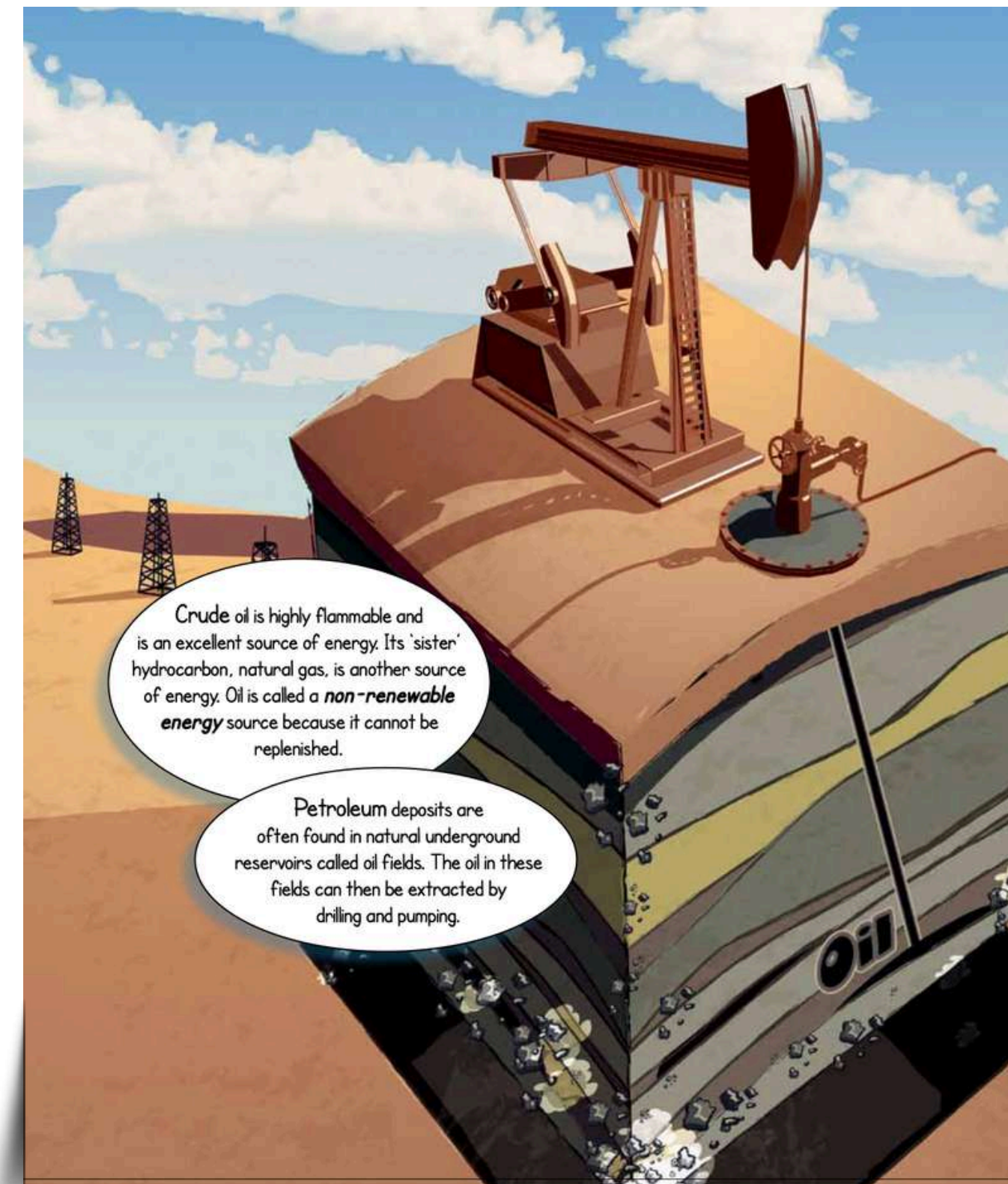
*Crude oil* is an organic liquid substance often found below the Earth's surface. It is made up of thousands of molecules composed of different hydrogen and carbon atoms. Such compounds are called **hydrocarbons**.

These hydrocarbons also contain different proportions of impurities like oxygen, sulphur, nitrogen and heavy metal atoms.

### WHAT IS PETROLEUM?

The word 'petroleum' is derived from the Latin *petra* (which means rock) and *oleum* (which means oil). It is commonly used to refer to crude oil, but it may also refer to other related hydrocarbons.

Some hydrocarbons are gaseous, rather than liquid. Methane is the most common example of these hydrocarbon gases. This is the kind of natural gas that we most often use in our kitchens at home.



A 3D illustration of an oil pumpjack on a desert surface. Below the surface, a cross-section shows an underground oil reservoir. A well is drilled into the reservoir, and a pipe labeled 'Oil' is shown. Two speech bubbles provide information about crude oil and petroleum deposits.

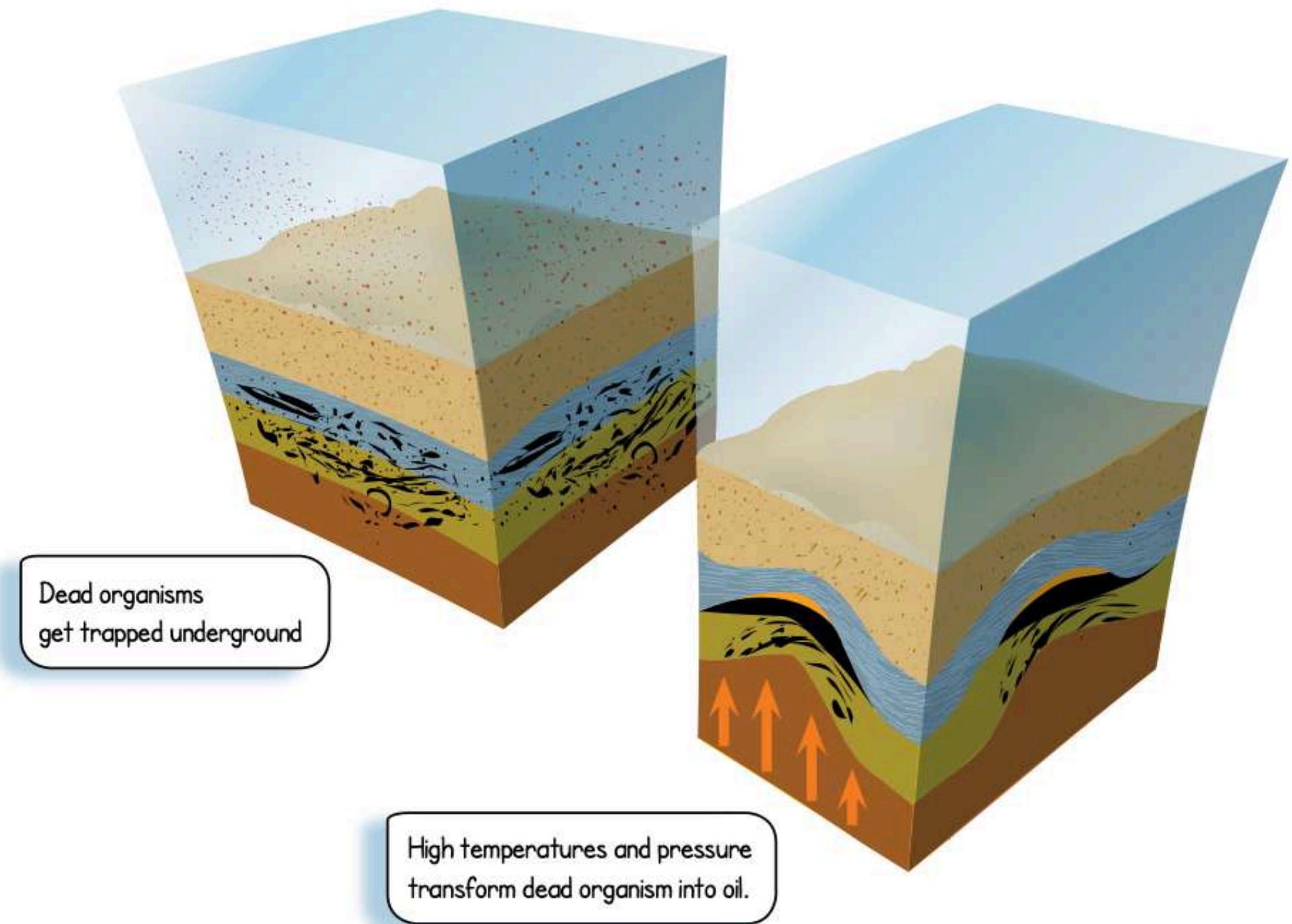
Crude oil is highly flammable and is an excellent source of energy. Its 'sister' hydrocarbon, natural gas, is another source of energy. Oil is called a **non-renewable energy** source because it cannot be replenished.

Petroleum deposits are often found in natural underground reservoirs called oil fields. The oil in these fields can then be extracted by drilling and pumping.



# How is Oil Formed?

- Oil is formed from the accumulation of hydrocarbons. Hydrocarbons accumulate naturally, thousands of feet below the surface of the Earth, from the decomposition of organic materials like plants and marine animals which died during the Palaeozoic Era (between 245 and 544 million years ago).
- Trapped beneath the ground under enormous pressure and high temperatures, these hydrocarbons were compressed and eventually transformed into crude oil after millions of years.





Underground oil accumulations are formed when three conditions are met.

## FIRST,

there must be a 'source' rock rich in hydrocarbons and buried deep enough so that the heat from the Earth's core can 'cook' them into oil.

## SECOND,

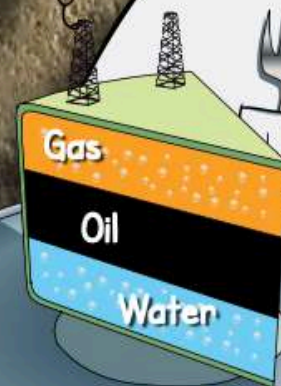
there should be a porous rock nearby in which oil can accumulate (it is often sandstone). If the holes in the rock are interconnected, then oil can flow easily out of the rock. This condition is called **permeability**. The porous rock must have good permeability, which is why studying the structure of rocks is an important step to finding oil.

## THIRD,

there is usually a 'cap rock' or seal to trap the oil in the underground reservoirs and prevent it from seeping to the surface. Within these reservoirs, hydrocarbons are typically organized like a three-layer cake—with a layer of water below the oil and a layer of gas above it.

Much of the oil that escapes to the surface often evaporates into the air. But it can leave behind deposits of residual hydrocarbons called **bitumen**.

Three-Layer-Cake



Underground Oil Accumulation

Underground Oil Accumulation



# Oil Properties

- Crude oil properties can vary widely depending on where the oil is found and under what conditions it was formed. Its different physical properties are used to design the right kind of refineries, classify the oil (for example, West Texas Intermediate or Oman) and determine an appropriate price for it.
- The properties of oil include its density, called the API gravity (named after the American Petroleum Institute), sulphur content, nitrogen content, carbon residue and distillation range.
- Each of these properties is important for different reasons. For example, the sulphur content of crude oil is important because it determines the kind of treatment that it will require at a refinery. The higher the sulphur level, the bigger the effect it will have on the environment—and the more corrosive effect it will have on equipment.

- API gravity is also important. It is essentially a measure of density. It determines whether a specific type of crude oil has a higher or lower boiling range (or distillate yields), which is important for separating and extracting different parts (or fractions).
- Different oil-producing areas produce different kinds of crude oil. And depending on its mixture of hydrocarbons, crude oil can vary in colour, composition and consistency. Crude oil properties can vary widely depending on where the oil is found and under what conditions it was formed. Its different physical properties are used to design the right kind of refineries, classify the oil (for example, West Texas Intermediate or Oman) and determine an appropriate price for it. The properties of oil include its density, called the API gravity (named after the American Petroleum Institute), sulphur content, nitrogen content, carbon residue and distillation range.

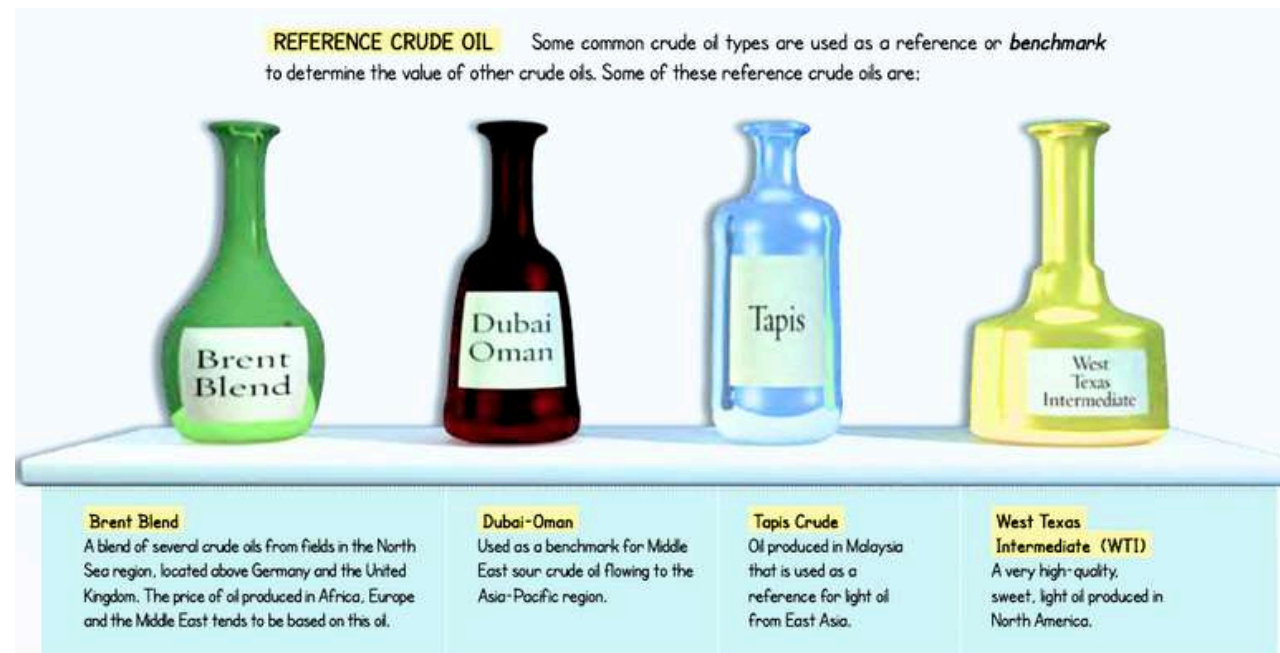


# Grades of Crude Oil

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## Light / Heavy

Crude oil can be classified as either **light** or **heavy** depending on its API gravity (or density). Generally, the higher its API gravity, the lower its density.

Oil that is lighter in colour, has a thin consistency and flows easily usually contains less metals and sulphur compounds. It is known as light oil.

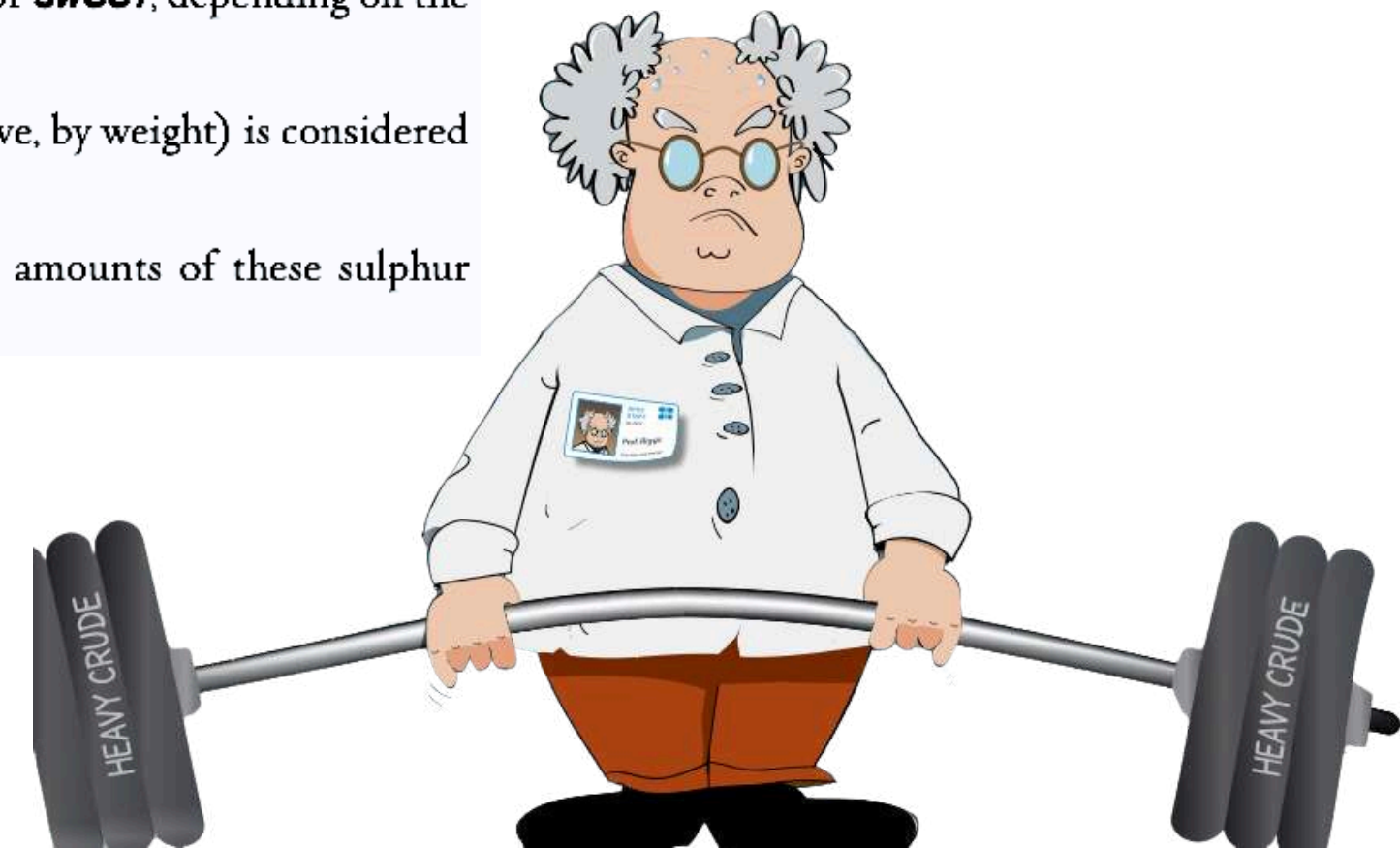
Oil that is high in metal and sulphur content is considered low-grade oil. It generally has too much carbon, not enough hydrogen and is more time-consuming to produce and hard to refine. It is known as heavy oil.

## Sour / Sweet

Crude oil can also be classified as either **sour** or **sweet**, depending on the amount of sulphur it contains.

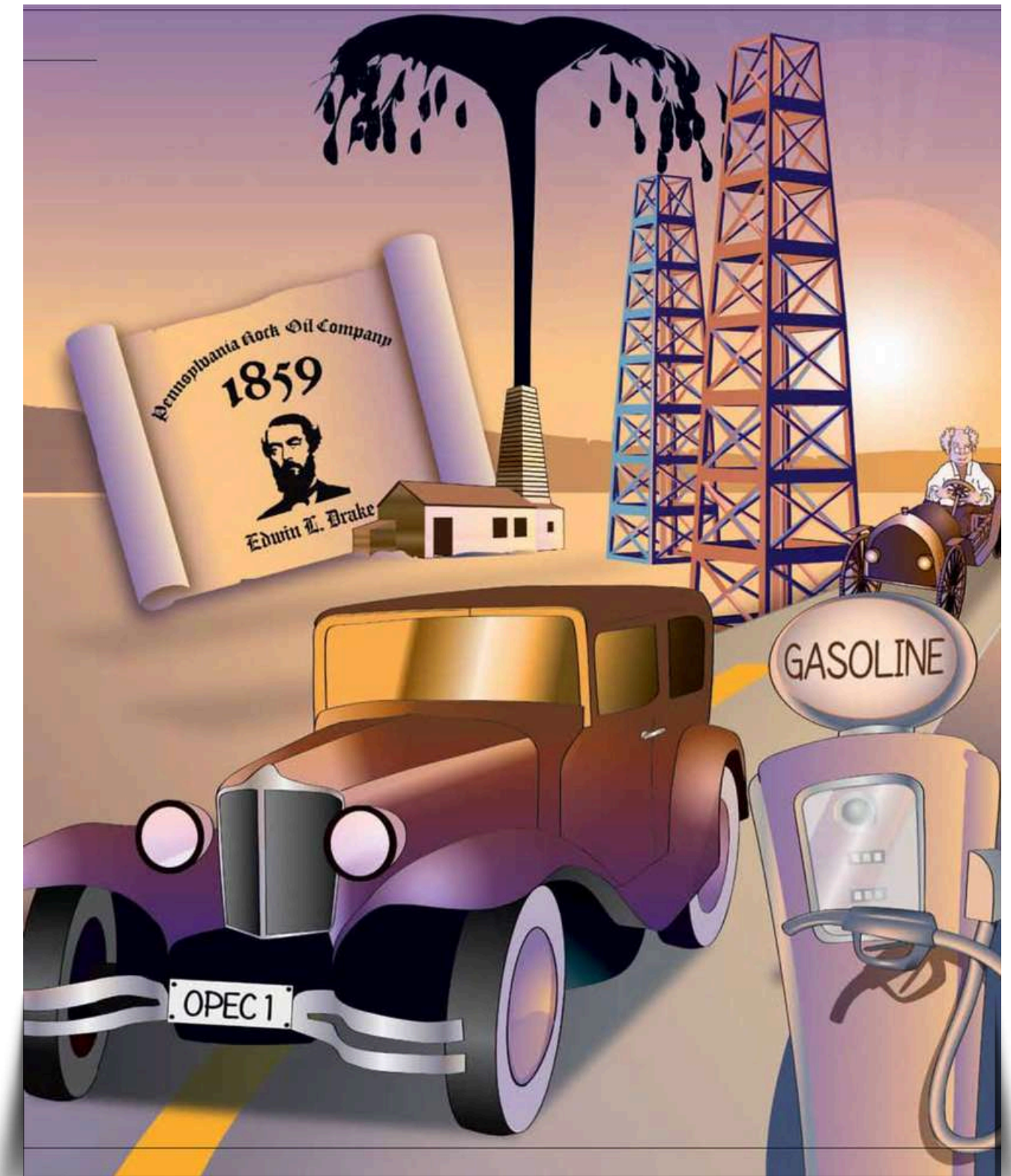
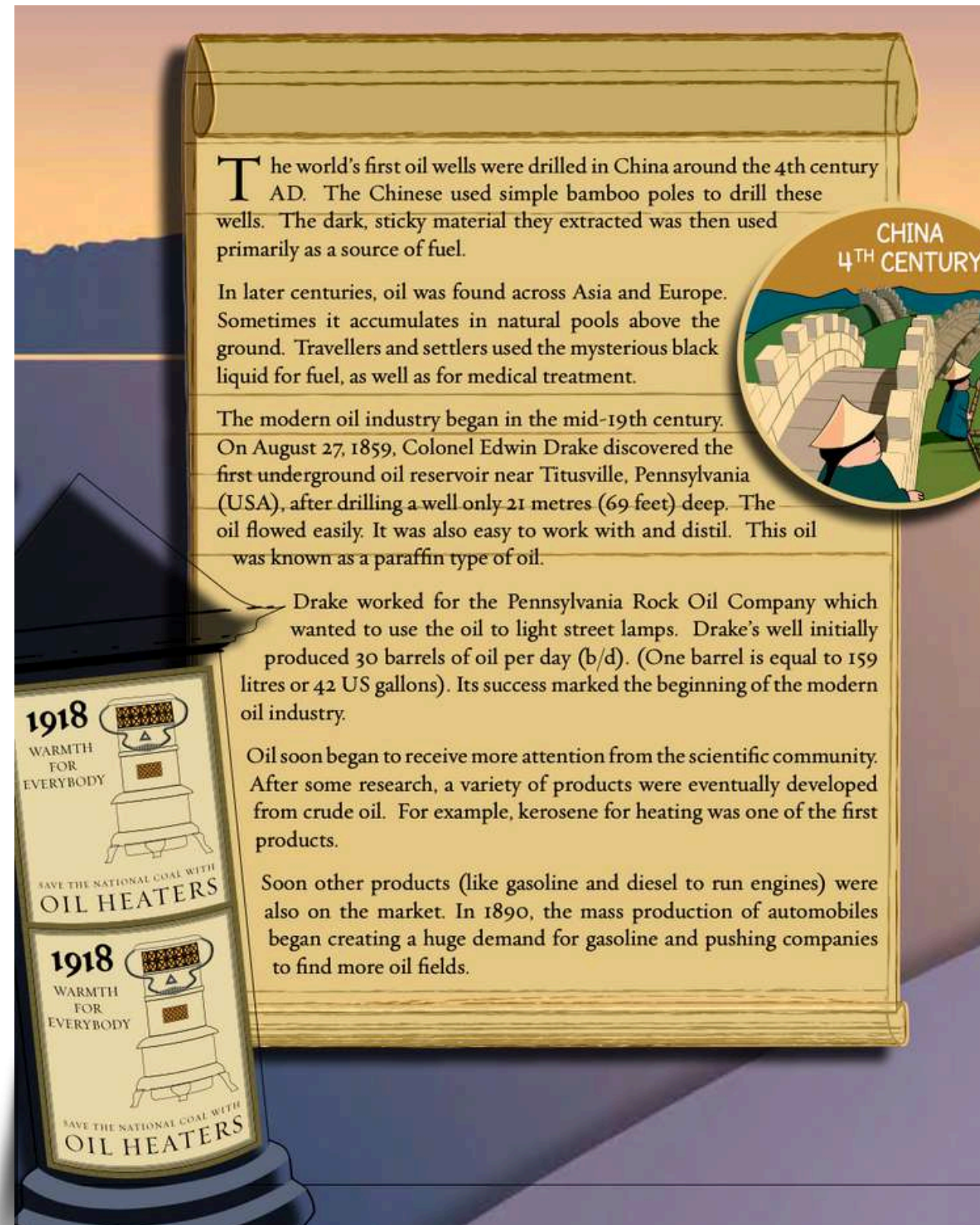
Oil with a high sulphur content (0.5% and above, by weight) is considered sour.

Sweet crude oil, on the other hand, has low amounts of these sulphur compounds.





# A Brief History of the Industry





# Finding Oil: **Upstream**

The upstream sector is the part of the oil industry involved with finding oil fields and bringing oil up from the ground. Upstream activities include exploratory work, such as the search for underground (or underwater) oil and gas reservoirs, and the initial drilling, followed by the production phase, which is the actual extraction of oil from the ground.

Upstream activities—or E&P—can be the most profitable, but are the most risky and capital-intensive part of the Energy sector. Huge investments can be lost entirely. Conversely, large discoveries of oil deposits can generate revenues for decades to come.





# Exploration

Exploration is almost like detective work. It requires looking for clues, careful observation of ground conditions, taking notes of different information and the evaluation of survey data. This is the job of petroleum geoscientists, who are experts on rocks.

Petroleum geoscientists working at oil companies begin looking for places where there might be oil. They look for signs that may indicate the presence of hydrocarbons underground and are responsible for determining the best places to drill.

They start by examining the shapes of different underground layers of rock. They have to use special tools in order to 'see' the rocks underground. Thus, using advanced technology—and special tools like aerial photography, satellite pictures and specialized machines that measure variations in the Earth's gravity and magnetic fields—geoscientists try to identify likely crude oil fields.



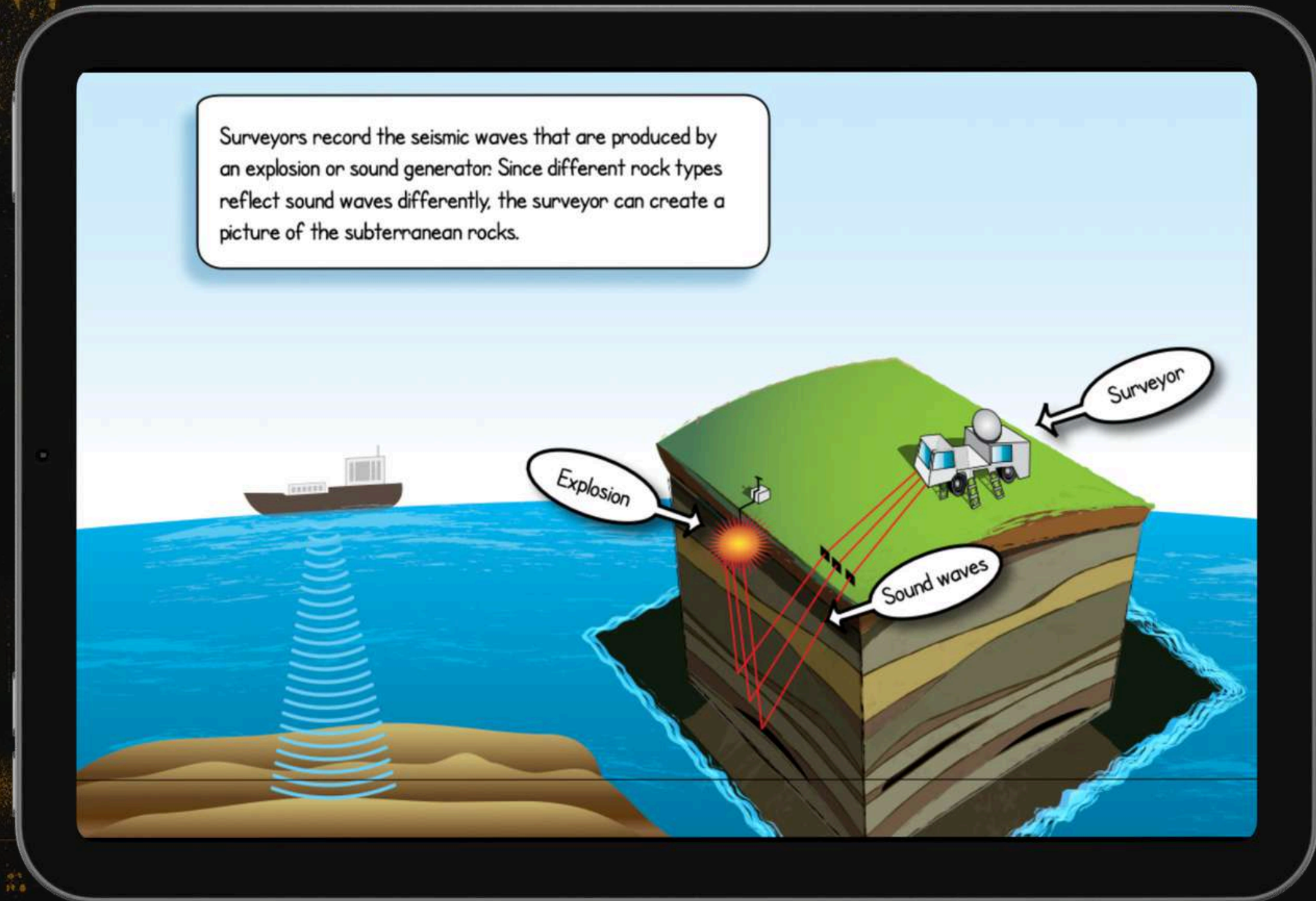


# Exploration

Oil can also be found underground at the bottom of the sea. In this case, special ships are used to look for these underwater oil fields. Geoscientists use several special technical tools—such as sound waves used in seismic technology—to form a clear picture of underwater rock layers.

But the only way to be absolutely sure that there is oil in the ground is simply to drill a well. This is a big gamble because not all wells result in the discovery of oil. It may take the drilling of many different wells until a new oil field is found. This costs a large amount of money because sophisticated equipment is needed and many people need to be hired.

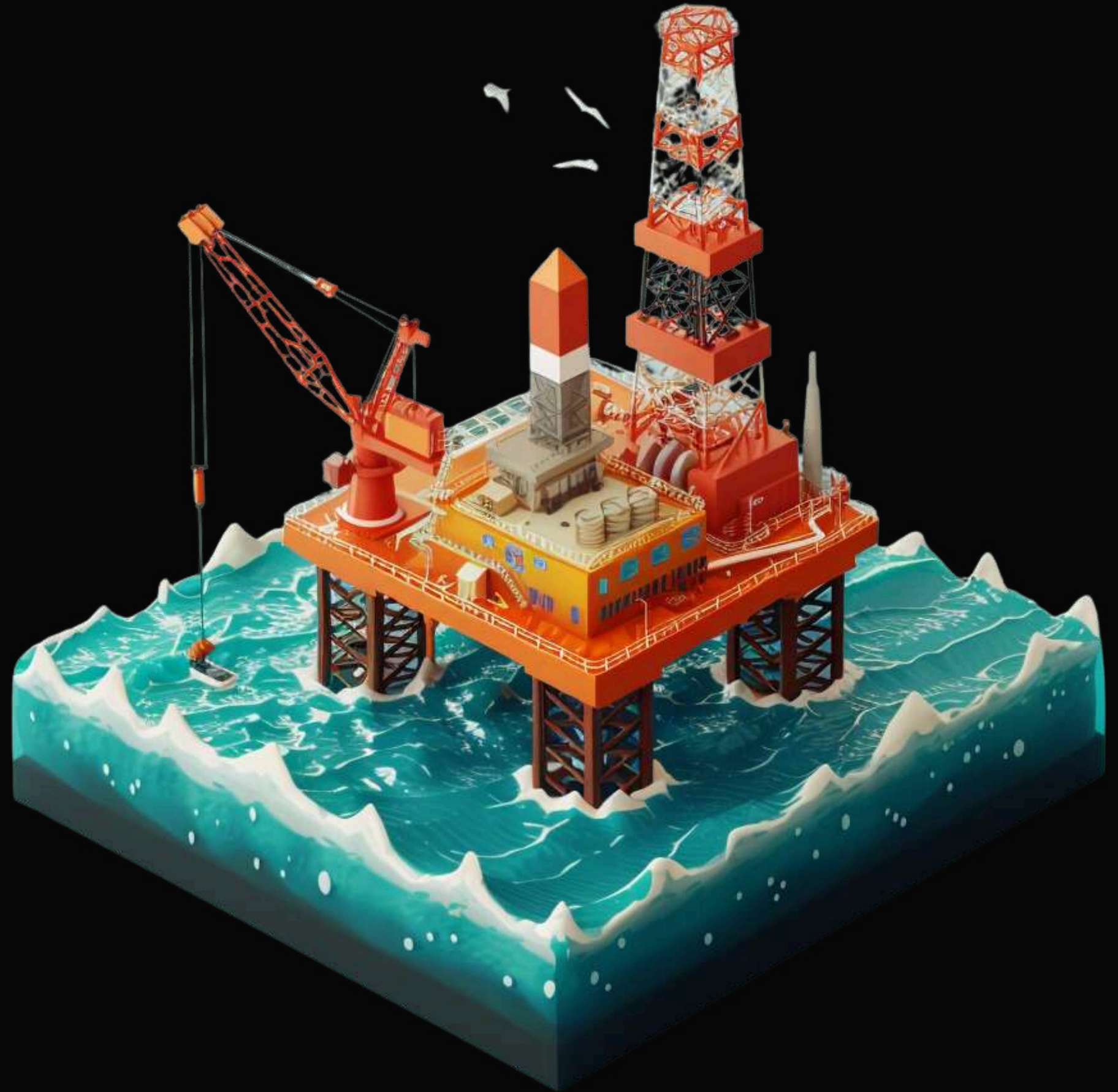
That is why the geoscientists then supply all their survey data to the economists and financial planners at the oil companies, who help make the decision whether or not to drill exploratory wells.





# Extracting Oil and Gas in Six Steps

1. Acquire the rights to explore for and develop oil and gas from the reserve holder. (A reserve holder is typically the owner[s] of the land a company wants to drill on.)
2. Conduct geological, geophysical, and seismic surveys to find oil and gas deposits.
3. Perform exploration drilling to test for deposits.
4. Conduct appraisal and development drilling to determine if the field contains commercially viable deposits.
5. Begin oil and gas production (or abandon the well if no commercial deposits are found).
6. Ensure payment to compensate reserve holders via royalties and/or production sharing agreements (PSAs).





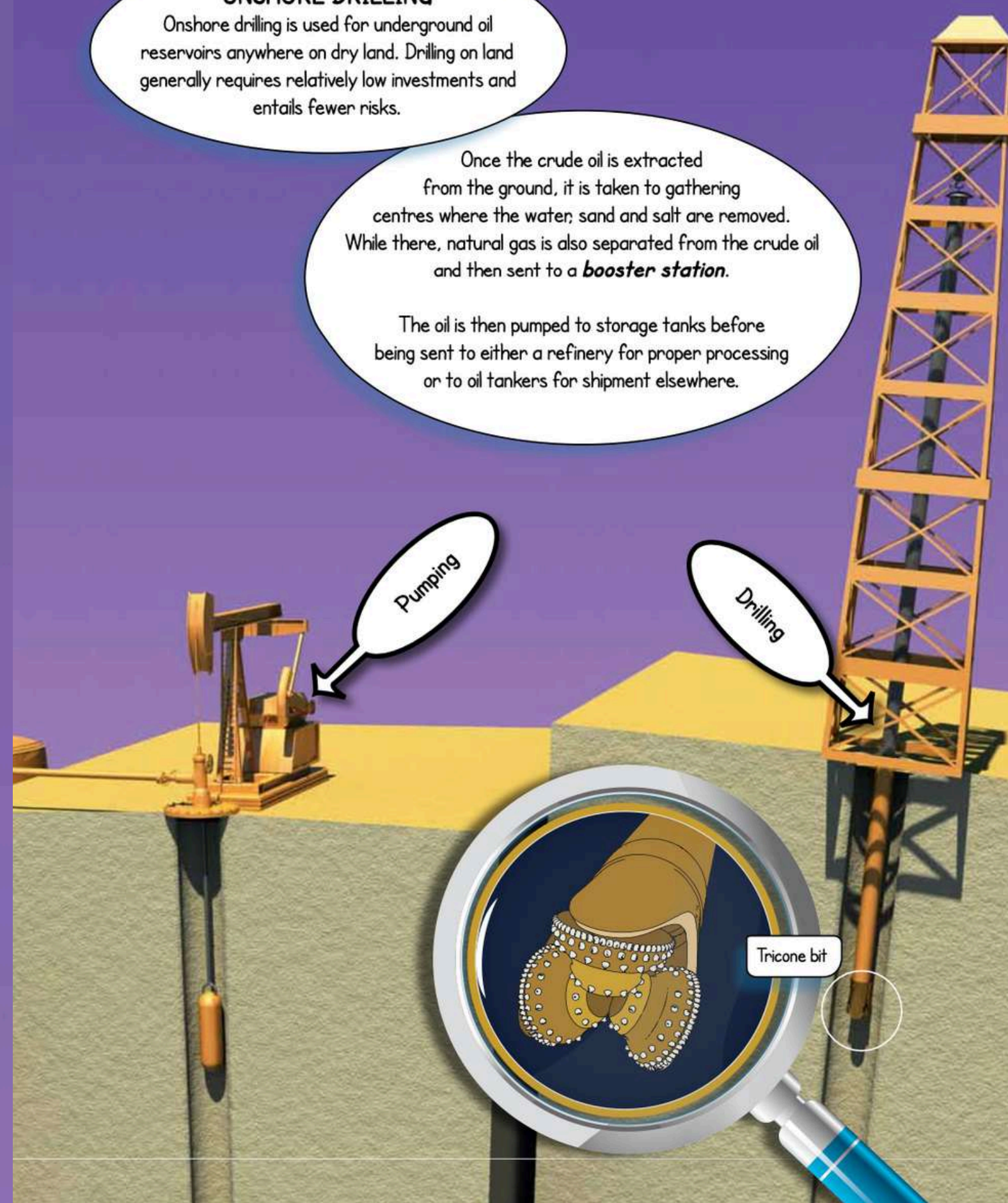
Once oil is found after the preliminary exploration phase and the drilling of exploratory wells, the production phase can begin. There are two forms of drilling: onshore and offshore.

Crude oil found underground is usually mixed with water, sand, salt and natural gas. As it is extracted, the pressure of these different substances must be kept at just the right levels. This is done by turning valves on and off at the surface level. Because these valves are often green in colour and the small dials are red, much like the pine trees decorated at Christmas time, this assembly is often referred to as a *Christmas tree*.



**ONSHORE DRILLING**  
Onshore drilling is used for underground oil reservoirs anywhere on dry land. Drilling on land generally requires relatively low investments and entails fewer risks.

Once the crude oil is extracted from the ground, it is taken to gathering centres where the water, sand and salt are removed. While there, natural gas is also separated from the crude oil and then sent to a **booster station**.  
  
The oil is then pumped to storage tanks before being sent to either a refinery for proper processing or to oil tankers for shipment elsewhere.







## OFFSHORE DRILLING

Offshore drilling is used to extract oil deposits buried under the ocean floor. Offshore drilling rigs are installed, operated and serviced on large platforms built out in the ocean. These platforms can either float or sit on tall 'legs' that touch the seabed. In this way, the offshore rig is able to remain stable and resist waves, wind and, in colder regions, floating chunks of ice.

Because of its difficulty, offshore drilling consumes more than 65% of planned investments even before oil is extracted, increasing the risk of financial losses, especially if wells come up dry!

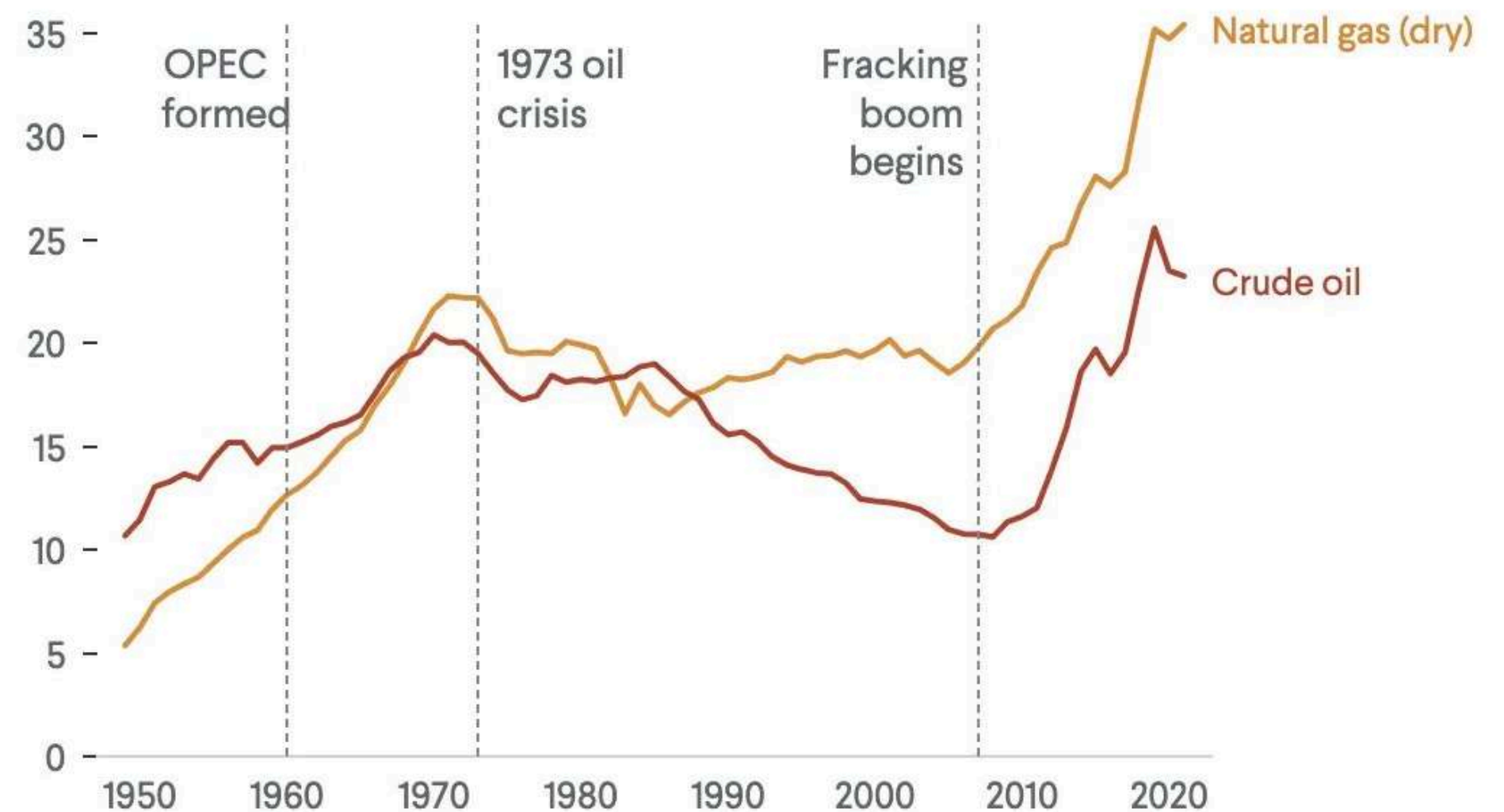


# American Oil Production

- American oil production reached a peak around 1970, and gradually began falling until new techniques became popularized to extract oil from “unconventional” sources. Two main innovations came into play.
- The first was a series of improvements in horizontal drilling, which allowed engineers to steer a drill bit sideways into a formation. Many oil formations were trapped in long, thin reservoirs not easily accessible by a vertical pipe. Horizontal drilling allowed for the extraction of these types of oil deposits.
- The second was hydraulic fracturing, or fracking, which aimed to extract natural gas out of shale rock. Shale formations harbored pores within which natural gas was stored. However, extracting sufficient quantities of gas was not possible with conventional vertical drilling techniques. Instead, methods of injecting pressurized fluids to crack the shale and create pathways for the natural gas to escape were developed and hugely improved during the 1990s

## U.S. Oil and Gas Production Soared With Fracking

Annual production, quadrillion British thermal units (Btu)



Source: U.S. Energy Information Administration.

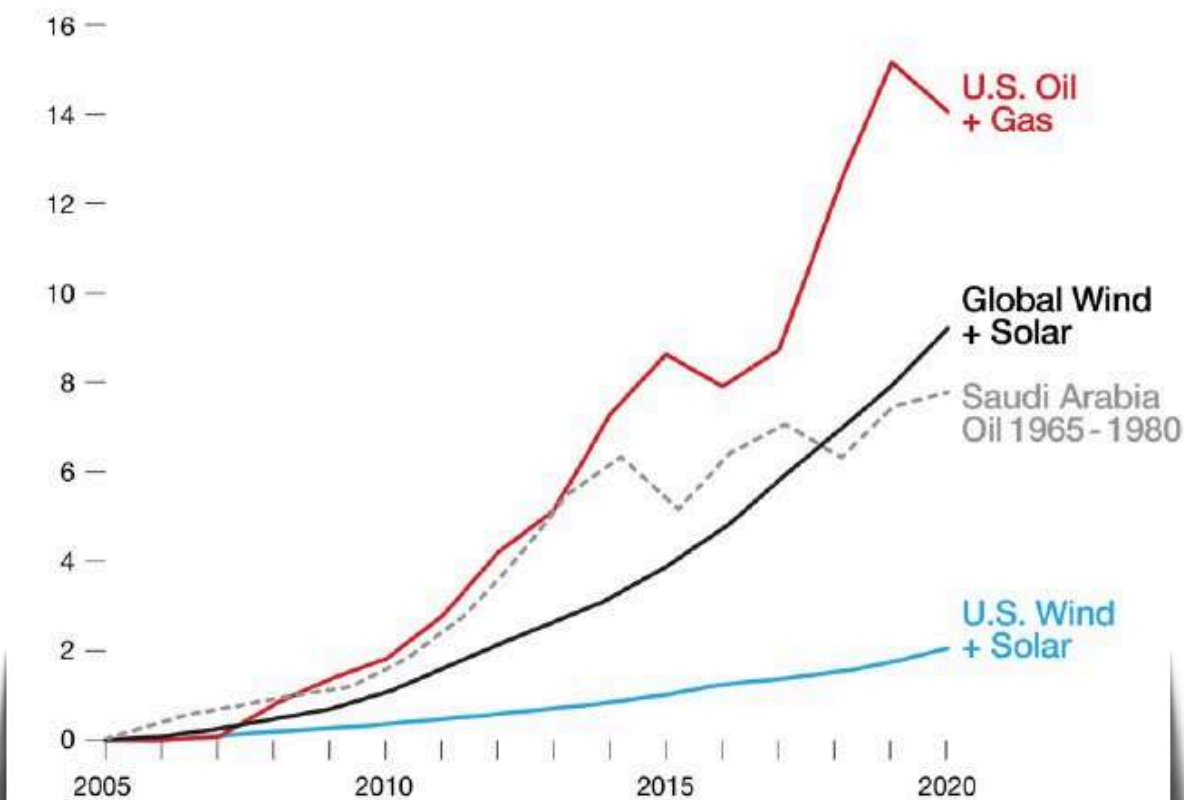
COUNCIL *on*  
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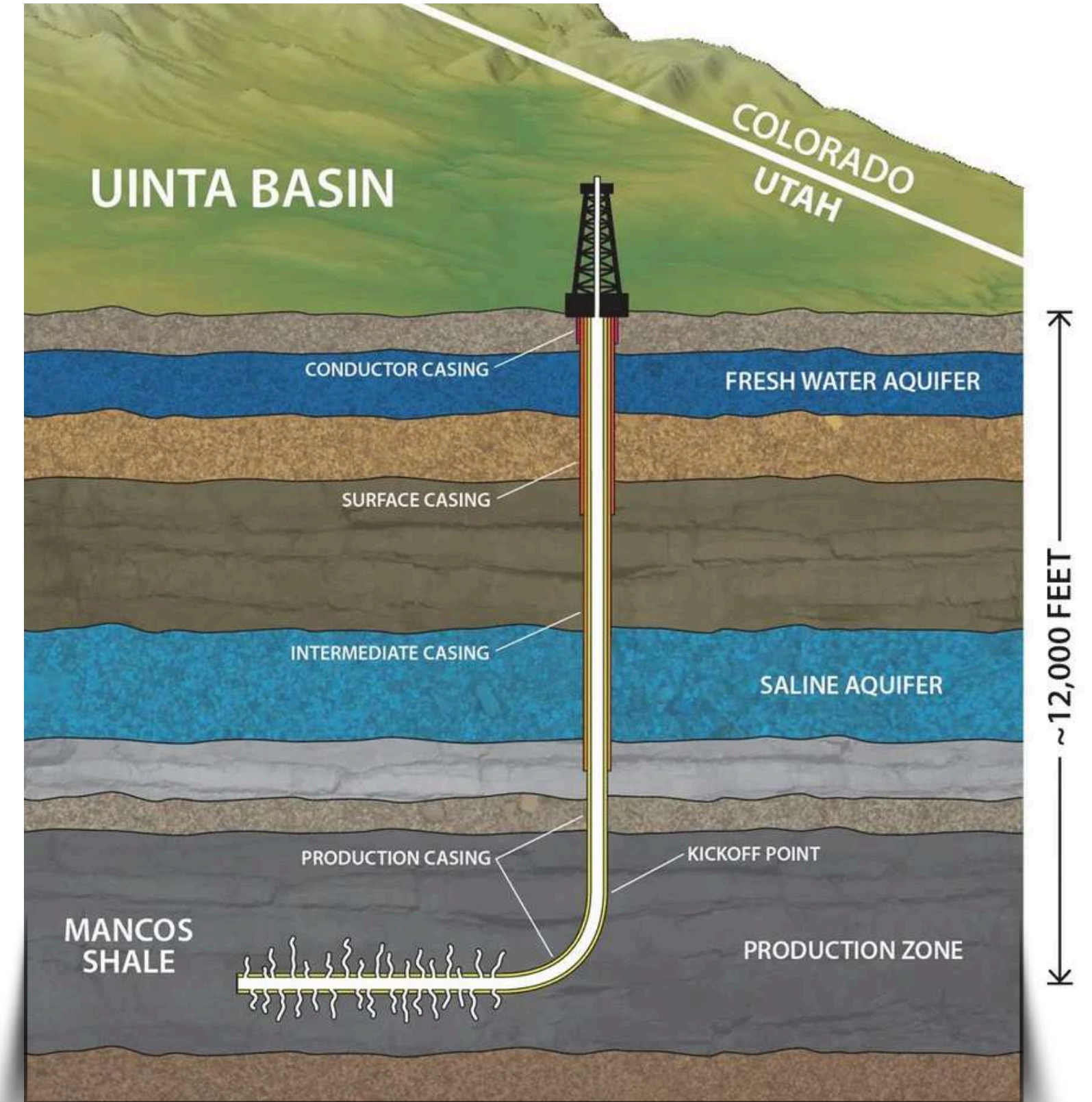
# American Oil Production

- The scale and velocity of the revolution that unlocked the vast amount of oil and natural gas in America's shale fields was history's fastest and greatest addition to global energy supply. The only comparable expansion happened from 1965 to 1980, with the opening of Saudi Arabia's giant Ghawar oil field, which led to the creation of OPEC and reshaped geopolitics. The pre-pandemic shale expansion added 800% more energy to the U.S. than did the (subsidized) growth of solar and wind combined, and shale energy expanded by nearly double the growth of global solar and wind power production.

Increased Production (Million BOE/Day)

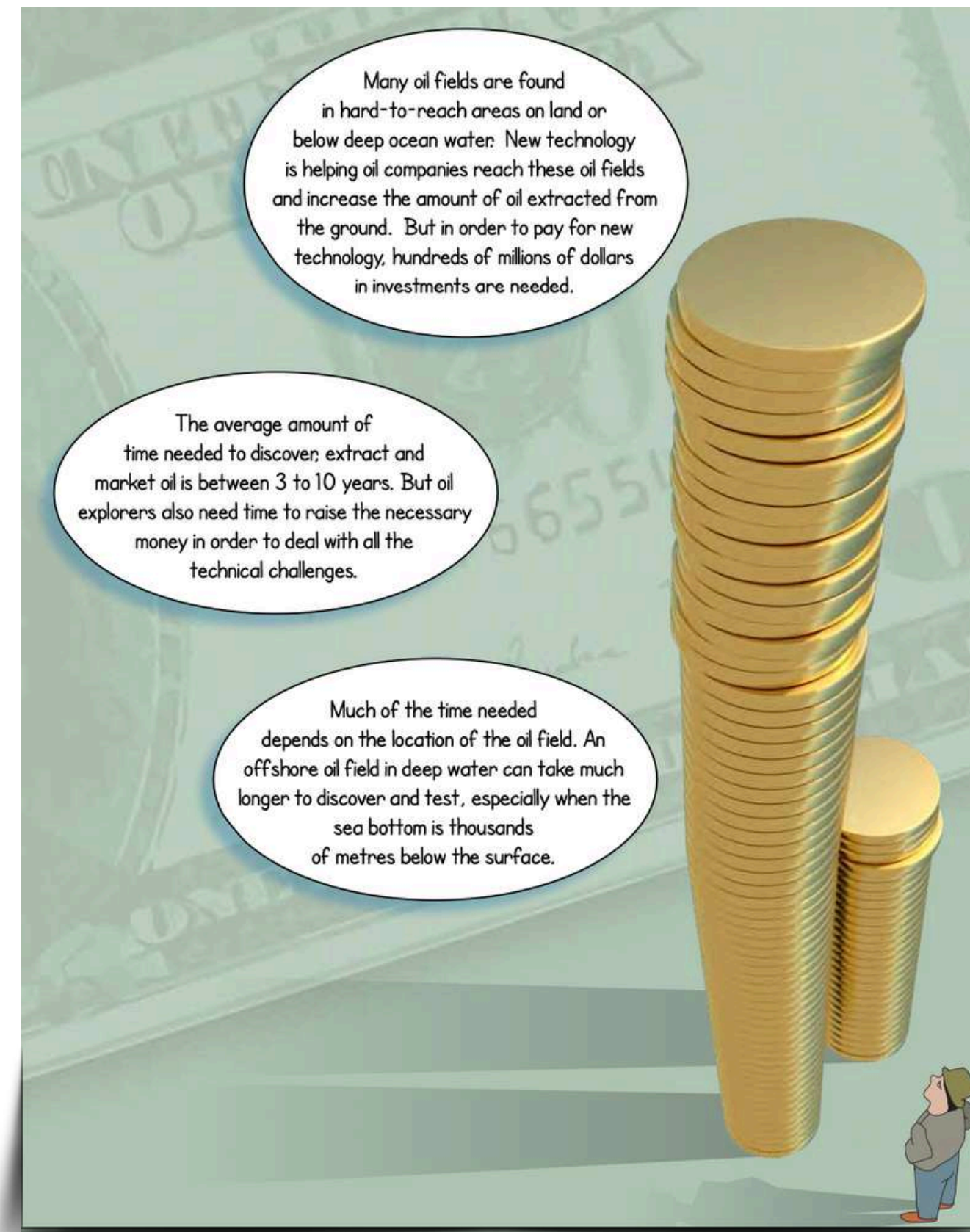
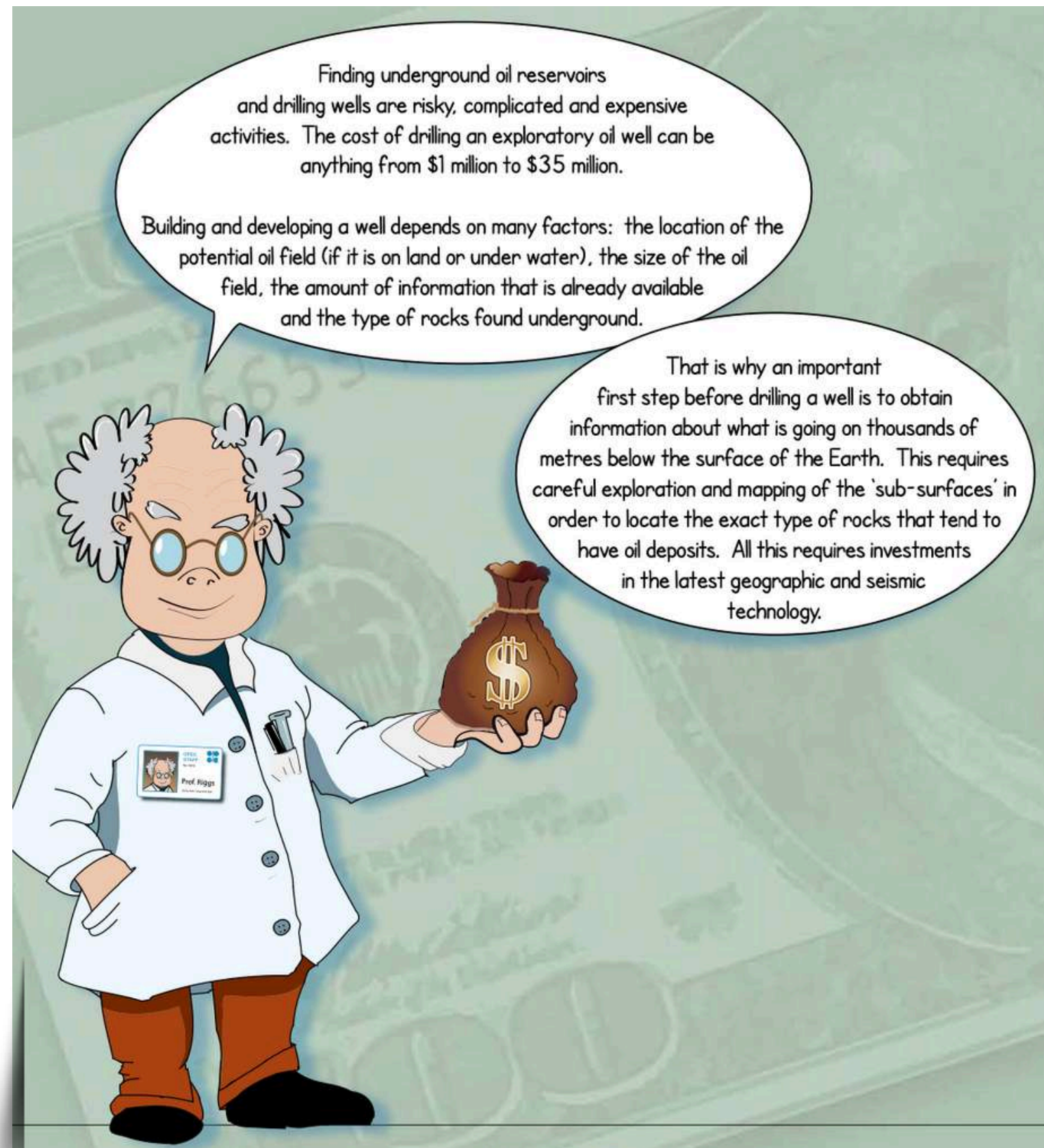


Source: BP, Statistical Review of World Energy 2022





# Huge Investments Are Required





# Upstream Segment

- The upstream segment is characterized by high risks, high investment capital, extended duration as it takes time to locate and drill, as well as being technologically intensive. Virtually all cash flow and income statement line items of E&P companies are directly related to oil and gas production.
- E&P companies are often valued by their oil and gas reserves; these untapped resources are the key to their future earnings. Reserve Replacement Ratio measures the ratio of a company's increase in reserves to its oil and gas production annually.
- So a reserve ratio greater than 100 percent shows a growth in reserves. Reserves usually grow or fall due to revisions, changing recovery rates at existing fields, extensions, new discoveries, and production for the year. Purchases and sales of reserves also apply to reserve replacement, but are not part of organic growth.





# Finding and Development Costs

- Finding and development (F&D) costs are usually expressed as dollars per barrel of oil equivalent (BOE). The technical definition is total costs incurred (purchases of properties, exploration costs, and development costs), divided by the summation of annual proved reserves on a BOE basis—attributable to revisions of previous estimates, purchases of minerals in place, and discoveries and extensions.





# Operating Costs and Exploration Expense

- Operating costs include labor and energy costs, repair, maintenance, taxes, insurance, and depreciation. Also called lifting or production costs, operating costs aggregate the field-by-field costs for extracting oil and gas from the ground.
- Exploration expenses are costs incurred when exploring a region for oil and gas resources, including drilling costs and non-drilling costs (such as seismic surveys). Exploration expenses can vary significantly due to the great differences in oil field characteristics (such as deepwater versus land).





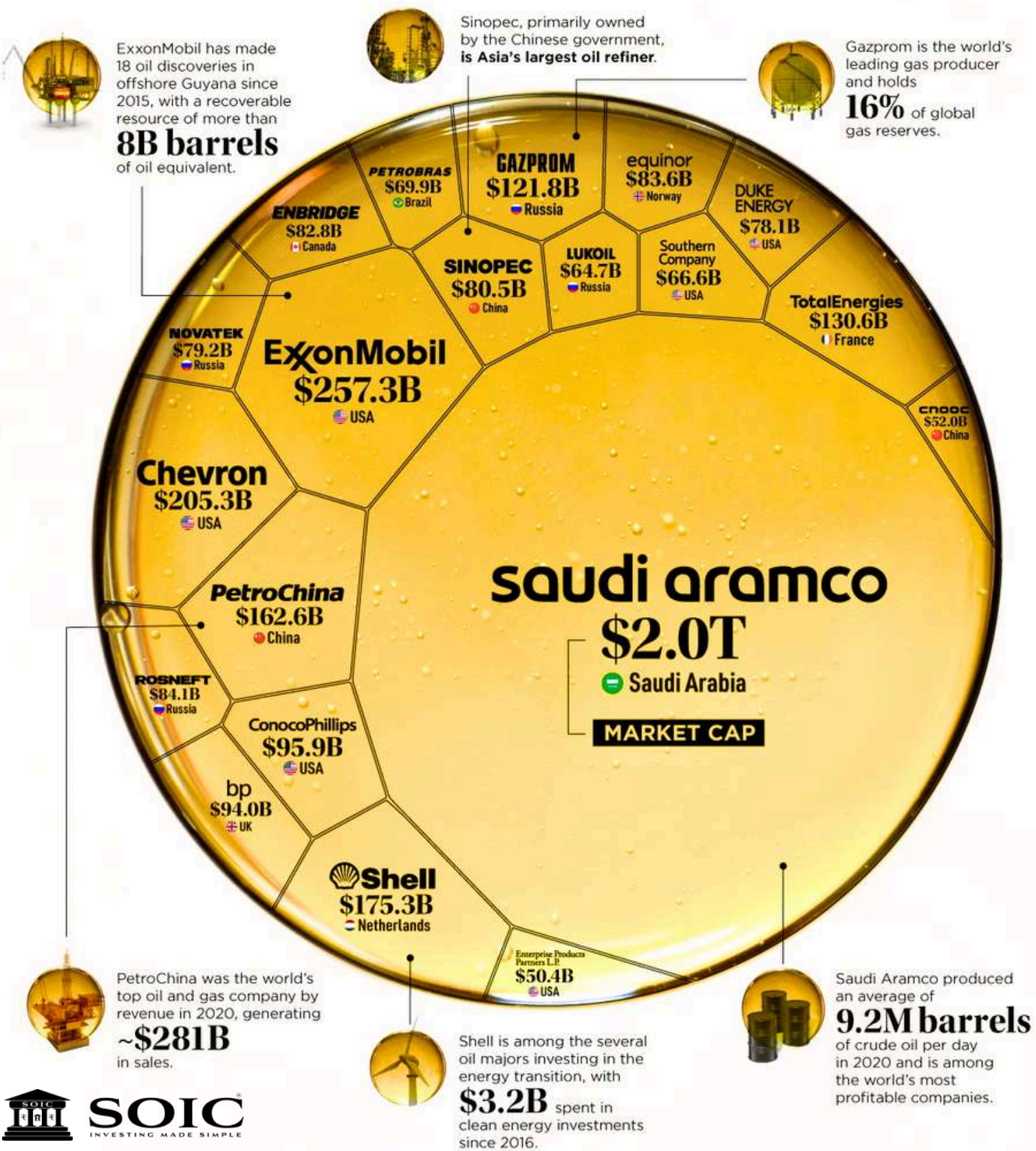


# TOP 20 OIL & GAS COMPANIES IN THE WORLD IN 2021

Despite the headwinds facing the oil and gas industry, oil still makes up the largest share of the global energy mix, fueled by the giant corporations that make up Big Oil.

Here are **the biggest oil and gas companies by market cap\*** in 2021.

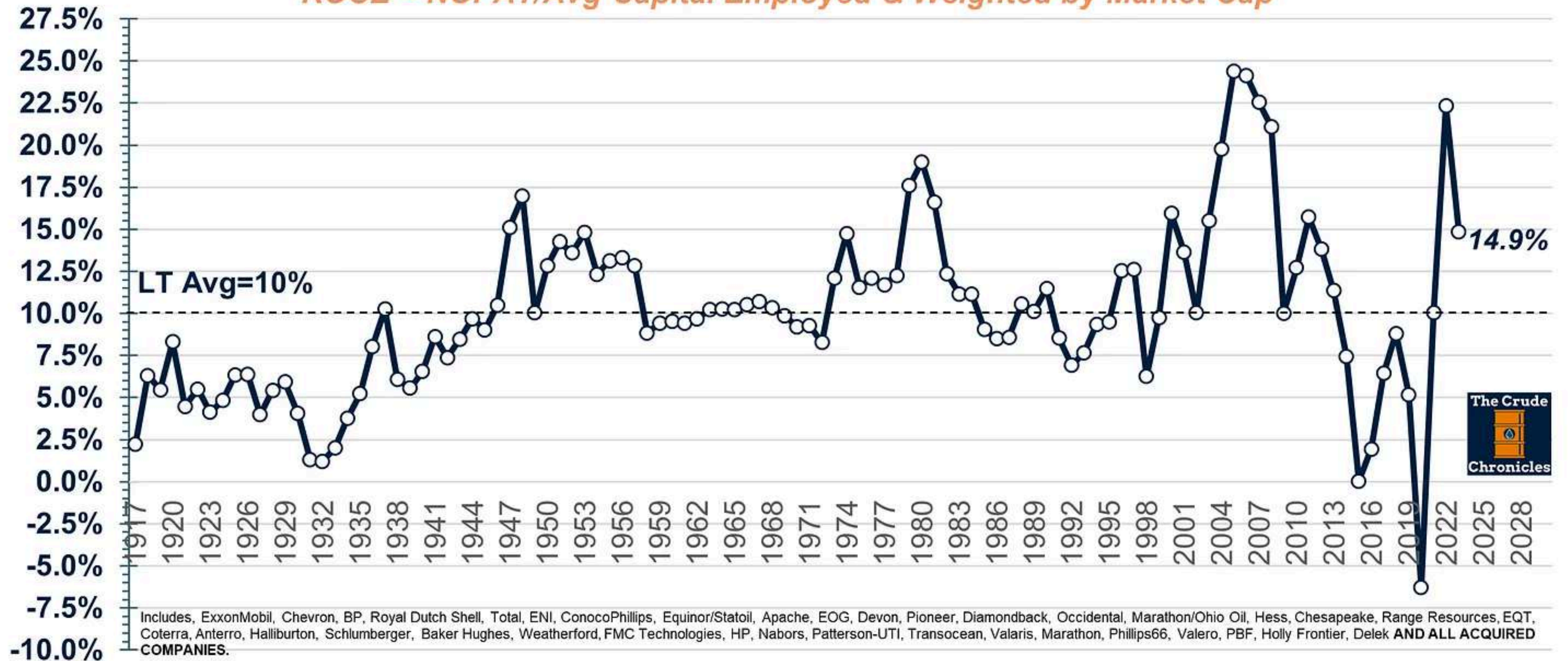
\*Data as of October 7, 2021. Figures have been rounded.





# Oil & Gas Industry Weighted Avg \* Returns on Capital Employed (ROCE)

*ROCE = NOPAT/Avg Capital Employed & Weighted by Market Cap*





# Oil & Gas Equipment & Services

Oil & Gas Equipment & Services firms like Schlumberger, Halliburton, and Baker Hughes provide all the equipment, services, and expertise required for oil field exploration, development, and production. They range from firms with specialized expertise in particular niches of the industry to total solution providers. Some key examples are:

- Drilling equipment: Equipment used for oil and gas drilling, such as drill bits, drilling fluids, mud pumps, drill pipes, and wellhead equipment.
- Pressure pumping services: Services include well cementing and stimulation, used to enhance production from wells.
- Wireline services: Data recording using electronic instruments lowered into wells.
- Directional drilling and measurement: Tools and services assisting in directional drilling and data recording.





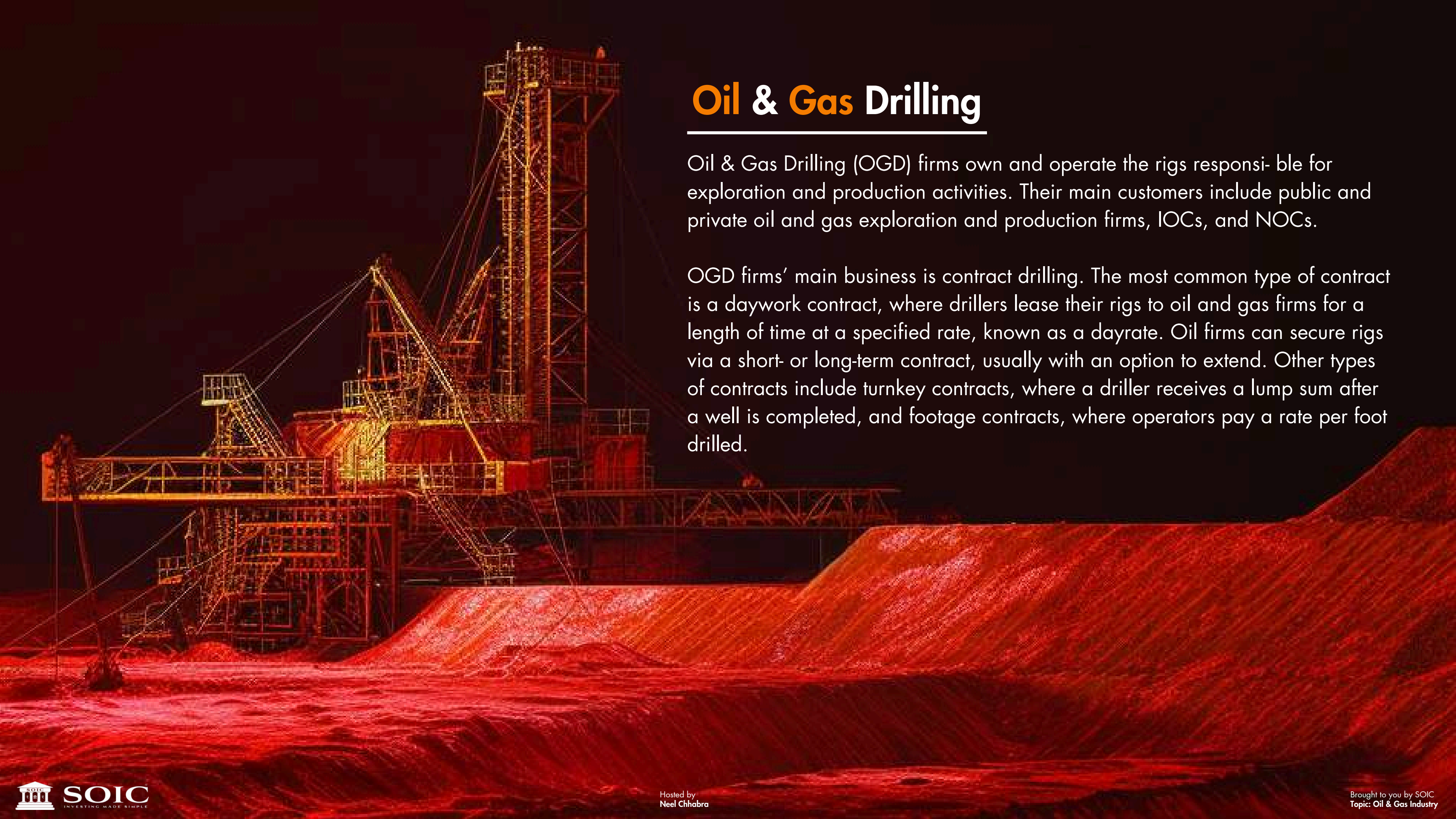


# Operating Costs and Exploration Expense

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- Seismic imaging and analysis: Data and analysis detecting the presence of oil and gas deposits using sound waves.
- Engineering and construction services: Designing, building, and operating massive oil and gas infrastructure such as refineries, LNG plants, rigs, and offshore platforms.
- Helicopters and boats: Transportation services like ferrying workers and equipment to and from offshore rigs.





# Oil & Gas Drilling

Oil & Gas Drilling (OGD) firms own and operate the rigs responsible for exploration and production activities. Their main customers include public and private oil and gas exploration and production firms, IOCs, and NOCs.

OGD firms' main business is contract drilling. The most common type of contract is a daywork contract, where drillers lease their rigs to oil and gas firms for a length of time at a specified rate, known as a dayrate. Oil firms can secure rigs via a short- or long-term contract, usually with an option to extend. Other types of contracts include turnkey contracts, where a driller receives a lump sum after a well is completed, and footage contracts, where operators pay a rate per foot drilled.



# Oil & Gas Drilling

OGD firms tend to have diverse rig fleets. Rigs vary by type but are mainly classified as either land or offshore. These also vary by drilling depth capacity and how far offshore they can drill. The following is a basic description of the main types of rigs:

1. Land rig. Designed to drill on dry land, categorized by size and drilling depth capability. Heavy-duty land rigs can drill over 30,000 feet.
2. Submersible. Offshore rig floating on water when moved, but is submerged so the lower part of the rig hits the seafloor. This rig typically operates in wetlands and swamps and is capable of operating in water depths up to 85 feet, drilling over 30,000 feet.
3. Jack-up. Offshore rig with legs extending to the sea floor, used to “jack up” the hull to sit above the height of the highest anticipated waves. This rig generally operates in shallow waters less than 400 feet deep but can drill over 30,000 feet.

4. Semi-submersible. Offshore rig submerged a few feet below the water’s surface, using air-filled steel floats on which the rig sits, held in place by anchors. This rig operates in intermediate to deepwater and can drill more than 30,000 feet, in water up to 10,000 feet deep.

5. Drill ship. Ship drilling while floating on the water’s surface. Drill ships can drill more than 30,000 feet, in water up to 10,000 feet deep.

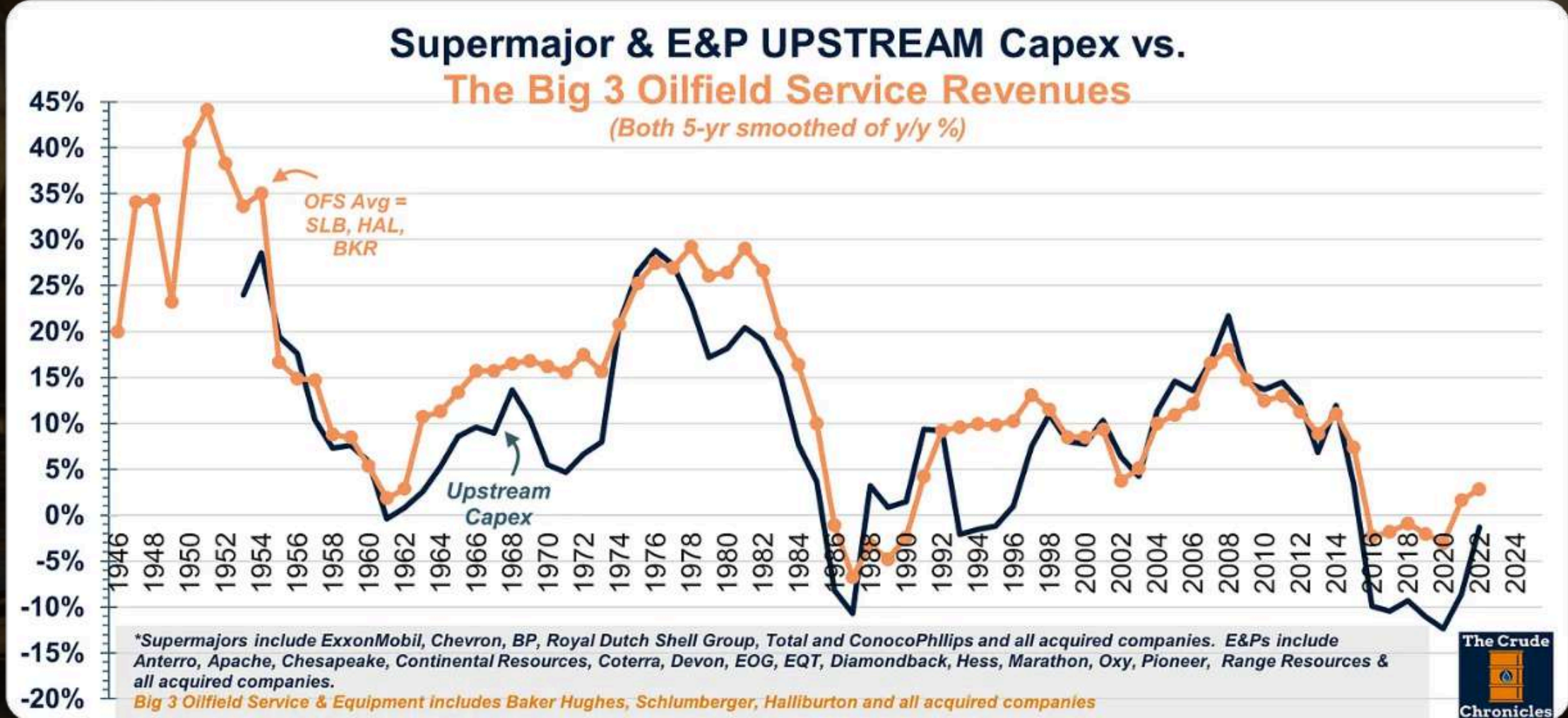
Another type of rig, a workover, is designed specifically to provide periodic maintenance as well as major repairs and modifications of oil and gas wells. Workovers use a variety of tactics to keep oil and natural gas flowing efficiently after a well is drilled, including modifying well depths, repairing leaks, and sealing off depleted zones.



# Oil & Gas Drilling

The following drivers are what we the most important specifically to the Energy Equipment & Services Industry:

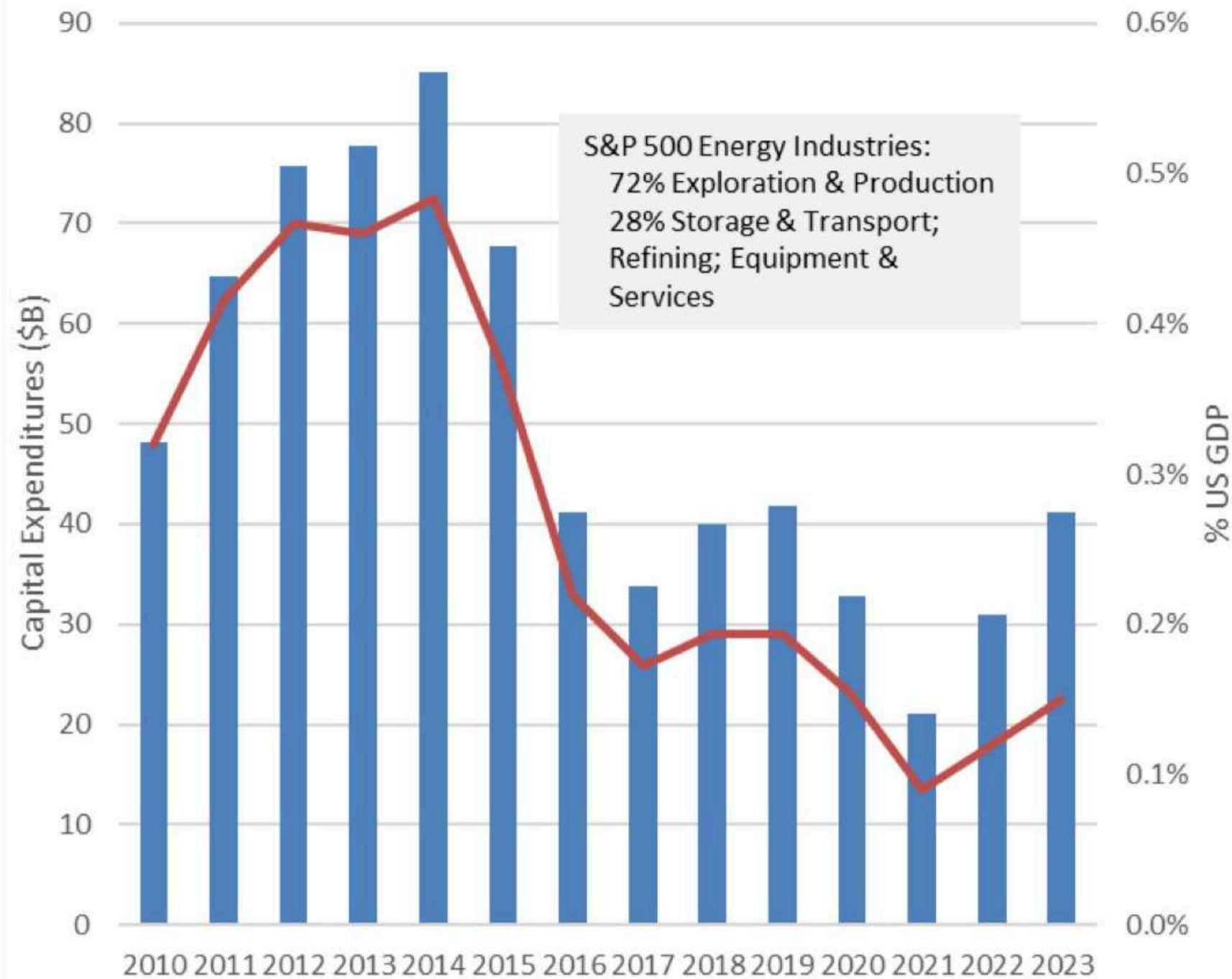
Upstream capital expenditures: Firms allocate their upstream capital expenditures between three main segments: exploration and development of new properties, acquisitions, and production from existing assets. In periods of rising oil and natural gas prices, firms tend to first increase drilling activity at existing sites to maximize production and profits. In periods of declining oil and natural gas prices, firms tend to reduce drilling activity and focus more on acquisitions and developing new reserves.





# Energy Sector

S&P Energy Sector Capital Expenditures



Source: Bloomberg

In 2022, the energy sector spent 64% less on capital expenditures than in 2014. Irrespective of oil companies' benefit from technology improvements, their equipment suppliers' revenues are down 31% from 2014, and capital expenditures are down over 50%. More importantly, their property, plant, and equipment (PPE) - their capacity to produce more of the product necessary for oil companies to produce more product - is also down over 50%.

## Halliburton Company (HAL) and Schlumberger Ltd. (SLB) Combined Selected Financial Data

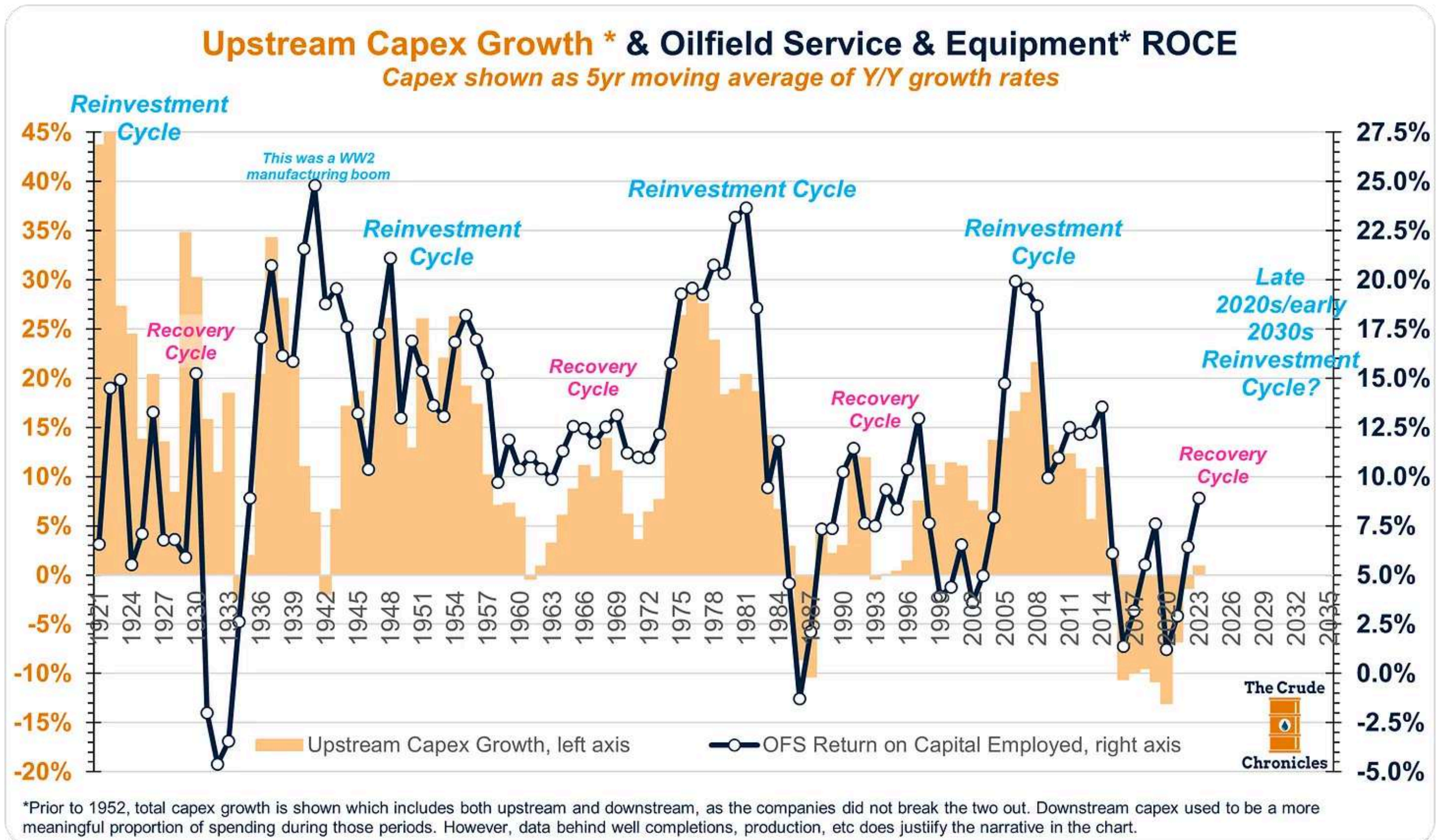
| (\$ mill) | Revenue         | PP&E            | Capital Expend. |
|-----------|-----------------|-----------------|-----------------|
| 2014      | \$81,450        | \$27,871        | \$7,259         |
| 2015      | \$59,108        | \$25,532        | \$4,594         |
| 2016      | \$43,697        | \$21,353        | \$2,853         |
| 2017      | \$51,060        | \$20,097        | \$3,480         |
| 2018      | \$56,810        | \$20,554        | \$4,186         |
| 2019      | \$55,325        | \$18,811        | \$3,254         |
| 2020      | \$38,046        | \$12,637        | \$1,844         |
| 2021      | \$38,224        | \$12,289        | \$1,940         |
| 2022      | \$48,388        | \$11,868        | \$2,629         |
| 2023      | <u>\$56,153</u> | <u>\$13,228</u> | <u>\$3,318</u>  |
| Change:   | (31.1)%         | (52.5)%         | (54.3)%         |

Source: Company reports





# Upstream Capex





# Oil & Gas Drilling

- World wide rig-count: Rig demand is driven primarily by oil and natural gas prices. At first, high commodity prices will increase drilling demand at sites already in production as oil firms hope to extract as much oil and gas as possible. Should prices stay high, firms may wish to also increase exploration for new supplies. Also, many fields in the world require more drilling to offset the effects of aging fields experiencing depletion. This phenomenon is likely to accelerate as peak production at the field, and more consequentially, on a global level, approaches. Sustained periods of low commodity prices have the opposite effect—reduced drilling demand.
- When rig demand increases or decreases, the industry's supply response is to build new rigs or take rigs out of service. Rig supply will vary greatly depending on the type of rig. (In general, the most common types of rigs are land rigs, followed by jack-ups, semi-submersibles, and drill ships.) It may take months to build a new land rig, but years to build a new drill ship
- Rig Utilisation: It is often characterized by strong rig demand, rising dayrates, and limited rig supply growth. Falling rig utilization is characterized by weak rig demand, falling dayrates, and rigs taken out of service.
- Day-rates: The vast majority of rigs work on dayrate contracts. Dayrates are driven by supply and demand for rigs and differ greatly depending on the rig type, age, and geography. In general, sustained periods of high oil and gas prices generally lead to higher rig demand and dayrates. Dayrates also tend to be an indicator of the future rig supply. When dayrates remain high or are rising for a certain rig type, it typically follows that more of those rigs will be built and enter service. Low or falling dayrates typically put on hold plans to build new rigs.



# Bullish and Bearish Drivers of Energy Equipment and Services

## Bullish Industry Drivers

Rising energy capital expenditures

Rising world rig count

Rising rig utilization

Rising dayrates

## Bearish Industry Drivers

Falling energy capital expenditures

Falling world rig count

Falling rig utilization

Falling dayrates



# IMPORTANT QUESTIONS TO ASK

**Rig Types.** Knowing the firm's rig mix will help determine which commodities and regions have the greatest effect on results. What's the breakdown of the firm's rig fleet (land or offshore, shallow or deepwater, etc.)? Do these rigs drill primarily for natural gas or oil?

**Contract Types.** Which type(s) of contracts does the firm use to contract its rigs (daywork, footage, or turnkey)? What percentages of its fleet use each? Generally, daywork contracts are viewed more favorably because they provide stable revenues, but this isn't always the case. Take a keen look at a firm's contract structure to determine its prospects for profitability.

**Dayrates.** What has been the trend of the firm's dayrates? What is the potential for dayrate expansion once its rig contracts expire? Dayrates can vary somewhat between rig operators but tend to follow industry trends.

**Contract Length.** When do the firm's current rig contracts expire? How many of the firm's rigs are due to come off contract in the short or long term? Depending on industry conditions, a short or long contract can be beneficial or harmful.

**Rig Quality.** Are the firm's rigs more technologically advanced than competitors? Do oil and gas firms pay a premium for the firm's rigs? The drilling industry is highly competitive, so having technologically advanced rigs can help in pricing negotiations.



# IMPORTANT QUESTIONS TO ASK

**Rig Location.** Where are the rigs located? What are the supply– demand dynamics in each region? Is it the firm’s strategy to a specific region or expand geographically? How easily can it move its rigs to more attractive locations? Rig demand varies considerably by region, thus firms able to adapt quickly to changing supply and demand dynamics have the ability to reap rewards.

**Rig Utilization.** What percentage of the firm’s rigs is in service? Is utilization growing, stable, or falling? This should help to gauge demand of the firm’s rigs and whether it has the capacity to take on more business.

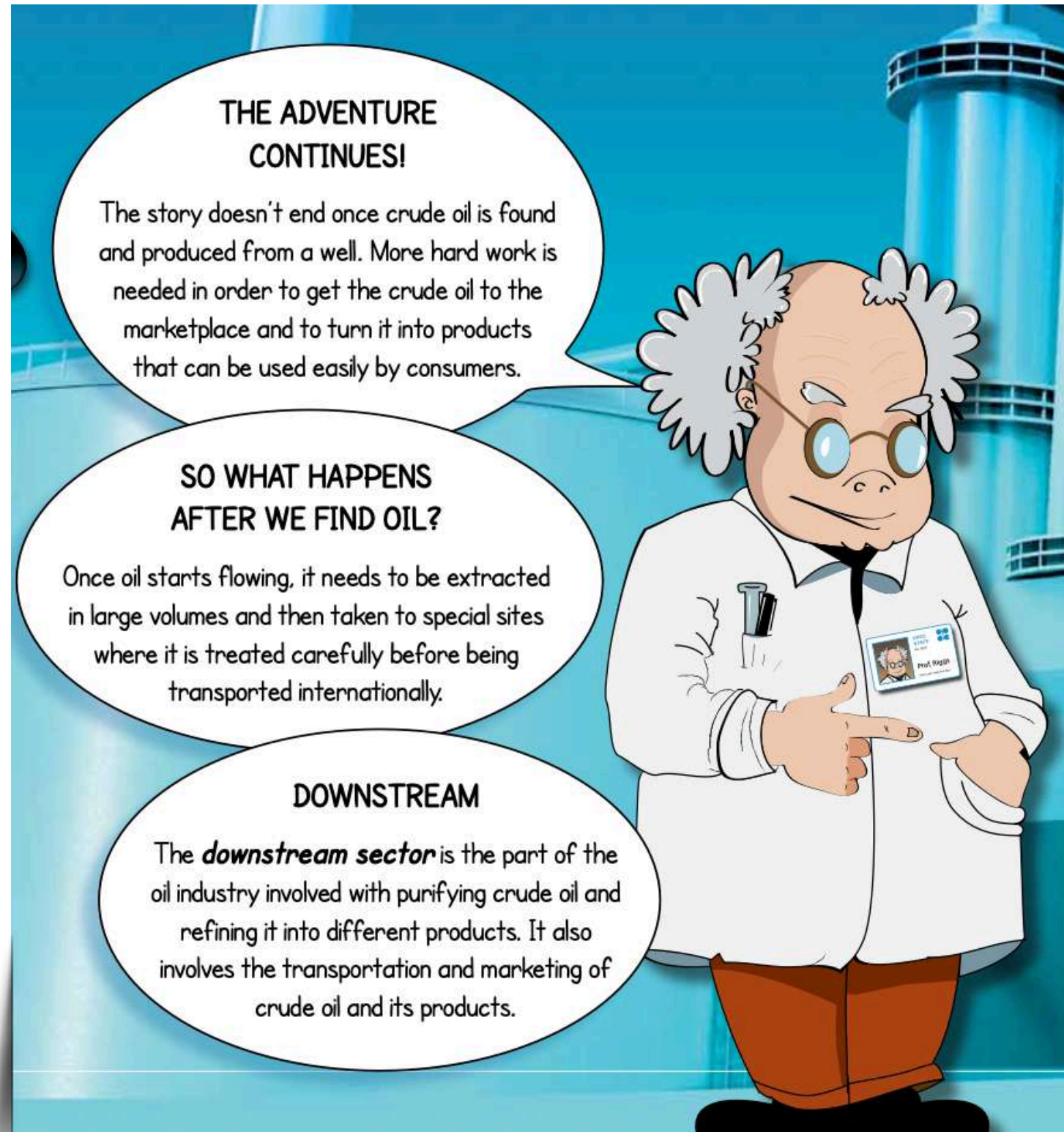
**Rig Growth Strategy.** What is the firm’s strategy to grow its asset base? Does it plan to build new rigs or acquire a competitor? At what pace can it build new rigs?

**Rig Fleet Age.** What is the average age of the firm’s fleet? Is it expanding its rig fleet or retiring old rigs? To what extent will the firm need to take rigs out of service for upgrades or maintenance?

**Operating and Maintenance Costs.** How do the firm’s operating and maintenance costs compare with peers? What is the firm’s strategy to mitigate industry cost inflation? Because rig firms often secure steady cash flows through contracts, their ability to control cost inflation will have a big impact on margins.



# Refining Oil: **Downstream**



**THE ADVENTURE CONTINUES!**

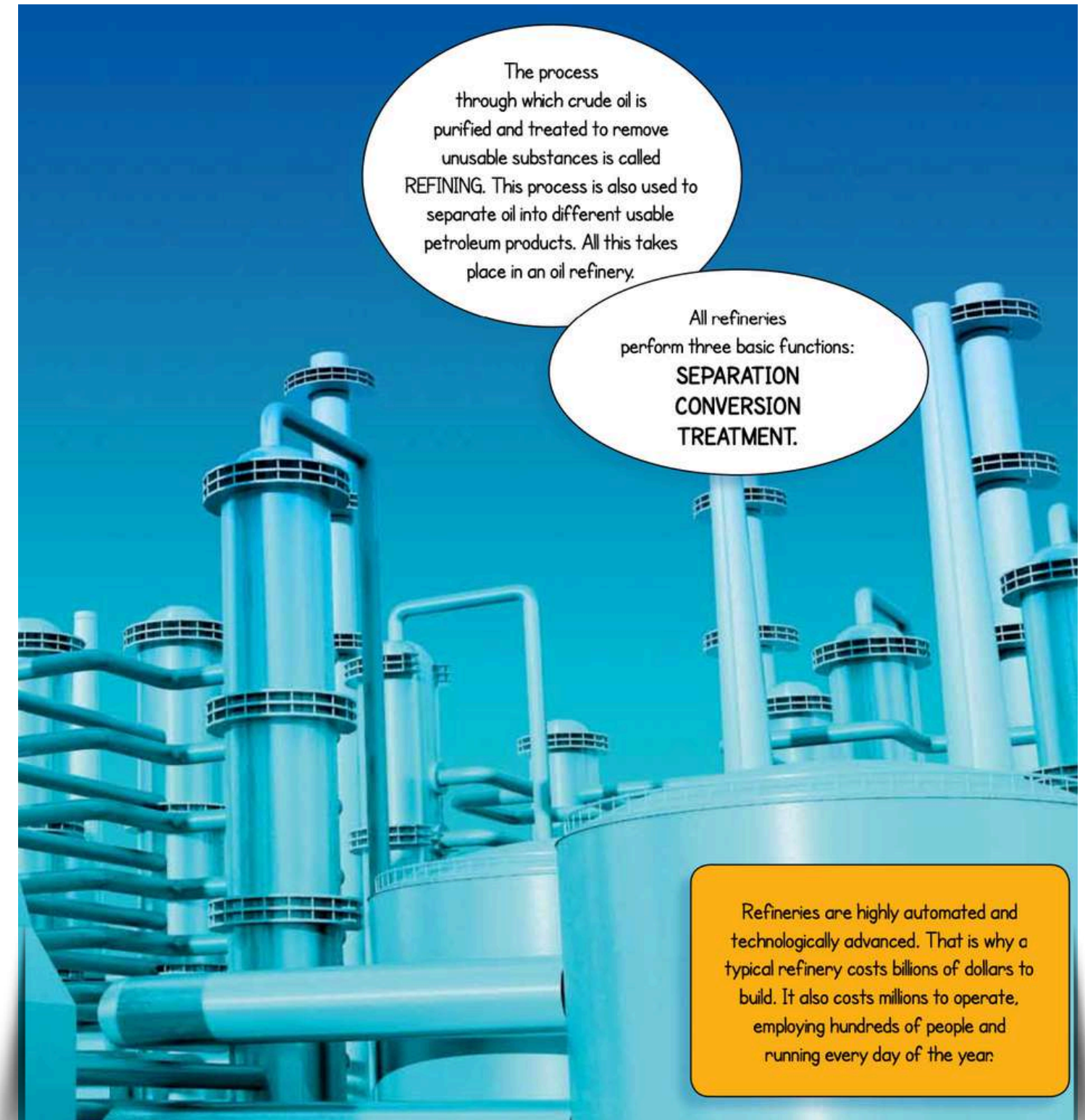
The story doesn't end once crude oil is found and produced from a well. More hard work is needed in order to get the crude oil to the marketplace and to turn it into products that can be used easily by consumers.

**SO WHAT HAPPENS AFTER WE FIND OIL?**

Once oil starts flowing, it needs to be extracted in large volumes and then taken to special sites where it is treated carefully before being transported internationally.

**DOWNSTREAM**

The **downstream sector** is the part of the oil industry involved with purifying crude oil and refining it into different products. It also involves the transportation and marketing of crude oil and its products.



The process through which crude oil is purified and treated to remove unusable substances is called **REFINING**. This process is also used to separate oil into different usable petroleum products. All this takes place in an oil refinery.

All refineries perform three basic functions:  
**SEPARATION  
CONVERSION  
TREATMENT.**

Refineries are highly automated and technologically advanced. That is why a typical refinery costs billions of dollars to build. It also costs millions to operate, employing hundreds of people and running every day of the year.



## SEPARATION

In this phase, a refinery heats crude oil to different temperature levels. Different parts of crude oil have different boiling points. As the temperature rises, these different parts or fractions are separated. This is done inside distillation towers.

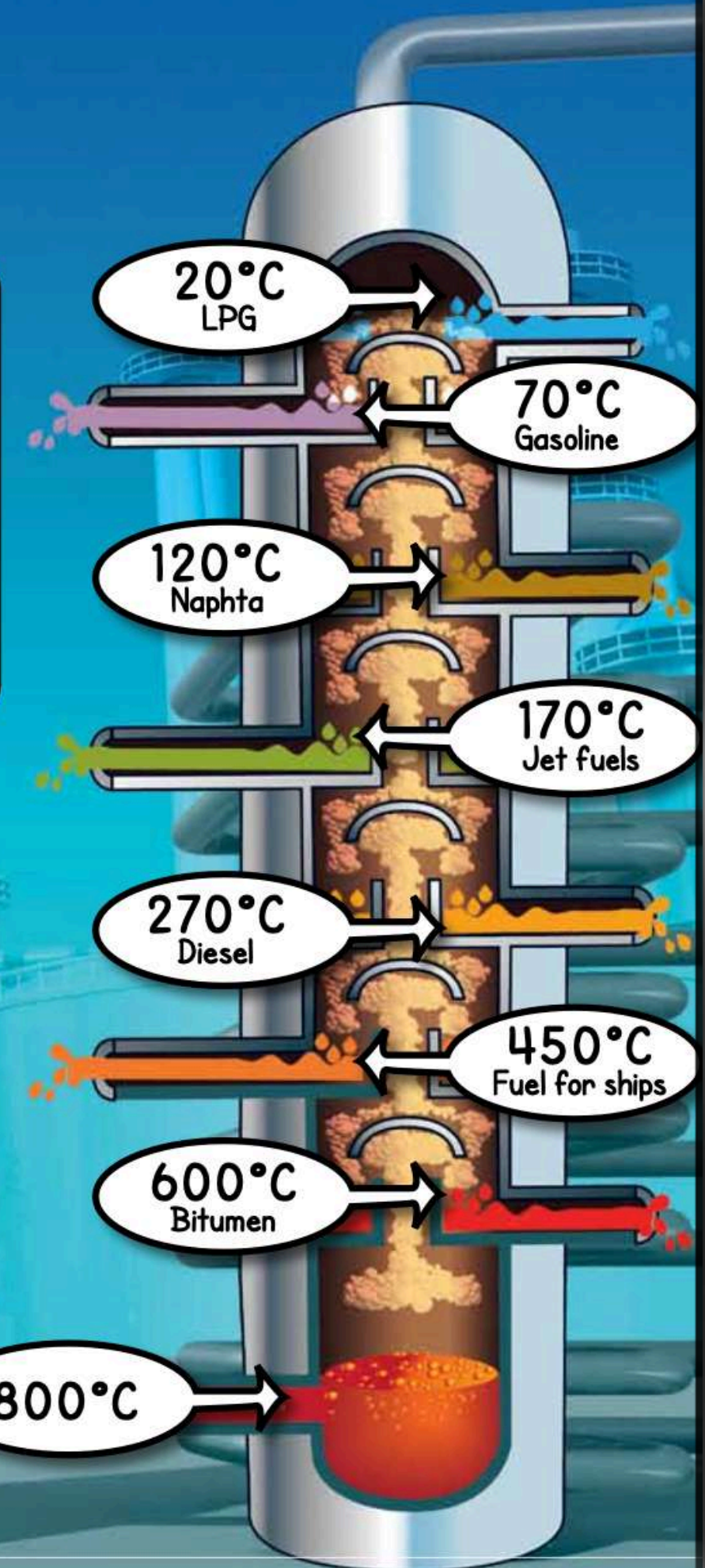
The lightest fractions, which include gasoline and Liquefied Petroleum Gas (LPG), vaporize quickly and rise to the top of the distillation towers. There they condense back into liquids.

The medium-weight fractions, which include kerosene and diesel oil distillates condense in the middle of the distillation tower.

The heaviest liquids (called residual oils) have the highest boiling points and leave through the bottom of the distillation tower.

Refineries also use chemical agents called **CATALYSTS**. These help refine oil further by either removing carbon or adding hydrogen.

Refining methods are constantly being improved. A variety of complex operations are now used which have improved their output. Refineries today turn more than half of every barrel of crude oil into gasoline. One barrel is equal to 159 litres (42 US gallons). This is a big advance from only 70 years ago when only 41 litres (11 gallons) of gasoline were produced from each barrel of crude.

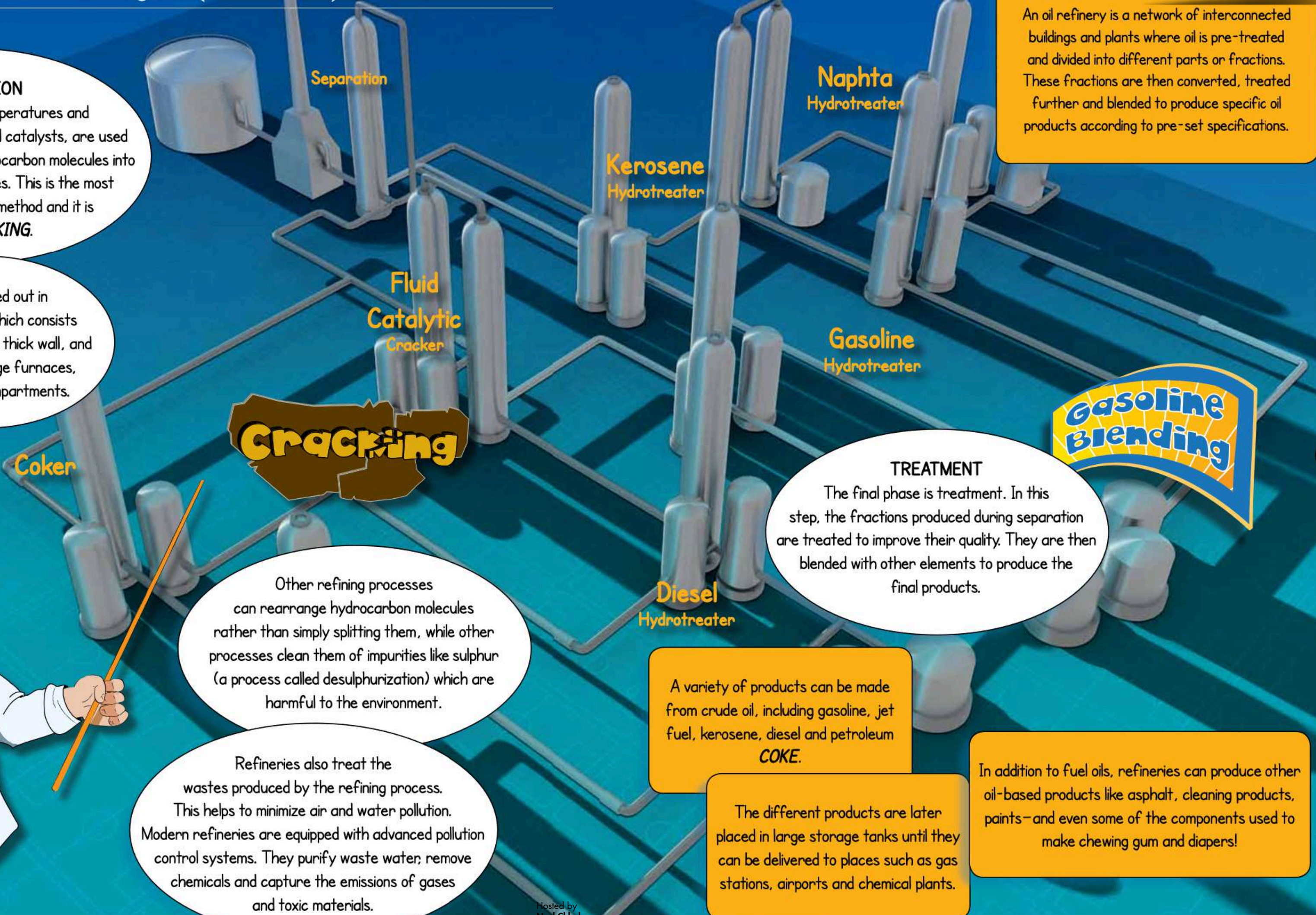




An oil refinery is a network of interconnected buildings and plants where oil is pre-treated and divided into different parts or fractions. These fractions are then converted, treated further and blended to produce specific oil products according to pre-set specifications.

**CONVERSION**  
In this phase, high temperatures and pressure, as well as chemical catalysts, are used to 'crack' or split heavy hydrocarbon molecules into smaller, more desirable ones. This is the most widely used conversion method and it is called **CRACKING**.

This is carried out in a cracking unit which consists of a reactor with a thick wall, and a network of large furnaces, dividers and compartments.



**TREATMENT**  
The final phase is treatment. In this step, the fractions produced during separation are treated to improve their quality. They are then blended with other elements to produce the final products.

Other refining processes can rearrange hydrocarbon molecules rather than simply splitting them, while other processes clean them of impurities like sulphur (a process called desulphurization) which are harmful to the environment.

Refineries also treat the wastes produced by the refining process. This helps to minimize air and water pollution. Modern refineries are equipped with advanced pollution control systems. They purify waste water; remove chemicals and capture the emissions of gases and toxic materials.

A variety of products can be made from crude oil, including gasoline, jet fuel, kerosene, diesel and petroleum **COKE**.

The different products are later placed in large storage tanks until they can be delivered to places such as gas stations, airports and chemical plants.

In addition to fuel oils, refineries can produce other oil-based products like asphalt, cleaning products, paints—and even some of the components used to make chewing gum and diapers!





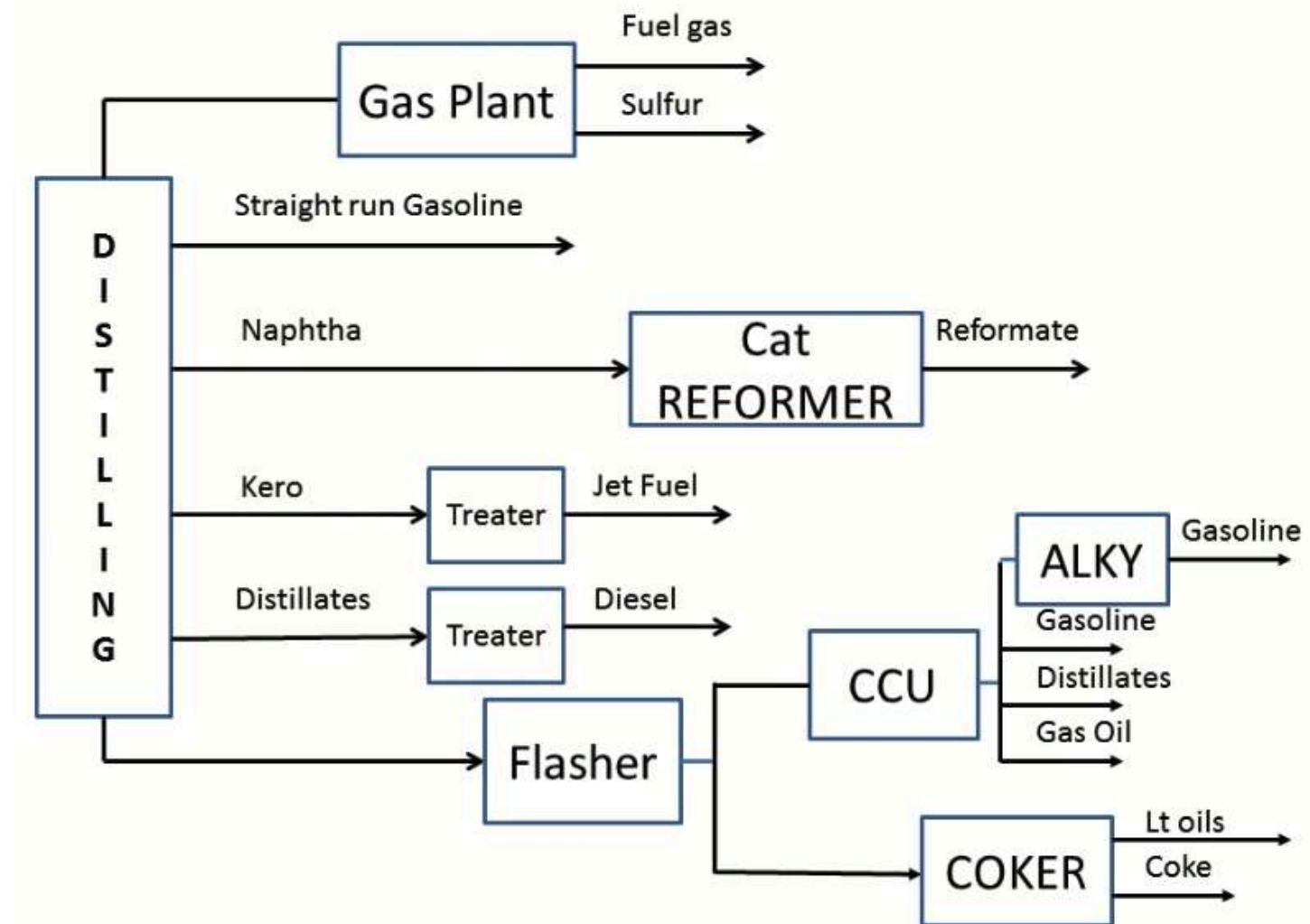
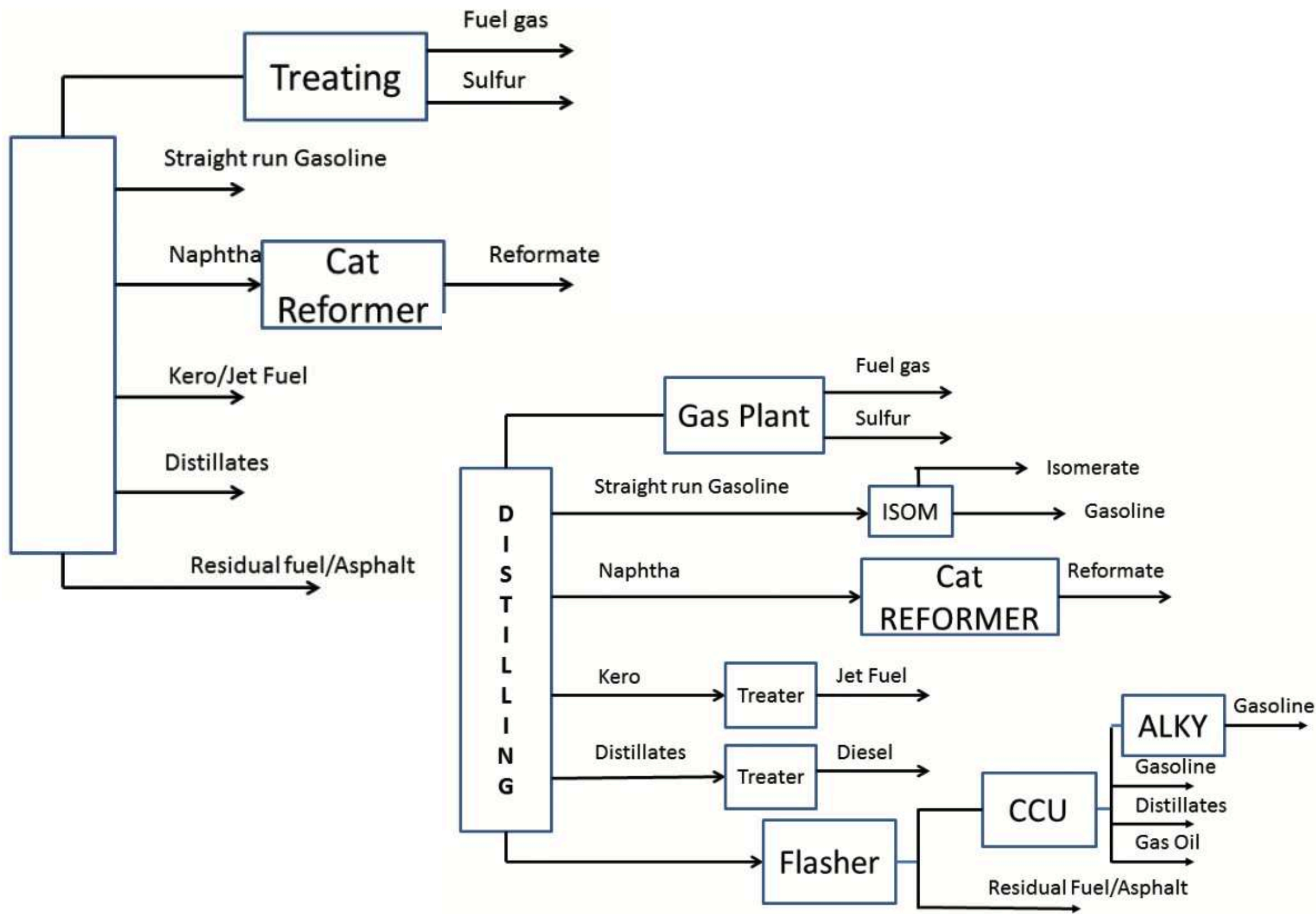
# Simple and Complex Refineries

In the 1980s, the businessmen running refineries made an intellectual breakthrough that coincided with the revolution underway in information technology. They recognized that the prices for crude oil, gasoline, distillates, and residual fuels connected to each other in a profound way. The kinds of hardware in refineries around the world not only impacted the profitability of refineries; they set, in large part, the margins between crude and products, and the differences between crudes themselves.

The constellation of relationships tying crude to products had a prerequisite. Prices had to be free to respond to market forces—the changing preferences of consumers, the sometimes capricious wishes of crude producers, and the influence of other fuels: natural gas, coal, and even renewables. The loss of oligopolistic control by the “Seven Sisters,” the assumption of it by OPEC, and the rise in spot markets during the 1970s provided the necessary setting.

The essence of the changing price relationships between crude and products and the resulting profitability was the relative ability of a refinery to produce various slates of products. The key concept was simple—the more valuable products a refinery makes, the more it can pay for crude oil. The more valuable products are, of course, the light oils, gasoline, jet fuel, and distillate fuels. The less valuable products are the heavy oils or black oils, i.e., residual fuel. From this theorem refiners adapted an old set of nomenclature—simple, complex, and very complex refineries.







**As you move from Simple to Complex to very Complex, the Gasoline Yield (and gain) goes up and the Resid Yield goes down.**

---

|                 | Percent Yields<br>from Medium Crude |         |              |
|-----------------|-------------------------------------|---------|--------------|
|                 | Simple                              | Complex | Very Complex |
| Gasoline        | 30                                  | 50      | 60           |
| Jet fuel        | 10                                  | 10      | 10           |
| Distillate fuel | 20                                  | 25      | 25           |
| Residual fuel   | 35                                  | 10      | —            |
| LPG             | —                                   | 3       | 4            |
| Coke            | —                                   | --      | 3            |
| Refinery fuel   | 8                                   | 12      | 13           |
| Gain            | (3)                                 | (10)    | (15)         |



# Refiners Margins

A refiner's margins are a function of his own refinery's processing units, the crudes he buys, and the prices he receives for his products. But the prices and costs are set by an elaborate constellation of industry refinery economics that mimic the refiner's own. The relationship between product prices and crude prices has more moving parts than just the refineries themselves. And that makes complexity watching vital as refiners try to catch early warning signals for significant profit-ability changes.

## The Crack Spread

In order to approximate refinery gross profit margins, a commonly used ratio is the 3-2-1 crack spread (the term crack spread comes from the stage of the refining process in which molecules are split—cracked apart—to separate what will become different final products). This ratio refers to the proportion of gasoline and heating oil produced for each barrel of oil—for three parts oil, two parts are made into gasoline and one into heating oil.

Refining margins are the difference between petroleum product prices and the cost of crude oil, usually quoted in dollars per barrel. They are the main earnings drivers for the downstream segment of the Oil & Gas industry.

Refining margins differ greatly depending on the country or region because petroleum product prices differ by country due to regional supply and demand conditions, tax policies, and government intervention (like gasoline price caps in several emerging market nations).



# Refiners Margins

In essence, Refineries vary in complexity—the ability to process different types of crude—thus the cost spread between light and heavy oil greatly affects earnings. The greater a refinery's ability to process cheaper, heavier crude, the greater the advantage it has during periods of large spreads in light and heavy crude oil.

The spread between light and heavy crude oil prices changes in response to changes in heavy oil production, primarily by OPEC. When OPEC decides to reduce production, it first reduces production of cheaper, heavier crude oil, reducing supply and increasing prices, thus closing the gap between heavy and light crude. When OPEC increases production, it raises supply of heavy crude, reducing prices and increasing the gap between light and heavy oil. So light oil tends to sell at a premium to heavy oil in periods of strong global oil demand.

## Bullish Refining Margin Drivers

- Rising petroleum product demand
- Falling crude oil prices
- Rising refining utilization
- Falling petroleum product inventories
- Falling petroleum product imports
- Rising light/heavy spread

## Bearish Refining Margin Drivers

- Falling petroleum product demand
- Rising crude oil prices
- Falling refining utilization
- Rising petroleum product inventories
- Rising petroleum product imports
- Falling light/heavy spread



# IMPORTANT QUESTIONS TO ASK

**Revenues and Earnings Breakdown.** How is the firm's revenue and earnings divided between exploration and production, refining and marketing, or other divisions like petrochemicals? This will reveal the firm's relative sensitivity to commodity prices and petroleum products.

**Production and Reserve Breakdown.** How are the firm's production and reserves split between oil, natural gas, or some other mix? This mix between oil, natural gas, and other hydrocarbons will show which commodity has a greater effect on the firm's operations. This mix can explain significant differences in performance among firms, as oil and natural gas prices can diverge widely.

**Oil and Gas Production Growth.** Is the firm consistently growing oil and gas production or experiencing declines? What is its production growth relative to peers? Is growth organic or due to acquisitions? What is the firm's strategy to grow oil and gas reserves?



# IMPORTANT QUESTIONS TO ASK

**Hedging.** To what degree does the firm use hedging in its operations? How exposed are the firm's operations to oil and natural gas prices? Firms' hedging strategies tend to show which ones have the most upside and downside relative to where commodity prices end up.

**Reserve Replacement.** What is the firm's history of reserve replacement? How many years can it sustain production with current reserves? Firms with relatively long life reserves may not need to take major risks to grow production in the future. This could be a distinct competitive advantage.

**Exploration and Production Costs.** How do the firm's finding and development costs compare with peers? How do operating costs and exploration expenses compare? Is the firm a particularly low-cost producer or is it engaged in high-cost operations like the oil sands, oil shale, or other unconventional resources? Typically, firms with particularly high-cost operations are more negatively affected by falling commodity prices as they have lower profit margins.



# IMPORTANT QUESTIONS TO ASK

**Geopolitical Risk.** What is the firm's geopolitical risk profile? Where are most of the firm's current and future planned production sites and refining assets? Are the sites in politically stable or unstable countries? What percentage of production comes from politically unstable regions? Does the firm have a solid history of operating in foreign territories? Has it ever had disruptions due to a geopolitical squabble?

While a firm with relatively high exposure to a geopolitically unstable region may face higher risks of government intervention or production disruption, the potential additions to production growth may warrant these risks.

**Government Ownership.** Are the firm's shares owned or controlled by a government, and if so, to what degree? Especially in emerging market countries, governments and powerful families can play a huge role in the Energy sector. Often, government-controlled Energy firms are subject to unique taxes, royalties, subsidies, and price controls. Such inhibitors can spring up unexpectedly—at the whim of the controlling government.



# IMPORTANT QUESTIONS TO ASK

**Cash Flow Use.** How is the firm spending its cash flow? To what degree is it buying back shares, paying dividends, spending on capital expenditures, or paying down debt? Depending on industry conditions, investors may prefer a firm that rewards shareholders with dividends and share buybacks over taking on risky and aggressive new drilling plans.

**Refining Complexity.** How complex are the firm's refineries? Does it do just one kind of refining, or many? How diversified are the firm's sources and types of crude oil? Is it dependent on one supplier or many? Firms capable of processing heavier crudes or a more diverse complexity may have a competitive advantage depending on the differences between light and heavy crude oil.

**Petroleum Product Mix.** A firm's production mix between transportation fuels versus chemicals or other products will determine to what extent factors like economic growth, driving trends, and weather will affect product demand. What is the firm's mix of petroleum products (gasoline/diesel/heating oil/chemicals)?



# Midstream: Transportation & Storage

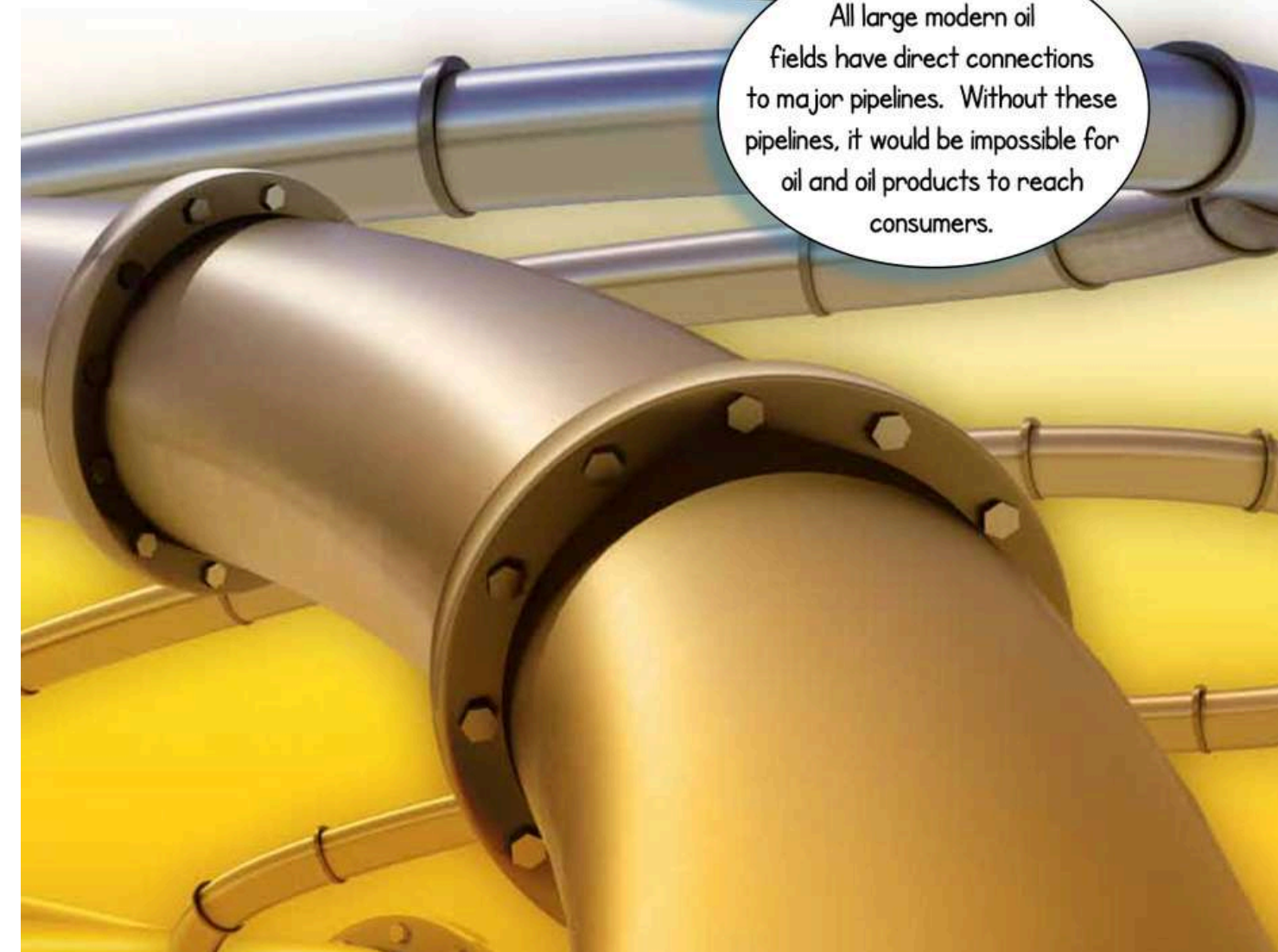
- Getting oil from a well to a refinery and getting the products to service stations requires a complex transportation and storage system.
- The midstream process involves the storage and transportation of hydrocarbons.
- It's essentially the "middle man" between producers and end users.
- Midstream assets include pipelines and crude tankers delivering oil, natural gas, and petroleum products from their origins to refiners.
- They also include storage terminals, where natural gas and oil are held until they're ready to be consumed or transported.

## PIPELINES

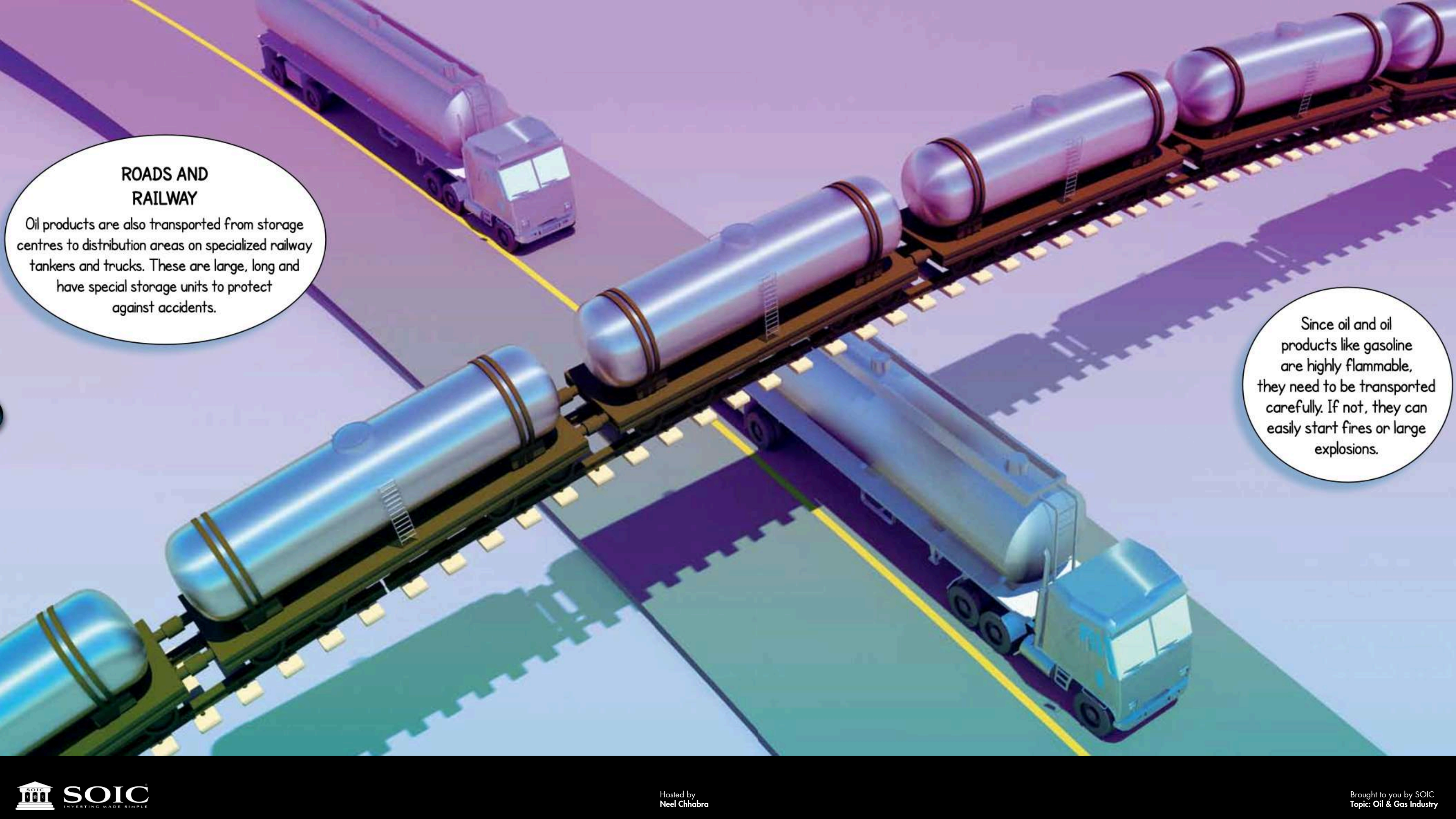
Oil pipelines are made from steel. They usually have an inner diameter of about 10 to 120 cm (about 4 to 48 inches) and can run for hundreds or thousands of kilometres. Advances in technology are constantly improving the size and strength of these pipelines.

Although pipelines are less flexible than other forms of transportation, they are the safest, most efficient and most economical way to move oil.

All large modern oil fields have direct connections to major pipelines. Without these pipelines, it would be impossible for oil and oil products to reach consumers.





An isometric illustration showing a road and a railway track. On the road, a yellow oil tanker truck is driving away from the viewer. On the railway track, several blue oil tanker cars are moving towards the viewer. The scene is set against a light blue sky and a green ground with yellow road markings.

## ROADS AND RAILWAY

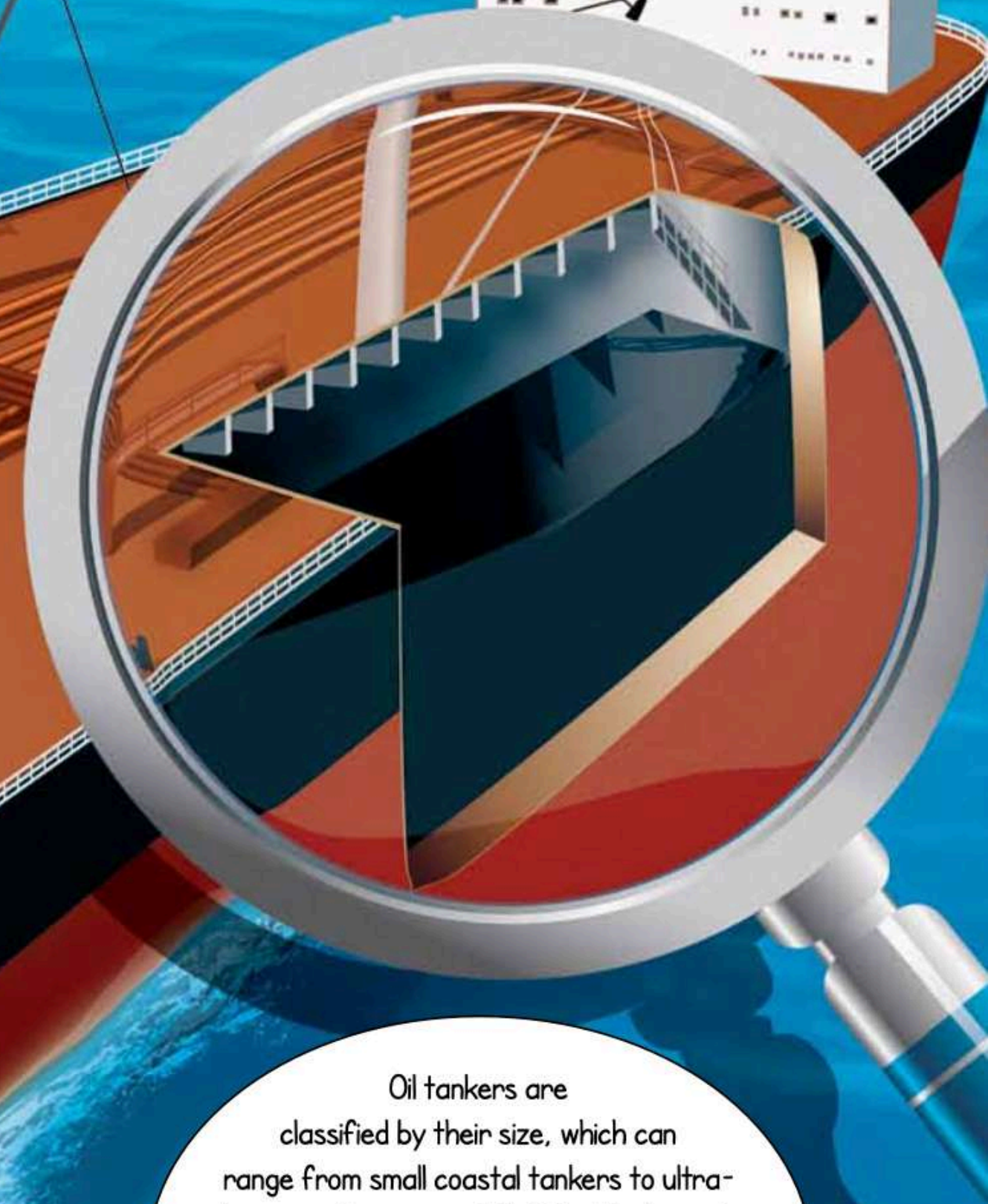
Oil products are also transported from storage centres to distribution areas on specialized railway tankers and trucks. These are large, long and have special storage units to protect against accidents.

Since oil and oil products like gasoline are highly flammable, they need to be transported carefully. If not, they can easily start fires or large explosions.



## OCEAN TANKERS

Oil tankers are large, specialized ocean-going ships specifically built to carry unrefined crude oil from one part of the world to another. They usually carry oil from where it is produced to where it is refined and consumed.



Oil tankers are classified by their size, which can range from small coastal tankers to ultra-large crude carriers (ULCC). The largest tankers often travel from the Gulf region and West Africa to Japan and other Asian countries. Other tankers travel to the northeastern United States and to countries across the Mediterranean region.



# Midstream: Transportation & Storage

- Midstream segment is not directly affected by commodity prices. Instead, Storage & Transportation services are generally fee-based and driven by volume. For example, pipeline firms charge fees for using their pipelines, earning greater revenues as the pipeline use increases. Thus, rising and falling commodity prices only affect revenues to the extent prices change the volume demanded.
- The storage business involves owning and operating fields and storage facilities for a variety of products including crude oil, natural gas, and petroleum products. For example, underground natural gas storage provides pipelines, local distribution firms, producers, and pipeline shippers with inventory management, seasonal supply backup, and access to natural gas needed to avoid imbalances between pipeline network receipts and deliveries.

## A few of the important drivers are:

- Hydrocarbon volumes
- Interest rates
- Demand for defensive/high income-yielding securities





# IMPORTANT QUESTIONS TO ASK

Revenues and Earnings Breakdown. Firms in the storage and transportation sub-industry can have incredibly diverse and complex business operations, making it important to understand what drives overall results. What percentage of the firm's revenues and earnings are regulated or unregulated? Does the firm have production or trading assets? Which of the two is more important to the bottom line?

Competition. What is the competitive landscape of the firm's peers? Does the firm operate in a region with significant barriers to entry? Pipeline operators generally face less competition than firms in the shipping business. This is because, in general, regulatory approval is needed to build new pipelines.

Balance Sheet. Many firms in this sub-industry tend to carry heavy debt loads, so financial health can vary widely. Those with greater financial resources could be more capable of funding future growth opportunities. Does the firm have the financial ability to make large acquisitions to fuel growth? Does the firm's balance sheet allow it to take on additional leverage? Debt isn't necessarily a bad thing—many firms generate an excellent return on borrowed funds. In either case, it's vital to understand the capital structure of a firm.



# IMPORTANT QUESTIONS TO ASK

**Regulation.** How are the firm's operations affected by regulation? Does the firm operate in a favorable regulatory environment currently? How might that change? What is the firm's history with gaining regulatory approval for its projects? Firms with more regulated assets are exposed to regulatory risks, but they may also have more stable returns.

**Interest Rates.** How sensitive are the firm's operations to interest rates? Are rising or falling rates good or bad for the firm's operations and share price? Firms with greater leverage tend to be affected by interest rate movements due to changes in interest expense. Firms with more regulated assets also tend to be more interest rate sensitive because their stable cash flows make them akin to bond proxies (since bond prices move inversely to interest rates).

**Distributions.** Does the firm have a reliable history of paying and growing distributions to shareholders? Because high cash flow distributions are one of the unique and most attractive characteristics of many pipeline master limited partnerships (MLPs), a stable history of distributions is an important attribute.

**Tax Implications.** Are there any tax implications for investing in the firm? Owning an MLP, for instance, can have unique tax implications, so it's a good idea to consult a tax adviser before investing.





**“When prices go up two processes will set in — an increased production and a decreased consumption. The effect will be a gradual fall in prices. If prices get too low two processes will set in — decreased production because man will not continue to produce at a loss and second, increased consumption. these two factors will tend to establish the normal balance.”**

**— Professor Newcomb — Ten years later I became rich by remembering those words.**



# About OPEC

The Organization of the Petroleum Exporting Countries (OPEC) is an **intergovernmental organization** currently made up of twelve oil exporting countries that work together to coordinate their petroleum policies.

The Organization was formed in response to the activities and practices of seven large international oil companies (IOCs) known as the “Seven Sisters”. The activities of these IOCs were often detrimental to the growth and development of the oil-producing host countries, whose natural resources they exploited.

The first move towards the establishment of OPEC can be traced back to 1949, when Venezuela approached four other oil-producing developing countries—Iran, Iraq, Kuwait and Saudi Arabia—to suggest that they explore avenues for regular and closer cooperation on oil matters.

But the main impetus for OPEC’s birth came a decade later, after the IOCs had decided to reduce the price of oil without first asking the governments. In response, several oil-producing countries decided to hold a meeting in Cairo, Egypt, in 1959. This was the First Arab Petroleum Congress. Iran and Venezuela were invited as observers.

The meeting adopted a resolution asking IOCs to consult the governments of oil-producing countries before changing the posted price of oil. But the IOCs ignored the oil-producing countries. And in August 1960, they once again lowered the price of oil.



- OPEC’s mission is to coordinate oil policies in its Member Countries and help stabilize oil markets.
- It also includes working together—to make sure there is always a regular supply of oil available to consumers, and to make sure a steady income is generated for producers



# The birth of OPEC

In response to this, five of the largest oil-producing countries held another conference on September 10–14, 1960. This time the venue chosen was Baghdad, the capital of Iraq. The countries that attended the conference were: Iran, Iraq, Kuwait, Saudi Arabia and Venezuela. These are the Organization's Founder Members. That was when OPEC was born.

Each country sent a delegate:  
Fuad Rouhani from Iran, Dr. Tala'at al-Shaibani from Iraq, Ahmed Sayed Omar from Kuwait, Abdullah Al-Tariki from Saudi Arabia and Dr. Juan Pablo Pérez Alfonzo from Venezuela.

At this Baghdad Conference, the delegates discussed the role of the IOCs and changes in crude oil prices. The oil-producing countries saw an urgent need to create an organization for the defence of their most important non-renewable natural resource, oil.

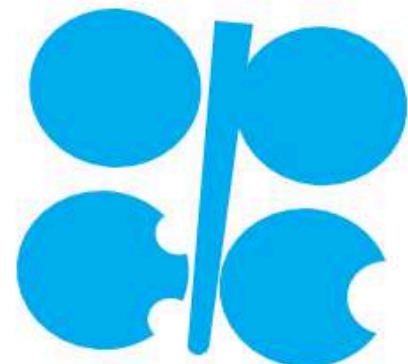
Thus, OPEC was established as a permanent intergovernmental organization. Its first headquarters were located in Geneva, Switzerland.

In April 1965, OPEC decided to move the Secretariat to Vienna, the capital of Austria. A Host Agreement was signed with the Austrian government and OPEC moved to Vienna on September 1, 1965.

After OPEC was created, the governments of OPEC's Member Countries began to have more control over their own natural resources. And in the years that followed, OPEC became increasingly aware of the important role it could play in the global oil markets.

## THE OPEC LOGO

The OPEC logo is the recognizable symbol of the Organization. It is the result of an international design competition held in 1969. An Austrian designer, Gertrude Svoboda, won the competition with her design, which combined the different letters of the Organization's name in a rounded design. This logo is still in use by OPEC.



HE Fuad Rouhani  
Head of Delegation, Iran



HE Dr. Tala'at al-Shaibani  
Head of Delegation, Iraq



HE Ahmed Sayed Omar  
Assistant to the Secretary of State  
Ministry of Finance  
Head of Delegation, Kuwait



HE Abdullah Al-Tariki  
Minister of Petroleum  
Head of Delegation, Saudi Arabia



HE Dr. Juan Pablo Pérez Alfonzo  
Minister of Mines and Hydrocarbons  
Head of Delegation, Venezuela





# Who are the members of OPEC?

OPEC is made up of twelve **sovereign nations**: six in the Middle East, four in Africa and two in South America.

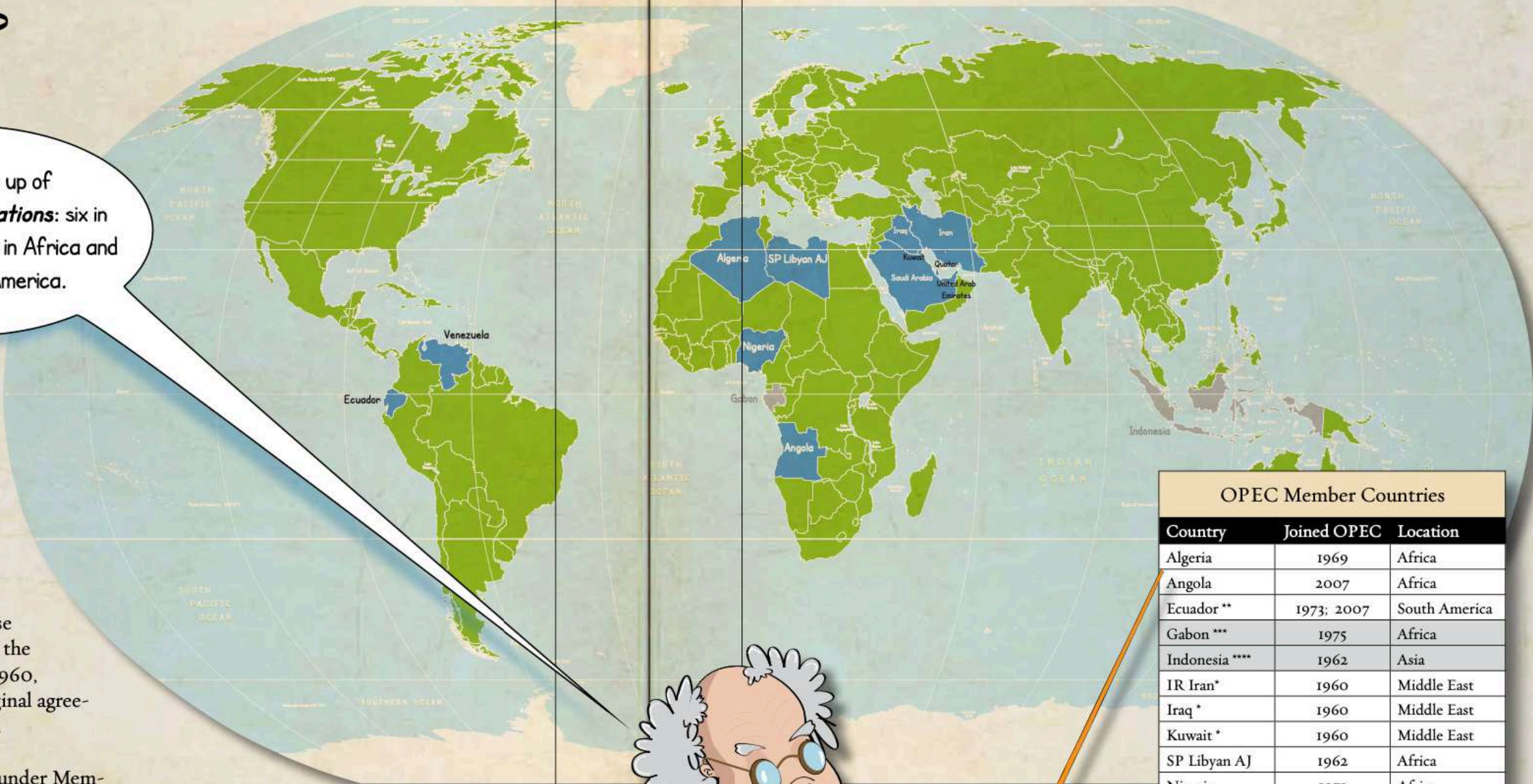
There are three categories of membership: Founder Members, Full Members and Associate Members.

**Founder Members** of the Organization are those countries which attended the Baghdad Conference in 1960, and which signed the original agreement establishing OPEC.

**Full Members** are all Founder Members, plus those countries whose applications for membership have been accepted by the Ministerial Conference.

**Associate Members** are countries which do not qualify for full membership (see p. 58 regarding Who can be a member?), but which are nevertheless admitted under special conditions.

The oil producing capacity of each Member Country is different. But when it comes to making decisions as an Organization, every country is equal. Decisions must be made unanimously with the agreement of all Member Countries.



| OPEC Member Countries |             |               |
|-----------------------|-------------|---------------|
| Country               | Joined OPEC | Location      |
| Algeria               | 1969        | Africa        |
| Angola                | 2007        | Africa        |
| Ecuador **            | 1973; 2007  | South America |
| Gabon ***             | 1975        | Africa        |
| Indonesia ****        | 1962        | Asia          |
| IR Iran*              | 1960        | Middle East   |
| Iraq *                | 1960        | Middle East   |
| Kuwait *              | 1960        | Middle East   |
| SP Libyan AJ          | 1962        | Africa        |
| Nigeria               | 1971        | Africa        |
| Qatar                 | 1961        | Middle East   |
| Saudi Arabia *        | 1960        | Middle East   |
| United Arab Emirates  | 1967        | Middle East   |
| Venezuela *           | 1960        | South America |

\* Founder Members

\*\* Ecuador joined OPEC in 1973 and suspended its membership from December 1992 to October 2007.

\*\*\* Gabon terminated its membership in 1995.

\*\*\*\* Indonesia joined OPEC in 1962 but suspended its membership effective January 2009.







Representatives of the five Founder Members at a Board of Governors meeting in the 1960s

## Who can be a Member?

OPEC membership is open to any country that exports crude oil and which shares the ideals of the Organization.

A country may become a Full Member if its request to join OPEC is accepted by a three-fourths majority of Full Members (including all Founder Members).

If most of the oil a country produces is consumed internally—to meet domestic demand—then it may not be able to join OPEC. Nevertheless, that country

may still be admitted as an Associate Member.

An Associate Member may be invited to attend any of the Organization's meetings. While an Associate Member may participate in these meetings, it does not have the right to vote.

In addition, other oil-producing countries that are not members of OPEC may also be invited to attend Meetings as Observers.

## Keeping the market supplied

Throughout its 50-year history, OPEC has seen periods of both extremely high and low oil prices. Each time OPEC has tried to respond in an appropriate manner.

OPEC does this by coordinating the oil policies of its Member Countries.

If demand suddenly grows and supplies fall short, OPEC can increase its oil production in order to keep the market well supplied. On the other hand, if demand suddenly falls and supplies grow, OPEC can slow down production in order to help maintain a balance in the market.

OPEC's actions at critical times have demonstrated the Organization's ability to keep the oil market well-supplied during different kinds of unexpected events, such as wars or natural disasters.

## Stabilizing oil markets

One of the most important parts of OPEC's mission is making sure that the oil markets remain stable. One way that OPEC tries to do this is by working to avoid price extremes (when oil prices are too low or too high).

Avoiding price extremes—and sudden changes in prices—is important for producers and consumers. When oil prices are too high or too low, problems can be generated for the global economy.

For example, with extremely high oil prices, it costs more to do things—like run a car, operate a machine or make products like plastics.

But extremely low oil prices also cause problems. If the price of crude oil is too low, then it doesn't make sense to spend money trying to find more oil. Investments in projects may then fall.



157th Meeting of the OPEC Conference in session in Vienna, Austria, October 2010.



# About OPEC

In the short term, oil supply is relatively fixed—at least for increasing supply. Once found, it takes years for new oil supplies to actually become available to the public

Meaningful, unexpected supply additions do not typically occur in a 12 to 18 month period. However, production capacity can decrease almost instantly due to forces like embargoes, hurricanes, strikes, wars, accidents, and so on. In an era of little to no spare production capacity, an asymmetric risk has developed—in the short run, there is little chance of an unexpected increase in supply, but there is significant risk of unexpected decreases.

## Important oil supply drivers are:

- World oil production
- Crude oil reserves
- Spare capacity
- Oil inventories
- Geopolitics



# What are some Examples of **Geopolitical Issues** that have Affected Oil Supply?

- Oil embargo (Middle East, 1973)
- War and the potential of war (Iraq invasion of Kuwait, 1990)
- Nationalization of oil reserves (Venezuela, 2007)
- Militant attacks against pipelines and oil workers (Nigeria, 2007)
- Contract renegotiations between governments and international oil firms (Russia, 2008)

One of the most significant geopolitical events in the past 30 years was the Iran-Iraq war beginning in September 1980. Iraq's invasion caused oil production in Iran to nearly halt, while Iraq's oil production was also severely cut.

With two of the largest oil-producing nations' supplies dramatically curtailed, prices around the globe soared. Oil prices behaved similarly following Iraq's invasion of Kuwait in 1990.

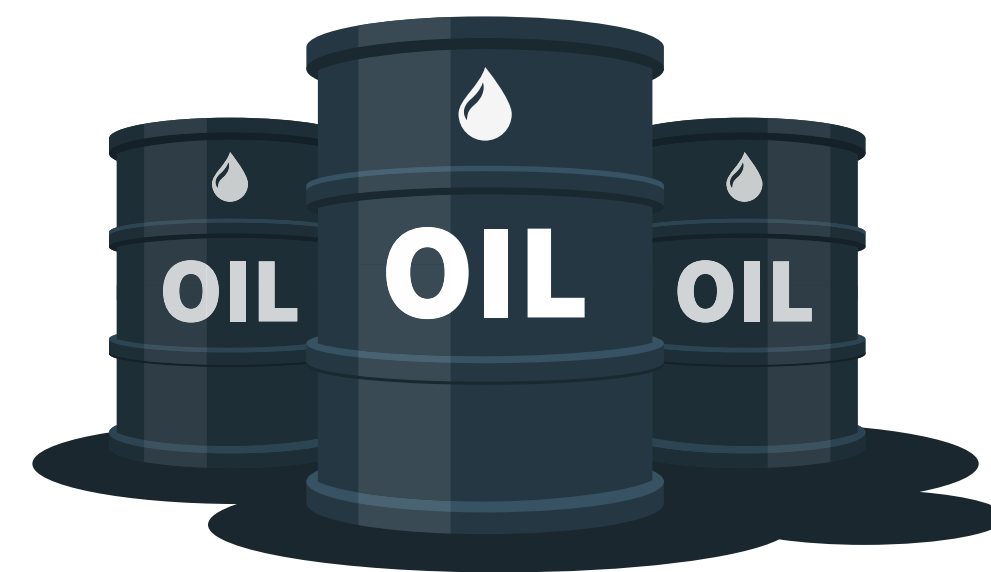
Geopolitical events are impossible to forecast, but they're unquestionably a force driving oil prices.



# OIL INVENTORIES

## OIL INVENTORIES

- Changing oil inventories around the world are an indicator of how oil supply balances with oil demand.
- Inventories held by the largest consuming nations are watched on a weekly basis to predict short-term price moves.
- The logic goes like this: Growing inventories might signal excess supply to meet demand, suggesting prices should fall. Conversely, shrinking inventories generally predict higher prices. (But, as always, seasonal factors must also be accounted for.)







# **BULLISH AND BEARISH DRIVERS OF OIL**

## **OIL INVENTORIES**

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### Bullish Oil Drivers

---

Rising oil demand  
Falling OPEC oil production  
Falling non-OPEC oil production  
Falling spare oil production capacity  
Rising global refining utilization  
Falling global oil inventories

---

### Bearish Oil Drivers

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Falling oil demand  
Rising OPEC oil production  
Rising non-OPEC oil production  
Rising spare oil production capacity  
Falling global refining utilization  
Rising global oil inventories

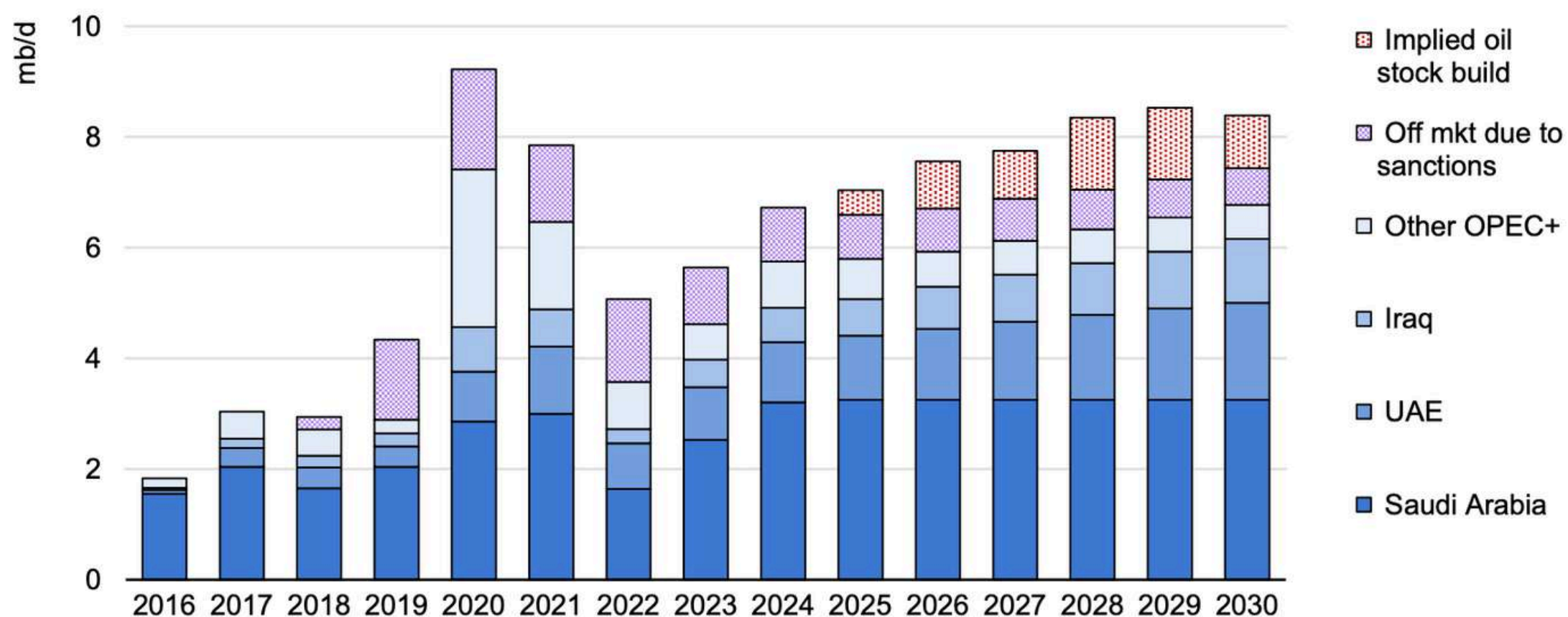
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# SURPLUS GLOBAL CAPACITY

**SURPLUS GLOBAL SUPPLY CAPACITY WILL REACH UNPRECEDENTED LEVELS BY 2030**

**OPEC+ spare crude production capacity and implied total oil stock build, 2016-2030**



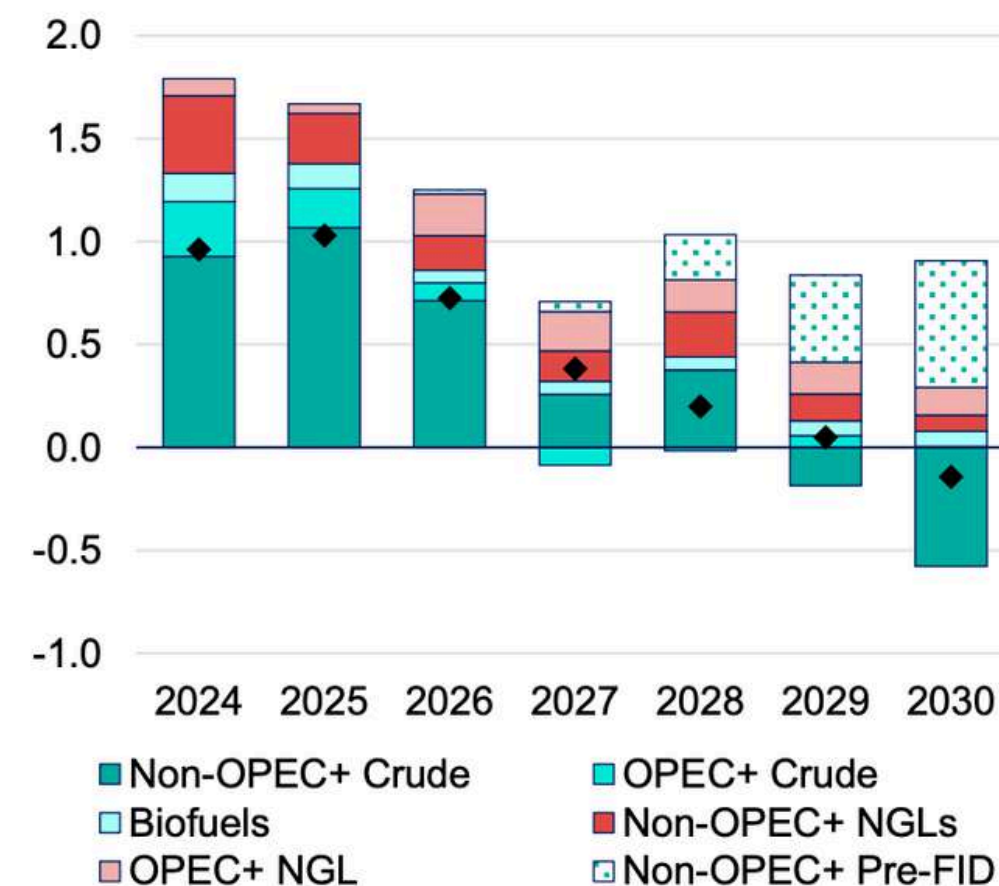
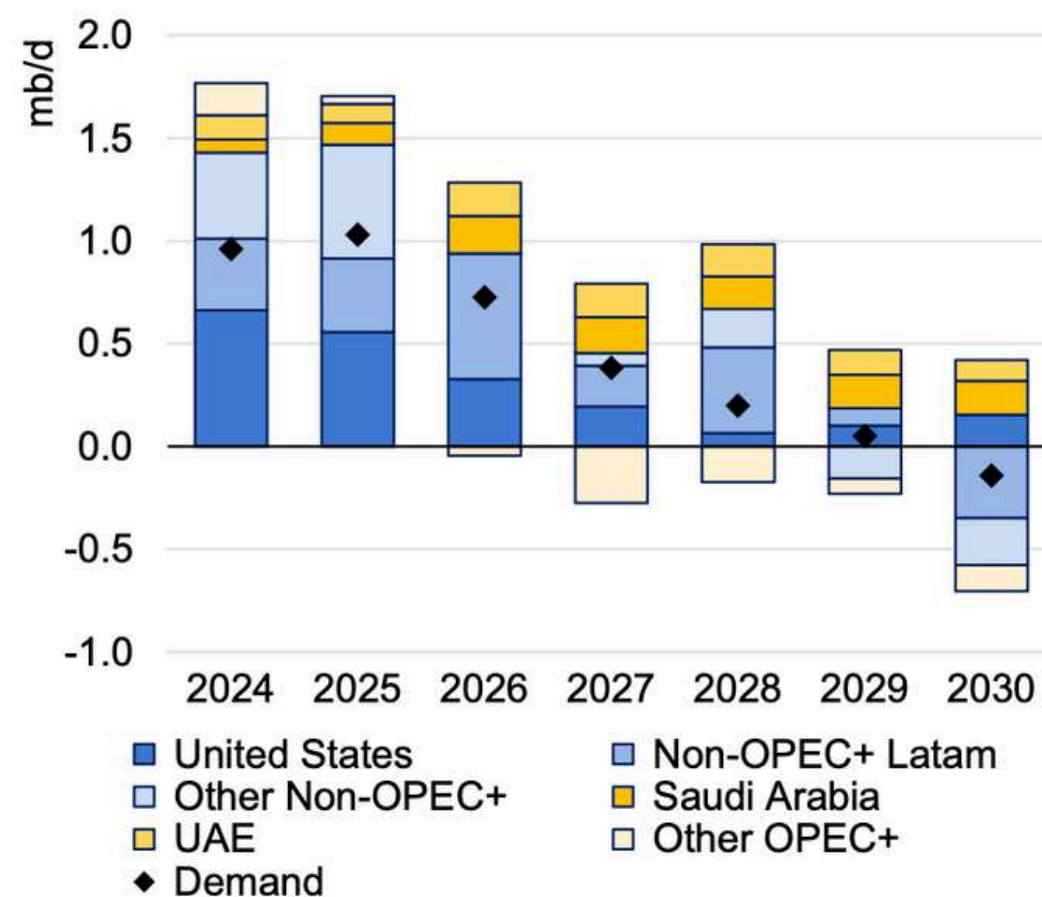
Total supply capacity rises by 6 mb/d to nearly 113.8 mb/d by 2030, a staggering 8 mb/d above projected global demand of 105.4 mb/d.



# SURPLUS GLOBAL CAPACITY

## SURPLUS GLOBAL SUPPLY CAPACITY WILL REACH UNPRECEDENTED LEVELS BY 2030

Global oil supply capacity forecast, year-on-year change, 2024-2030



IEA. CC BY 4.0.

Notes: Assumes Iran and Russia remain under sanctions. OPEC+ NGLs include condensates. Crude includes processing gains and non-conventional volumes. Right-hand chart includes pre-sanctioned projects, listed in the Tables section.



# WORLD OIL DEMAND

## WORLD OIL DEMAND TEMPERED BY CLEAN ENERGY TRANSITION

- Global oil demand will level off at around 106 mb/d towards the end of the decade amid the accelerating transition to clean energy technologies. Surging EV sales and continued efficiency improvements of vehicles, and the substitution of oil with renewables or gas in the power sector, will significantly curb oil use in road transport and electricity generation.
- Total oil demand is nevertheless forecast to rise by 3.2 mb/d between 2023 and 2030, supported by increased use of jet fuel and feedstocks from the booming petrochemical sector. Indeed, consumption of naphtha, liquified petroleum gas (LPG) and ethane will climb by 3.7 mb/d over the forecast period, fuelled also by growth in LPG use for clean cooking.

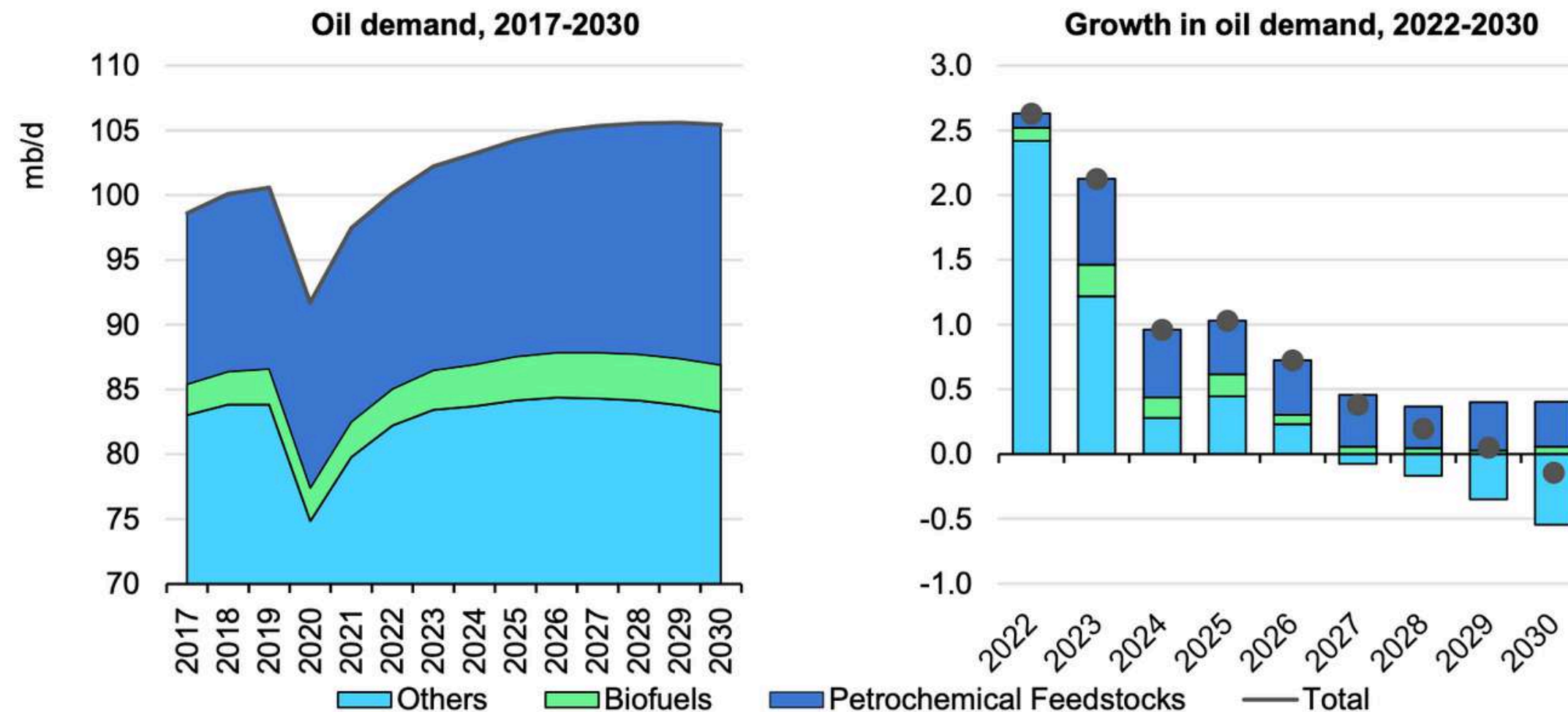




# WORLD OIL DEMAND

## WORLD OIL DEMAND TEMPERED BY CLEAN ENERGY TRANSITION

### World oil demand dominated by growth in petrochemical feedstocks

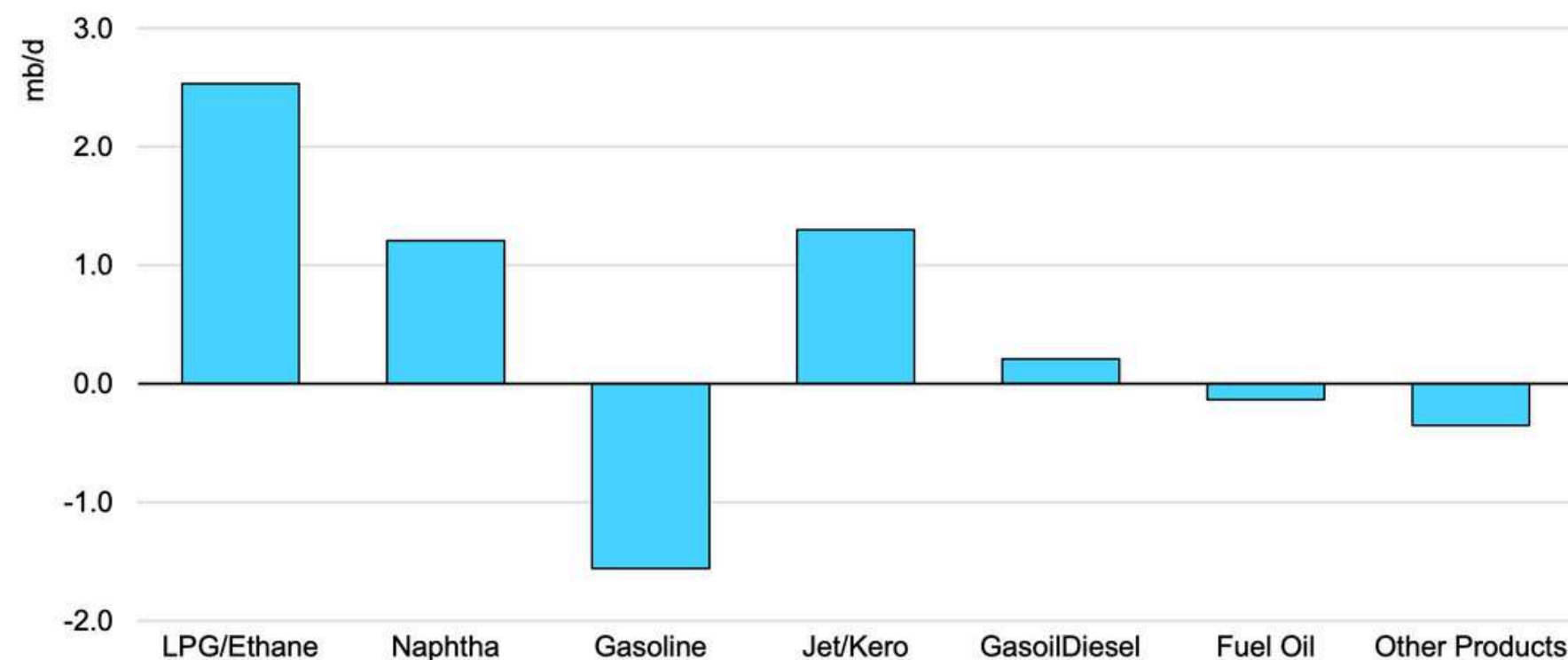




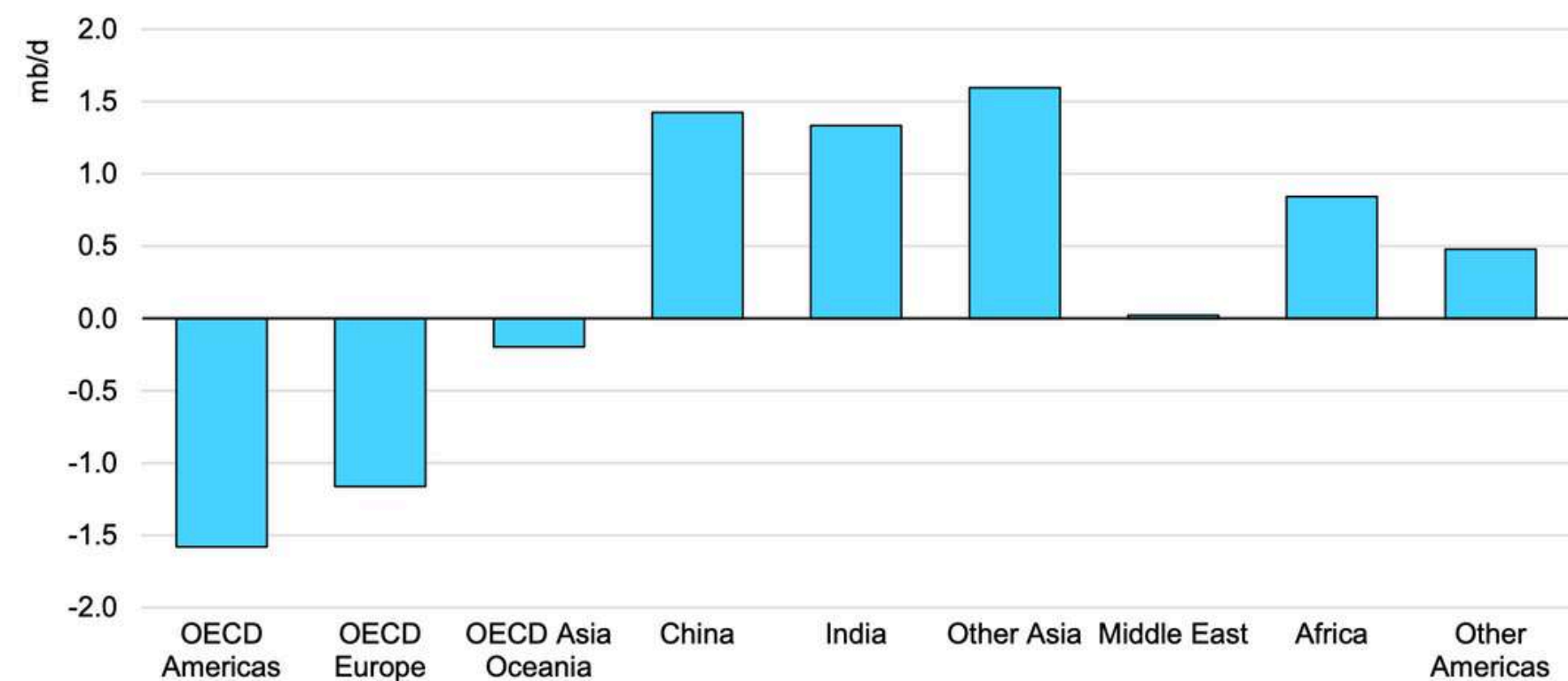
# WORLD OIL DEMAND

## WORLD OIL DEMAND TEMPERED BY CLEAN ENERGY TRANSITION

Growth in world oil demand by product, 2023-2030



Growth in oil demand by region, 2023-2030

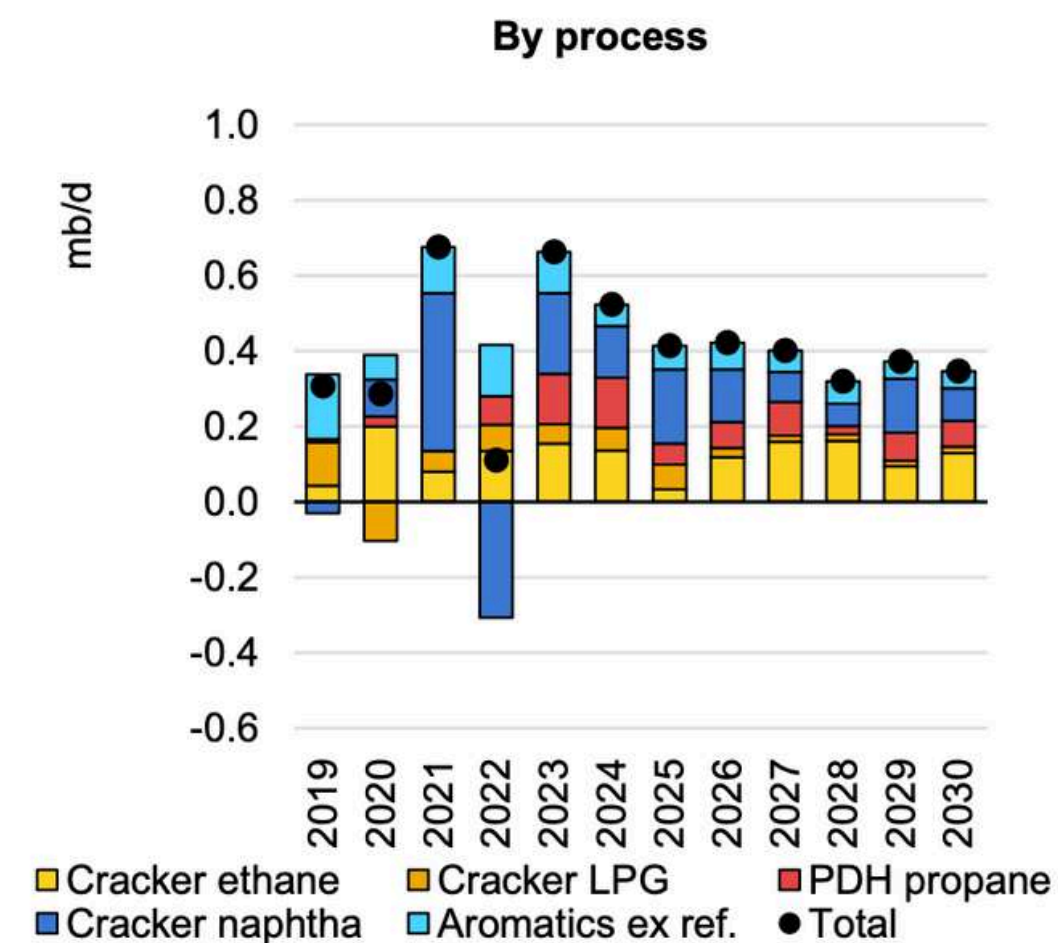
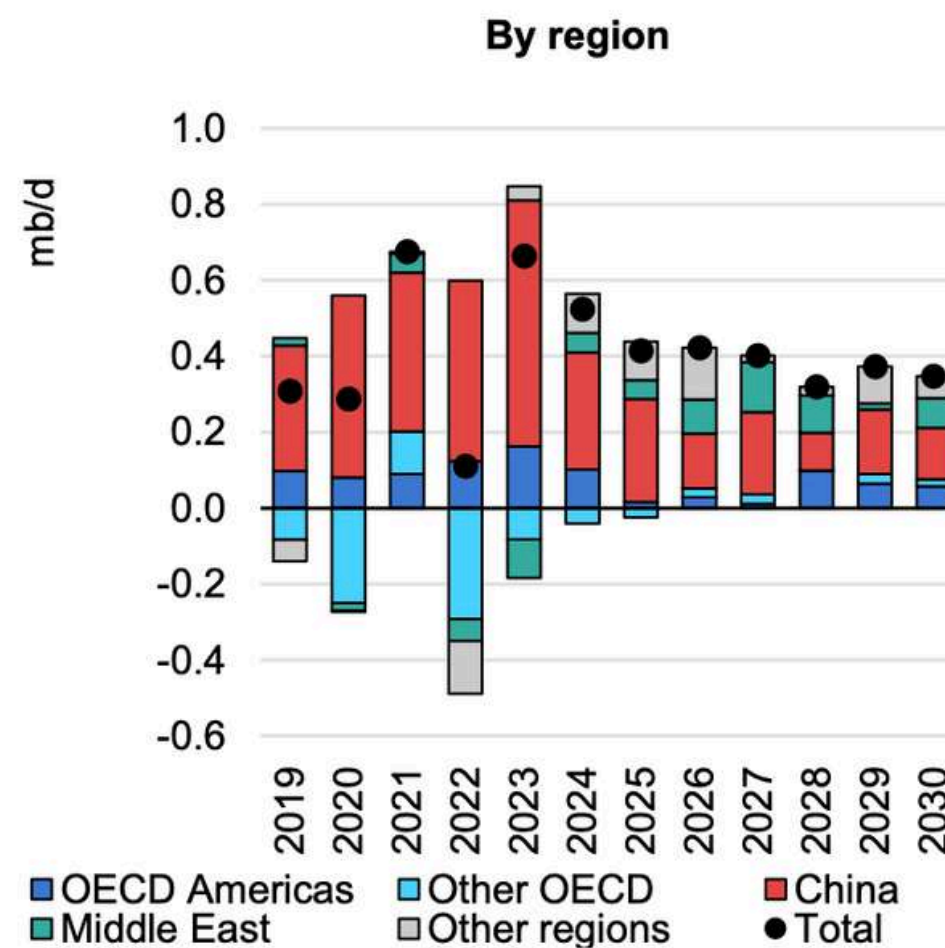




# PETROCHEMICAL FEEDSTOCK DEMAND

## PETROCHEMICALS FEEDSTOCK

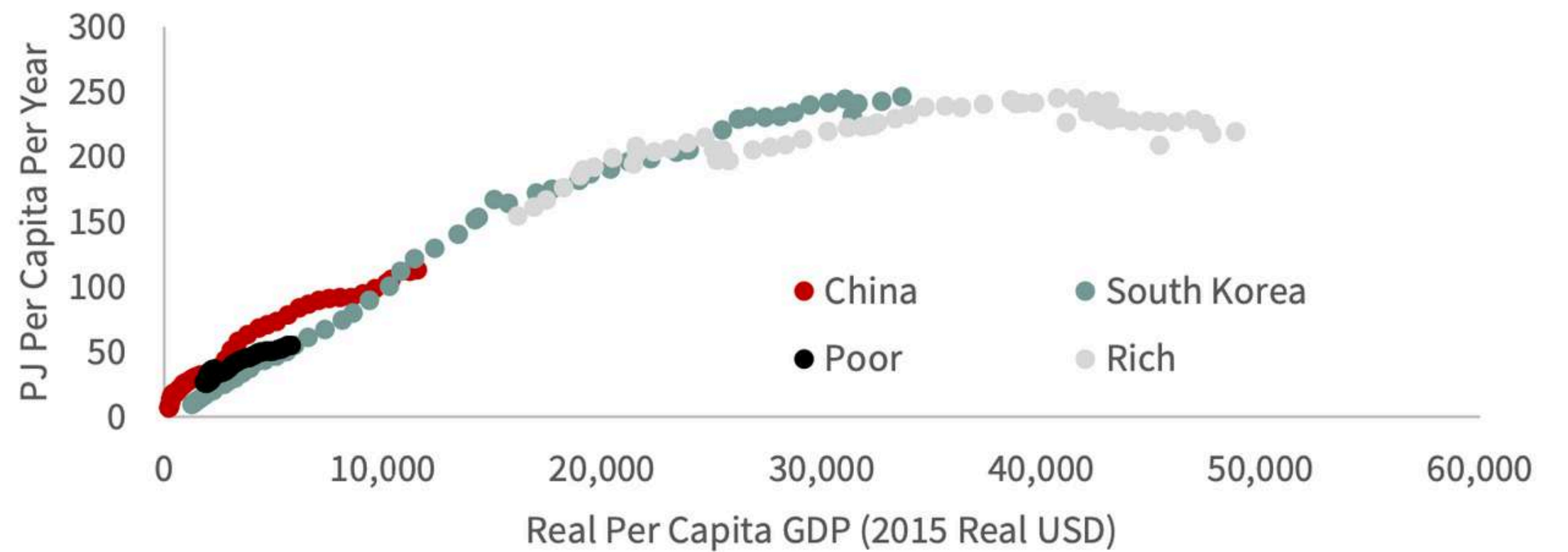
### Petrochemical feedstock demand growth, 2019-2030





# S-CURVE OF DEMAND

## S-CURVE

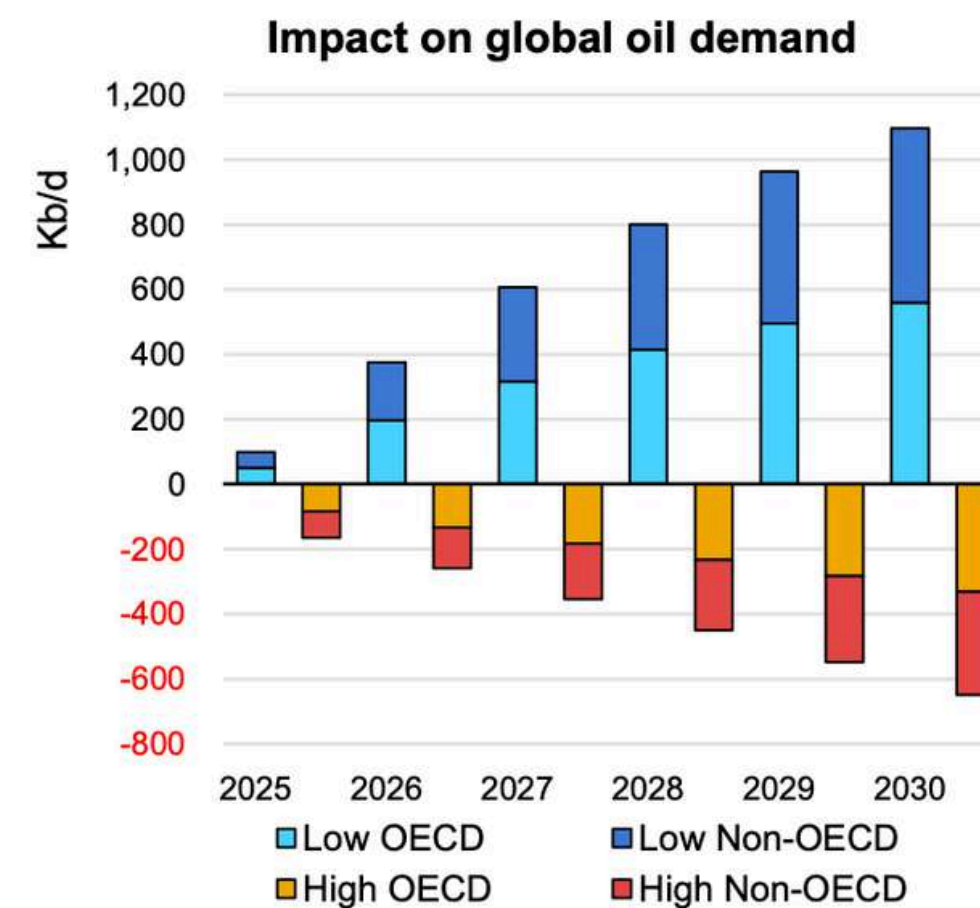
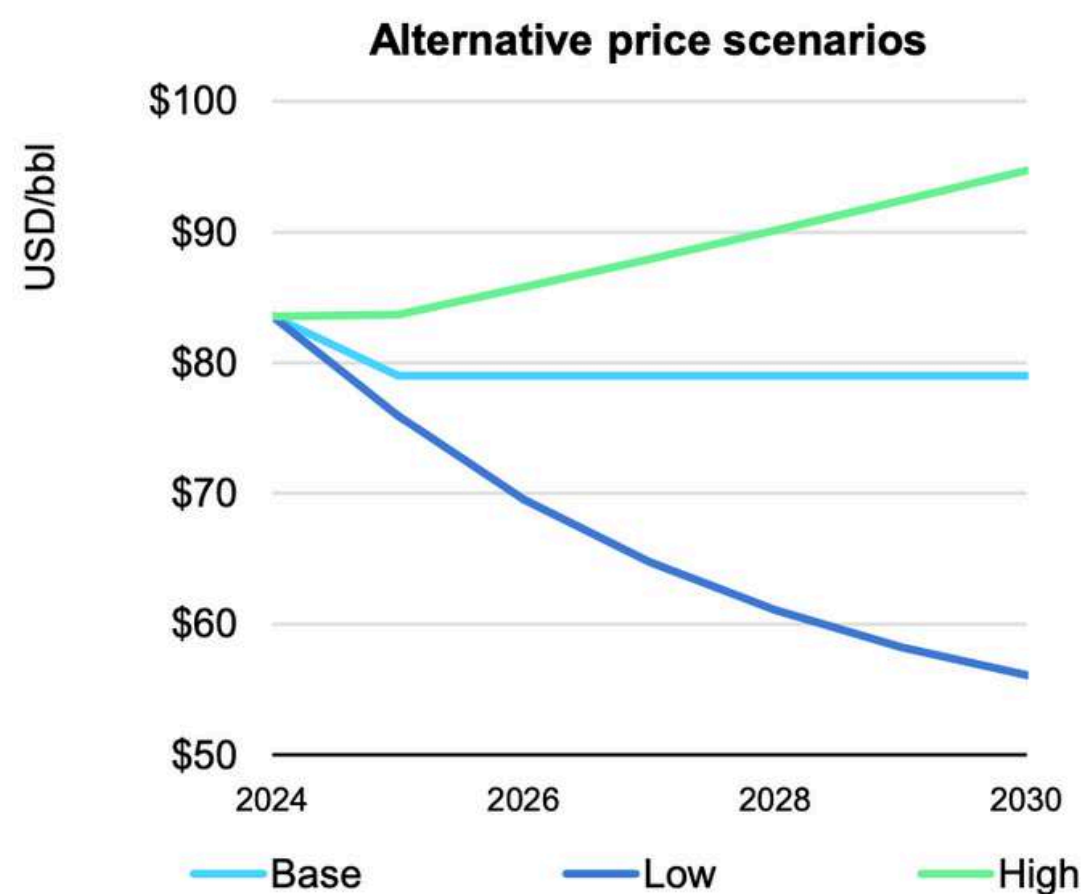




# ALTERNATIVE REAL OIL PRICE

## ALTERNATIVE REAL OIL PRICE

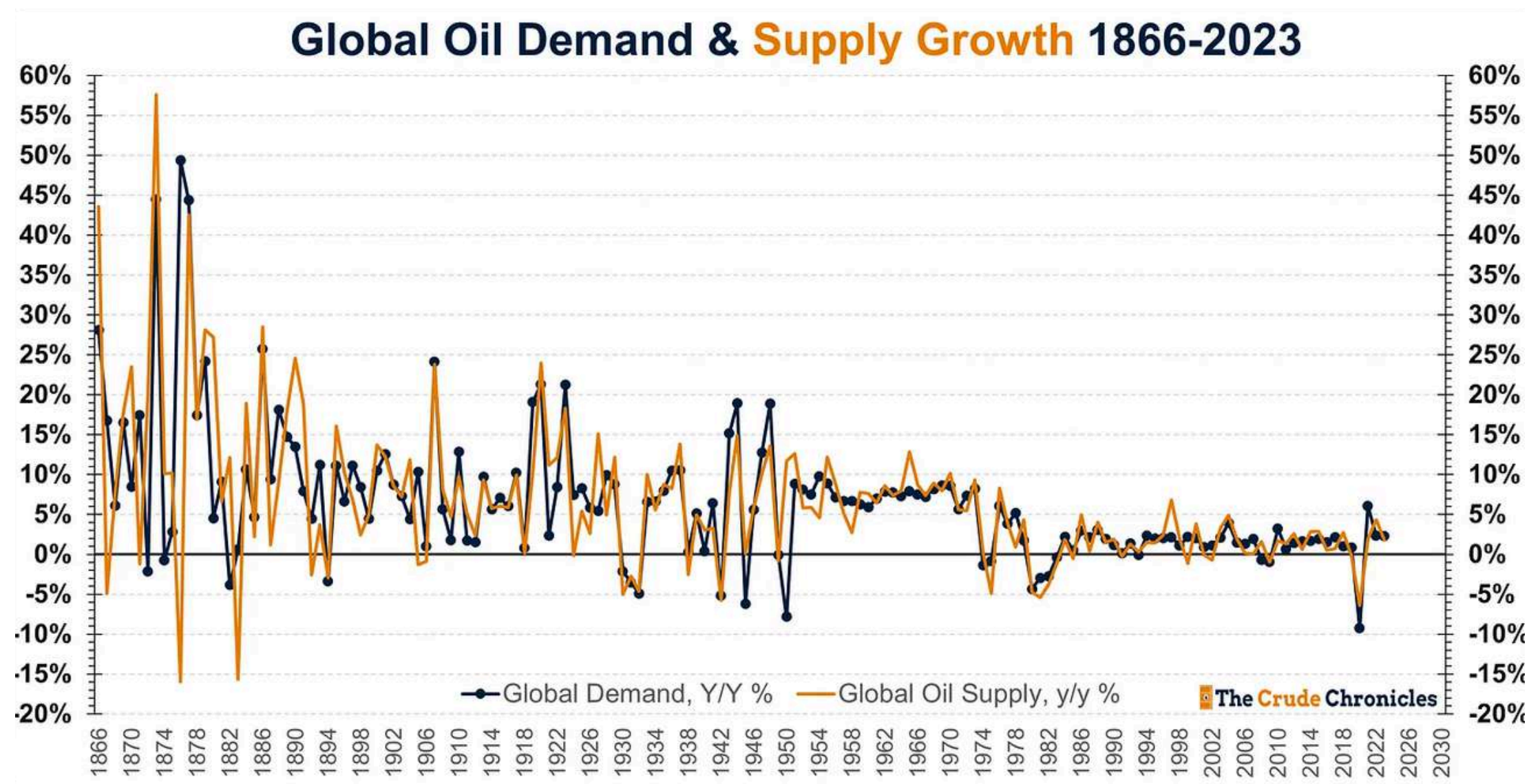
### Alternative real oil price scenarios and impact on global oil demand, 2024-2030





# GLOBAL OIL DEMAND

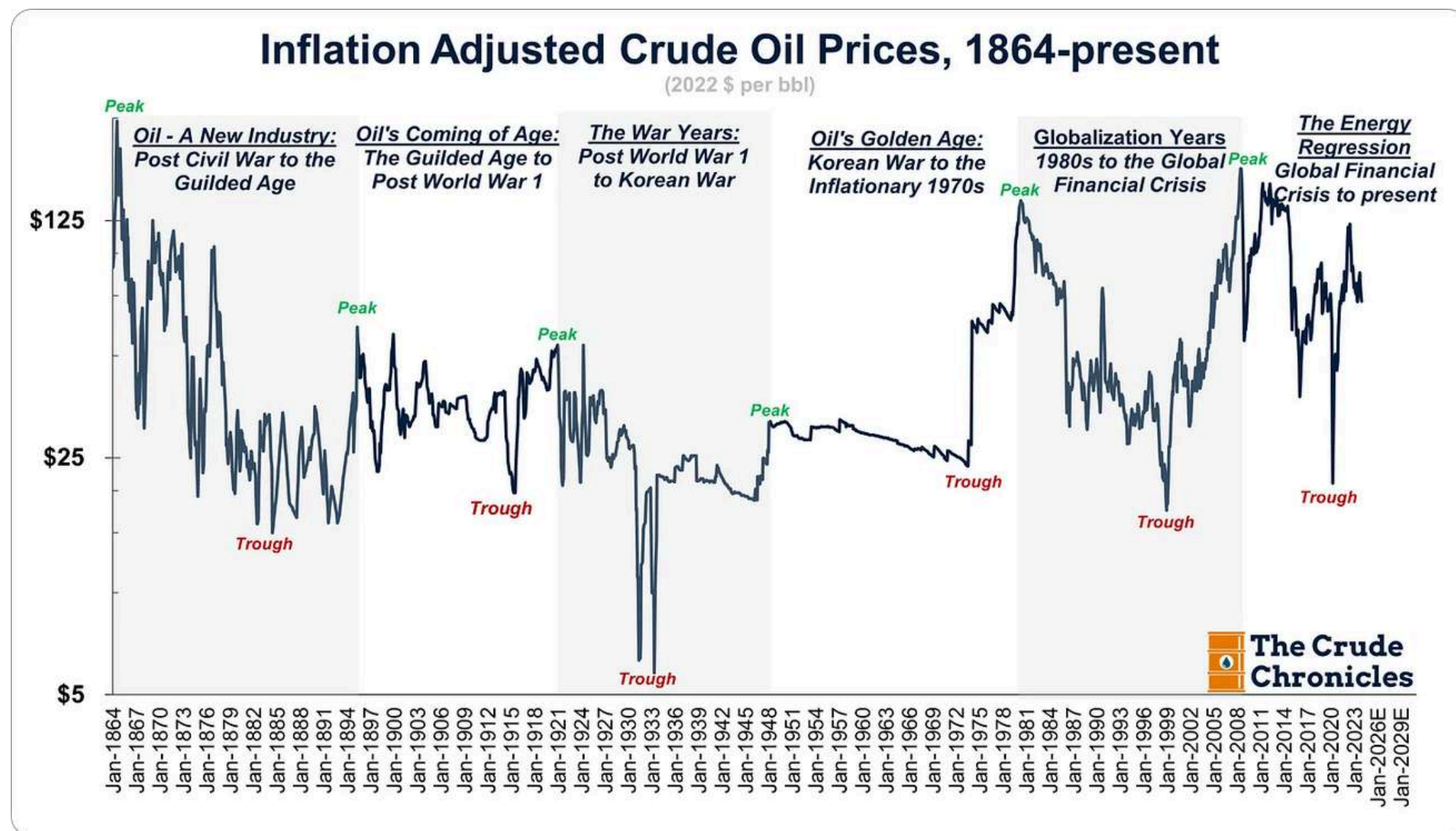
## ALTERNATIVE **REAL OIL PRICE**





# GLOBAL OIL DEMAND

## ALTERNATIVE REAL OIL PRICE





# STRUCTURAL DEFICIT?

## STRUCTURAL DEFICIT?

When the realization dawns that oil and gas demand is not in free fall, investors will be forced to confront how little the industry has invested to offset declines. Global oil markets have already fallen into a “structural deficit” masked by massive releases from government-controlled strategic stockpiles.

### Remembering 1970 and 2000

*WTI Crude Oil*

*1/1/1970: \$3.18*

*2/15/1981: \$34.30*

*1/1/2000: \$25.55*

*7/14/2008: \$145.18*





# OIL MARKET

## THE OIL MARKET IS AGAIN ENTERING A PERIOD SIMILAR TO 1971 AND 2003.

Crude markets are at an inflection point similar to 1970 and 2003, yet investors remain more complacent than ever. In the early 1970s, the energy weighting in the S&P 500 bottomed at 15% before reaching an all-time high of 35% in 1981. In 1999, energy's share of the S&P 500 bottomed at 6% before reaching a high of 15% in 2008... With crude at \$78 per barrel – nearly seven times higher than in 1999 – the energy weighting of the S&P 500 cannot break 4%. Investors are convinced energy stocks remain “uninvestable.”

“In the 1970s, oil advanced tenfold as the world experienced two distinct oil crises: the Arab embargo of 1973 and the Iranian revolution in 1979. In the 2000s, oil advanced thirteenfold, stopped only by the onset of the global financial crisis in 2008, the most severe economic dislocation since the Great Depression...”

Both bull markets were driven by predictable, albeit under-appreciated, changes in non-OPEC oil supply. In the 1970s, those changes took place in the United States. By 1970, the US was by far the world's largest oil producer. Production peaked in 1970 at 11.3 mm b/d – significantly greater than the combined production of Saudi Arabia and Russia, the world's second and third-largest oil producers. Between 1965 and 1970, US production growth represented 30% of total non-OPEC growth. Although US production had spent the preceding sixty years growing steadily, it was about to roll over and spend the next forty years declining. Hubbert predicted that peak fourteen years earlier in his dinner speech at the American Petroleum Institute in 1956. In that presentation, Hubbert presented two possible scenarios. In the second scenario, which he confirmed in 1962, he predicted US production would peak in 1970 at 10 mm b/d.

Production peaked in 1970 exactly as Hubbert predicted, and by 1976 production had already fallen 15% or by 1.6 mm b/d. Lost supply and surging demand meant the US's need for imported oil surged. Between 1970 and 1976, US net imports more than doubled, rising from 3.4 m b/d to nearly 8 m b/d by 1976. The dramatic increase in US net imports gave OPEC and Saudi Arabia a significant advantage in market share and pricing power. Between 1970 and 1974, OPEC went from representing 46% of global crude production to 52%. OPEC first leveraged its improved market share in 1973, when the Arab oil producers engineered a tripling of oil prices in only six months in retaliation against the United States' support of Israel in the Yom Kippur War. After the Iranian revolution in 1979, OPEC again exerted its influence, causing prices to double. Using its new price power related to market share gains, OPEC was able to produce a ten-fold increase in oil prices in ten short years and to change the geopolitical orientation of the world for years to come.





# OIL MARKET

## THE OIL MARKET IS AGAIN ENTERING A PERIOD SIMILAR TO 1971 AND 2003.

Over the last thirteen years, the US has provided almost 90% of total non-OPEC supply growth, far more than the US in the period leading up to 1970 or the North Sea and Mexico in the period leading up to 2003. Similar to the US in 1970 and the North Sea and Mexico in 2003, it is expected that the US shale production is about to roll over.

Hubbert believed that production would decline once an oilfield had produced half its ultimate recoverable reserves. This underpinned his prediction that the US would peak in 1970. Similarly, the North Sea and Cantarell reserves too, would roll over in the early 2000s. It is expected that the shales have produced over half their recoverable reserves. If our modeling is correct, production disappointments in the shales are rapidly approaching.

Except for King Hubbert in the 1970s and Colin Cambell and Jean Laherrère in the 2000s, no one predicted that the US, North Sea, or Cantarell would roll over.

Shale growth has been slowing for several years. The slowdown in non-OPEC supply has already impacted oil markets. The US has been forced to orchestrate a 320 mm barrel release of strategic petroleum reserves and OPEC is flexing its regained market share and pricing power. On April 3, OPEC renewed its production cuts, even though many agencies, including the IEA, anticipate crude deficits later this year.

Crude markets are at an inflection point similar to 1970 and 2003, yet investors remain more complacent than ever. In the early 1970s, the energy weighting in the S&P 500 bottomed at 15% before reaching an all-time high of 35% in 1981. In 1999, energy's share of the S&P 500 bottomed at 6% before reaching a high of 15% in 2008. As we write, with crude at \$78 per barrel – nearly seven times higher than in 1999 – the energy weighting of the S&P 500 cannot break 4%. Investors are convinced energy stocks remain “uninvestable.”



# OIL MARKET

## THE OIL MARKET IS AGAIN ENTERING A PERIOD SIMILAR TO 1971 AND 2003.

A similar situation developed in the 2000s. Throughout the late 1990s, Saudi Arabia and Venezuela (both OPEC member countries) waged a price war over disagreements around OPEC production quotas. Crude prices collapsed to \$11 per barrel in the first quarter of 1999. Demand was then materially impacted following the September 11, 2001 terrorist attacks, putting further downward price pressure on crude. At the same time, non-OPEC oil supply grew sharply in the early part of the 2000s. Russian production rebounded as companies, such as Yukos, began adopting modern Western drilling and oil service techniques. Surging non-OPEC supply and weak demand produced a universally bearish oil market psychology. Although it is hard to believe, most investors in the early 2000s failed to appreciate the coming surge in Chinese oil demand growth.

Despite the bearish outlook, forces were at work in non-OPEC supply that would significantly tighten the market and severely disrupt the bearish narrative, much like what happened back in the early 1970s. Investors and analysts willing to do the original research and apply various geological theories to the modeling of various hydrocarbon basins would recognize what these forces were and how they would impact non-OPEC supply.



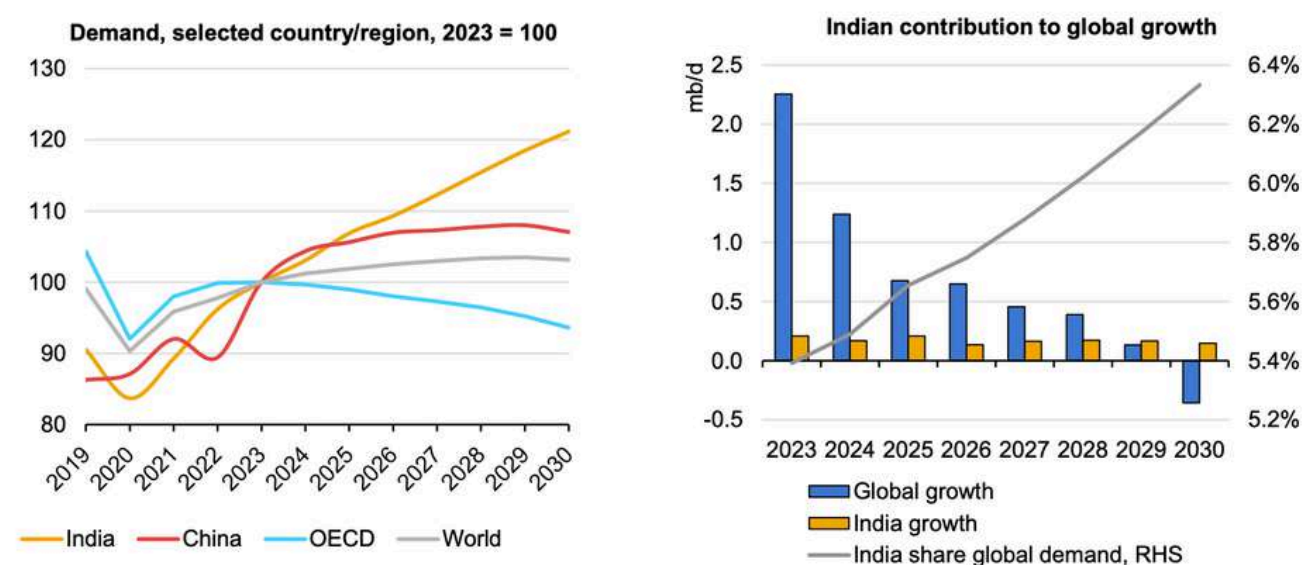


# INDIAN OIL MARKET

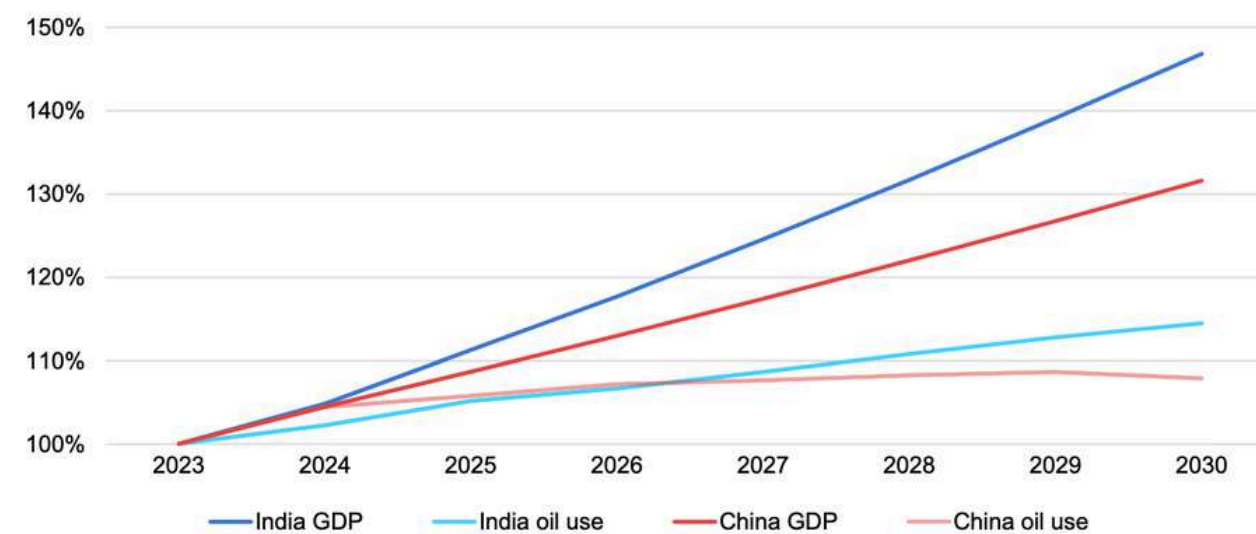
## INDIAN OIL MARKET

India ranks as the world's second largest net crude importer and sixth largest product exporter.  
India's economic engine powers global oil demand through 2030

India oil demand growth in a global context, 2019-2030



India and China per capita GDP and oil use, index (2023 = 100)



IEA. CC BY 4.0.

Source: GDP growth estimates from Oxford Economics.

Hosted by  
Neel Chhabra





# INDIAN OIL MARKET

## INDIAN OIL MARKET

Indian retail energy (and food) prices have historically been controlled by extensive government oversight and interventions. However, 2014 saw a deregulation push that introduced a more market-based mechanism. This resulted in a much closer relationship between pump prices and their underlying market drivers such as global oil prices and currency rates. Between 2015 and mid-2022, the monthly correlation between domestic gasoline/diesel prices and their wholesale equivalents (FOB Mideast Gulf) was about 60%. Even stronger was the correlation between pump prices and the Indian rupee (INR) versus the US dollar exchange rate, at about -80%.

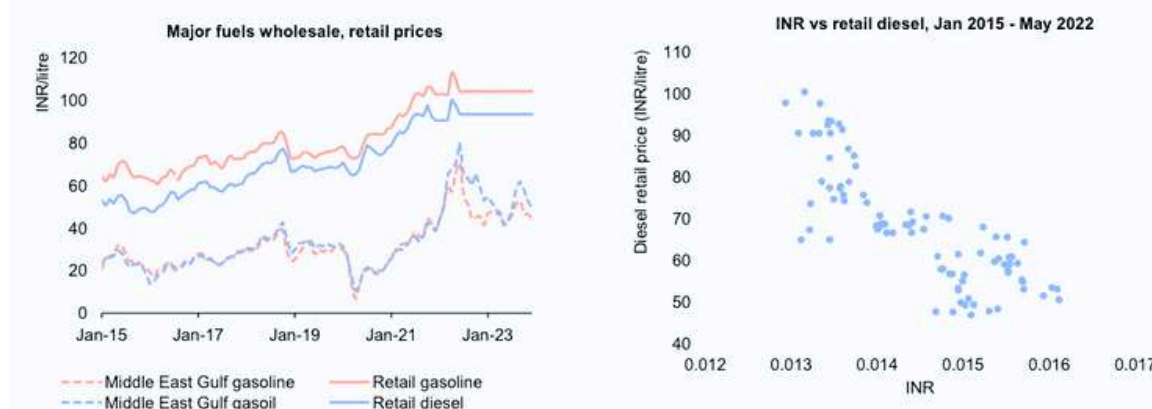
This firm inverse relationship between the local currency and the price of oil is a feature that India has in common with other emerging market commodity importers such as Türkiye, Pakistan and Egypt. Combined with the rupee's recent relative weakness, this causes persistent upward pressure on domestic pump prices. Higher oil prices tend to be accompanied by weaker currencies in commodity



# INDIAN OIL MARKET

## INDIAN PUMP PRICES DECOUPLE FROM GLOBAL OIL MARKET

importing countries, contributing to consumer inflation and worsening trade and fiscal balances. The rupee's slump is structural – it has declined against the US dollar in 12 out of past 13 years, losing about half of its value in total. India's annual consumer inflation averaged 6.1% over this period, accompanied by persistent current account and budget deficits (averaging 1.8% and 5.3% of GDP, respectively).



Sources: Argus Media Group, GlobalPetrolPrices.com.

The linkage between pump prices and market drivers halted in May 2022 when Indian oil marketing companies froze retail petroleum prices in the wake of Russia's Ukraine invasion and ensuing spike in commodity markets, at INR 104.18/litre for gasoline (USD 1.33/litre) and INR 93.48/litre for diesel (USD 1.19/litre) – significantly below the then-market levels. While the freeze sheltered India's consumers from inflationary pressures and economic hardship that gripped other countries like Pakistan and Egypt, state-run refiners initially incurred large losses. These were partially recouped in early 2023 by the payment of USD 3.7 billion in government capital support. Lower oil prices in 2023 allowed refiner margins to recover further, while the import of massive volumes of heavily discounted Russian crude also bolstered profit margins.

Indian pump prices have now remained unchanged for 20 months. A prolonged absence of price signals might dampen incentives for the country's refining sector to continue to invest. A return to market-based pricing would be opportune now that oil markets have normalised, and price volatility has receded after the turmoil in 2022.



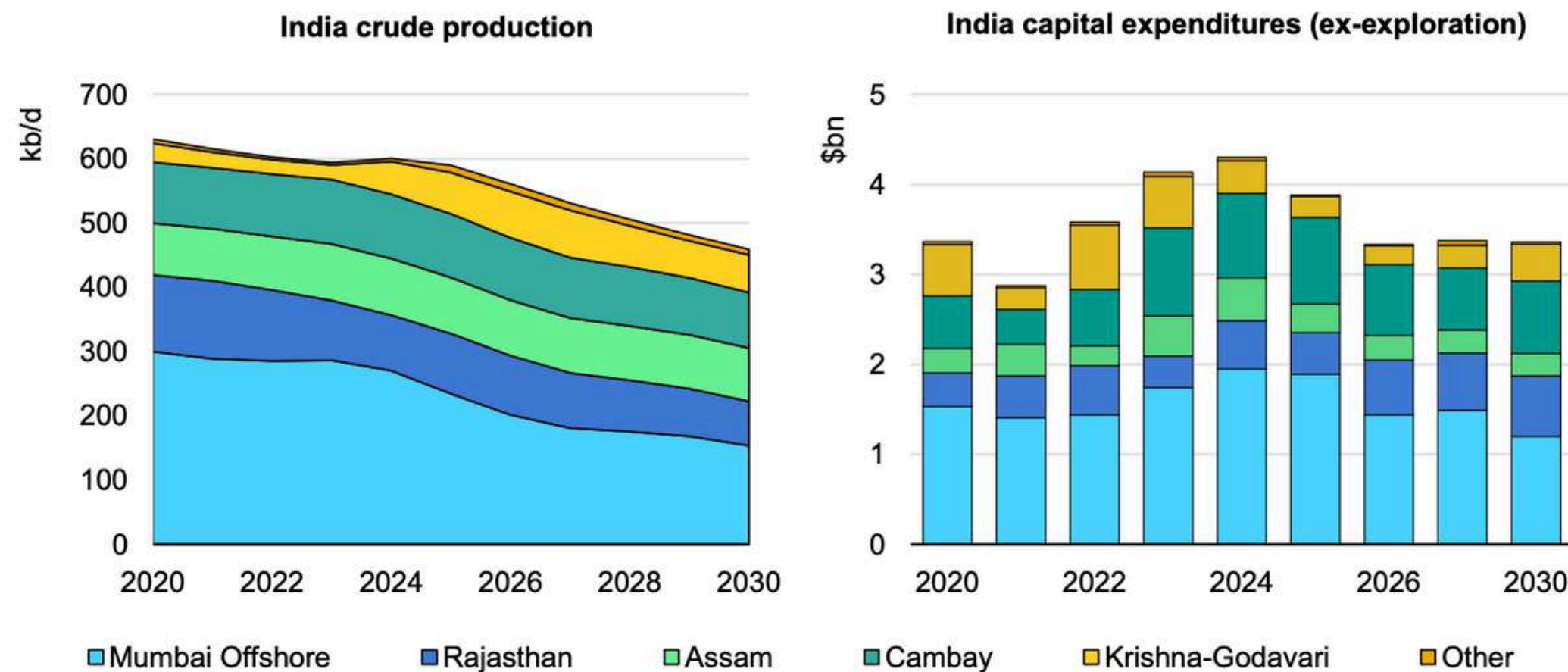


# OIL SUPPLY

## OIL SUPPLY

Indian oil production has been on a managed decline for more than a decade, following its peak at just over 900 kb/d in 2011. In 2023, output was just under 700 kb/d, of which crude oil production was 600 kb/d and the remainder natural gas liquids (NGLs). Output is overwhelmingly concentrated in the western part of the country, with the Mumbai Offshore Basin accounting for 60% of output and the Rajasthan Basin near 20%.

### Indian crude oil production and capital expenditures by basin, 2020-2030



IEA. CC BY 4.0.

Note: Assumes the KG-D5 project comes online in 1Q24 and ramps up over the next two years.

Source: IEA analysis based on data from Rystad Energy UCube.



# OIL SUPPLY

## OIL SUPPLY

Domestic oil production accounts for just 10% of India's oil supply needs, with the country's high dependence on imports a major concern for the government. A key energy policy objective is to cut India's oil import dependency by accelerating domestic oil exploration and production activities. Since 2016, the government has implemented a series of policy and fiscal reforms, and improved contract terms for private and foreign companies in an effort to attract more investment aimed at reversing steep decline rates and raising crude oil production to narrow the gap with refinery intake that is currently met by imports.

Production declined by around 4% per year between 2018 and 2023, lower than the global average annual rate of 7%. Field output declines are expected to continue at a similar rate through 2030, bringing total oil output down to 540 kb/d, and crude production to 460 kb/d, as growth in the Krishna-Godavari and Mumbai

Offshore Basins help support legacy fields. In the longer term, exploration successes from the recent Open Acreage Licensing Program (OALP) bid rounds may help stem the decline.

State-owned Oil and Natural Gas Corp. (ONGC) produces approximately 70% of all oil in India at 380 kb/d. The second largest company, Vedanta Resources (Cairn India), produces just under 20% at 130 kb/d and has built up a large position during recent open acreage licensing rounds. Oil India Ltd (OIL), also state-owned, rounds out the top three producers, with 2023 volumes averaging 60 kb/d. BP is currently the only major international oil company (IOC) with a presence in the Indian upstream, via a joint venture (JV) with Reliance Industries Ltd.

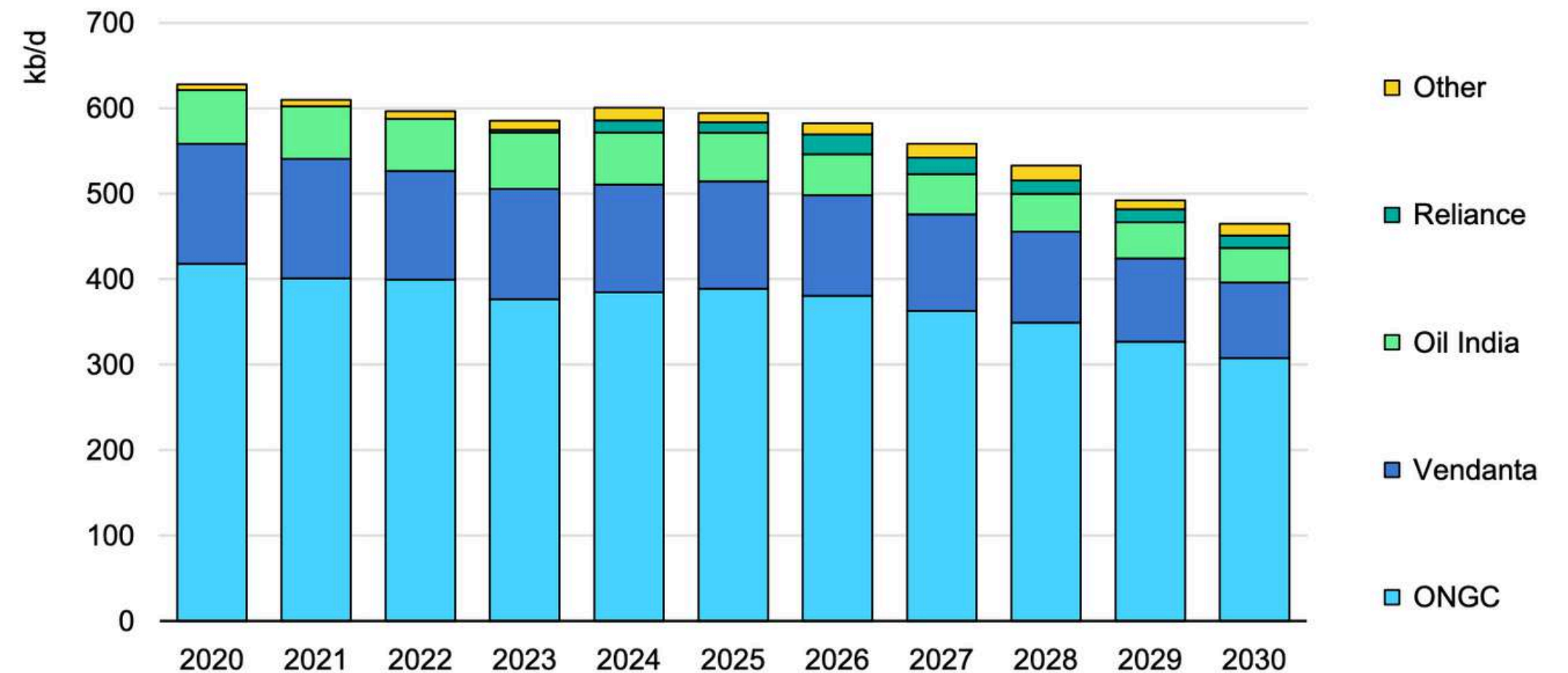
ONGC signed an exploration memorandum of understanding (MoU) with Chevron and ExxonMobil in 2022. In an effort to attract additional IOCs, the Directorate General of Hydrocarbons (DGH) is collaborating with the University of Houston to better understand the country's offshore basins. This comes after a series of attractive fiscal reforms and licensing revitalisation efforts. Additionally, private data providers have seen an uptick in interest and purchases of new Indian offshore data on offer.



# OIL PRODUCTION

## OIL PRODUCTION

Indian oil production by company, 2020-2030



### Indian Refineries Set for Continued Growth

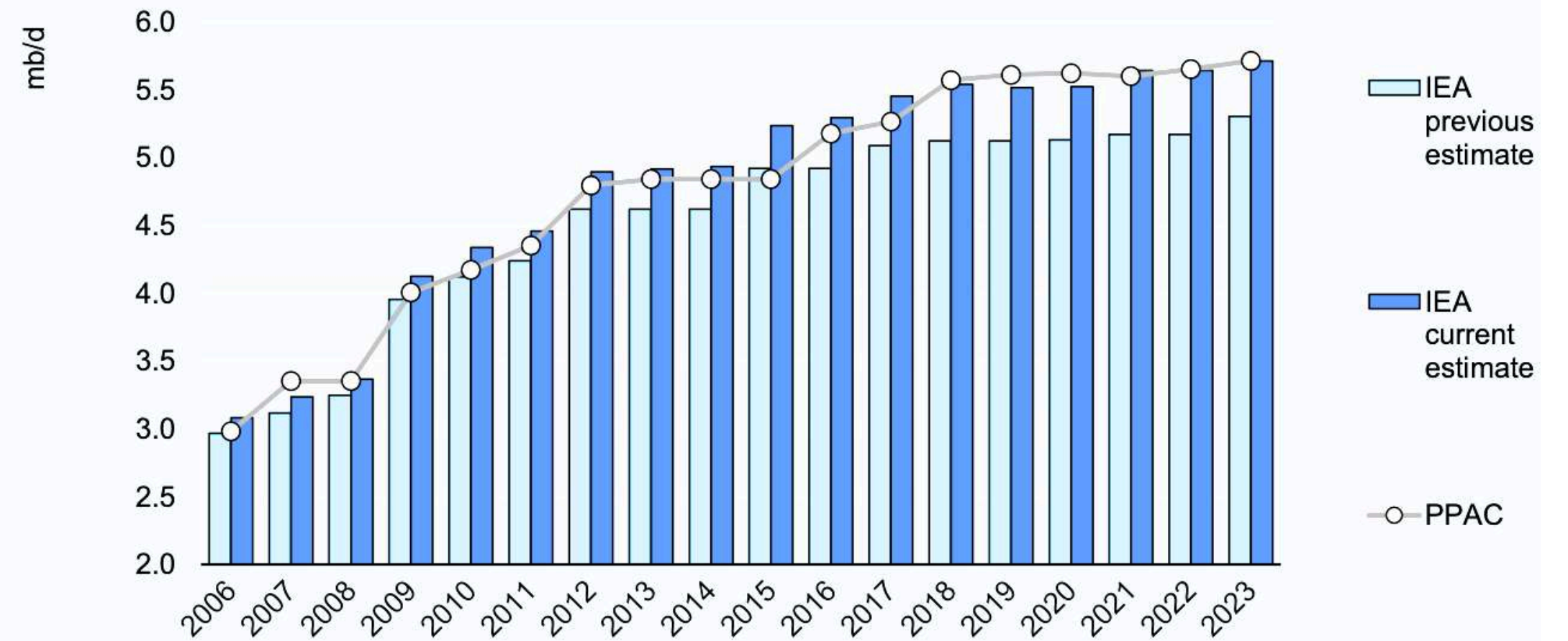
Indian refining capacity is currently assessed at close to 5.8 mb/d and is set to expand by 1 mb/d by the end of the decade. Growth will be dominated by public sector undertaking (PSU) refineries, as they prepare for continued rising domestic demand and an increased share of petrochemical production. By contrast, there is little prospect for private refineries to substantially expand their refining operations over this time frame. However, public and private refinery operators will both continue to expand existing crude processing capacity through the debottlenecking of existing facilities – likely in tandem with the addition of new, or increased hydrotreating capacity, petrochemical unit additions or residue upgrading.



# OIL REFINERY CAPACITY

## OIL REFINERY CAPACITY

Indian refinery capacity, 2006-2023





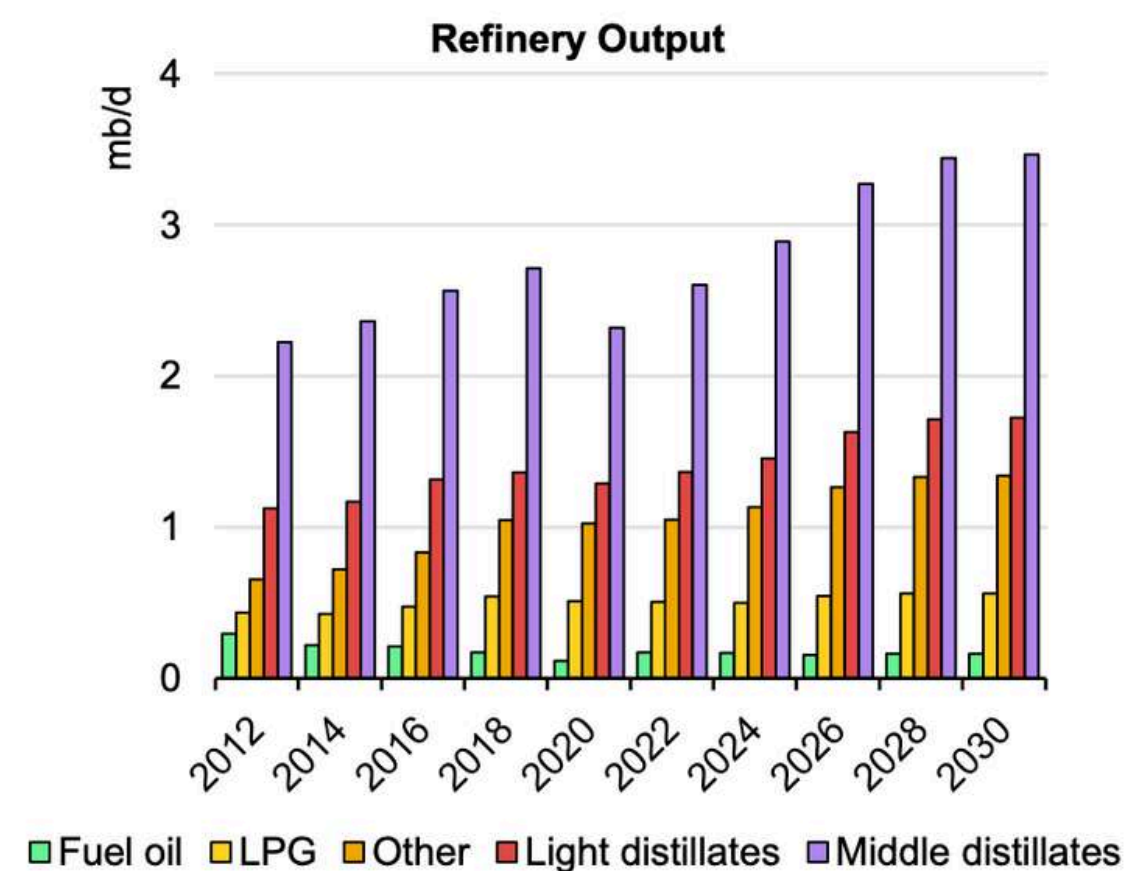
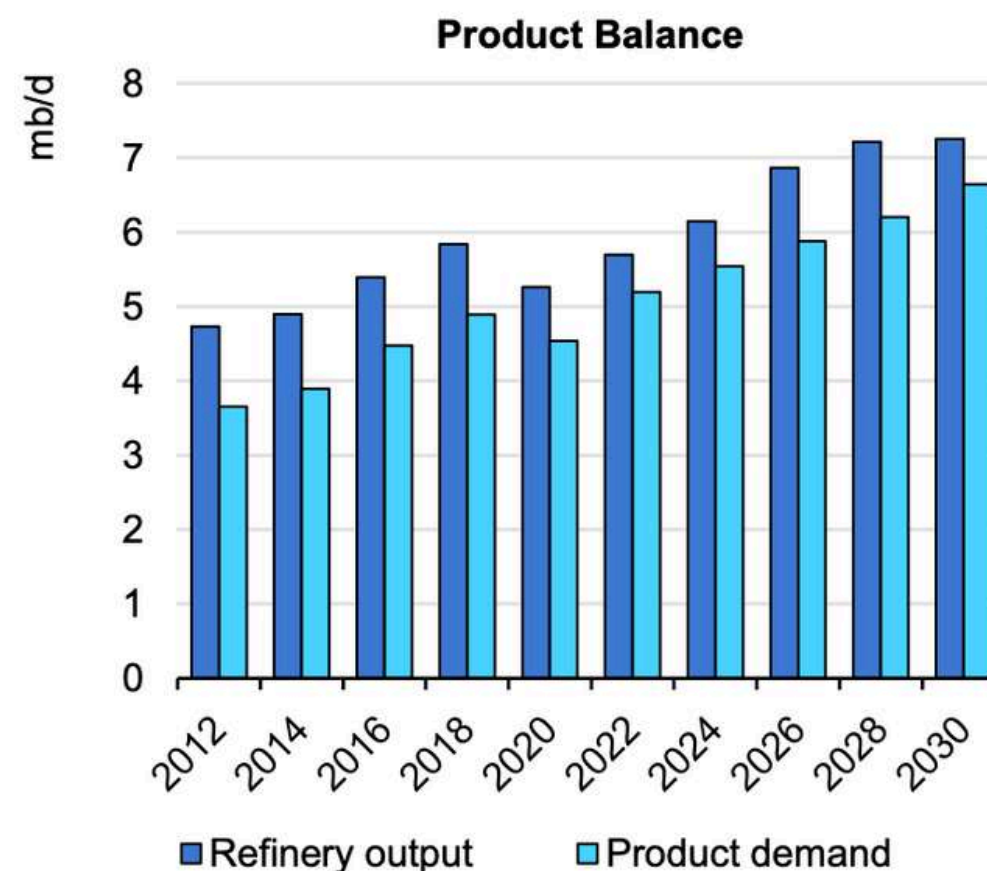
# OIL REFINERY CAPACITY

## OIL REFINERY CAPACITY

Over the past two decades, Indian refining investments have raised average complexity levels substantially, via the addition of coking and other residue upgrading technologies. This has facilitated a pivot away from light and medium sweet crude feedstocks towards heavier, sour crudes from the Middle East, Latin America and, on occasion, from Canada and Europe. Simultaneously, the share of light and middle distillates production has increased, with fuel oil yields halving over the past decade, despite average crude imports dipping towards 30 API.

This trend is set to continue with two vacuum residue hydrocrackers due to start up this year, with an associated loss of fuel oil output and an increase in petrochemical feedstocks, as well as light and middle distillate production.

### Indian product demand and refinery output, 2012-2030





# **Refineries focus on petrochemical integration and increased light and middle distillates**

**Across all the PSU refineries, petrochemical integration is targeted to rise from 7% currently to 25% by 2030. The increased exposure will be achieved by adjustments to refinery operating modes. Primarily, these revolve around increased fluid catalytic cracker (FCC) severity to generate higher yields of LPG and olefins. Elsewhere, naphtha reforming will be reconfigured to increase the yield of aromatics. Average refinery production yields will change over the forecast period. The combined impact of new refinery start-ups, investment in additional gasoline-focused units, e.g. alkylation units, and the addition of coking and vacuum residue hydrocracking will boost light and middle distillate output. Concurrently, we expect Indian refineries to target the acquisition of even greater volumes of heavy Middle East crude, as well as increased extra heavy crude from the Americas. Consequently, we see naphtha yields increasing at the expense of gasoline, and middle distillate yields benefitting from the reduction of residue output.**



# **Investment plans and capacity**

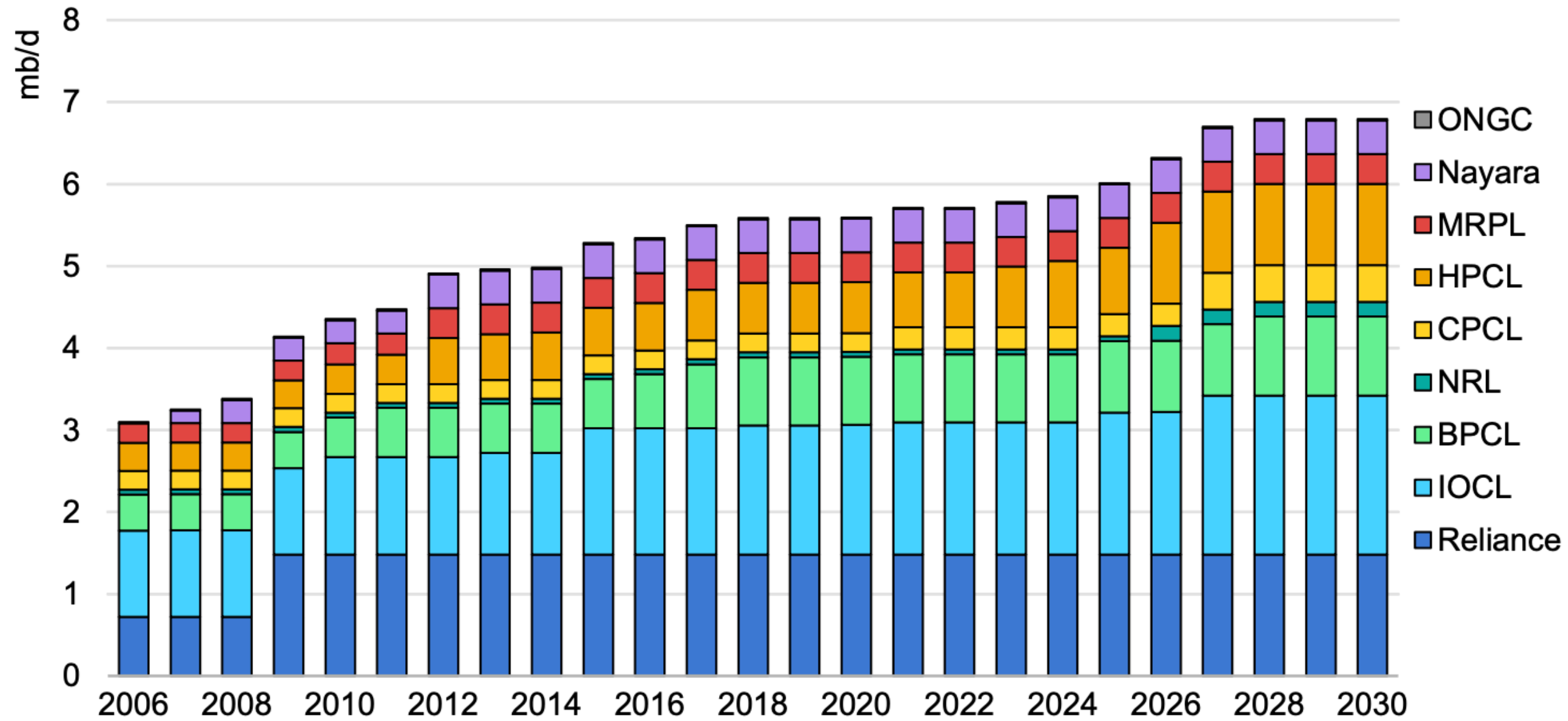
**India has significantly expanded its nameplate refining capacity, from 3.1 mb/d in 2006 to 5.8 mb/d by 2023, pegging the country as the fourth largest refiner in the world. This 2.7 mb/d capacity increase has been accompanied by notable improvements in refinery configurations and, in recent years, a strategic emphasis on integrating petrochemicals.**

**There are presently 23 operating refineries in India, with an additional greenfield project slated to be commissioned by 2026. Between 2023 to 2030, 1 mb/d of new distillation capacity is expected to come online with an average yearly addition of around 150 kb/d. Net capacity will therefore increase from 5.8 mb/d to 6.8 mb/d by 2030, with the majority of those additions starting up by 2026.**

**Indian Oil Corp. Ltd (IOCL) commands a significant position in the country's downstream sector, operating and holding approximately 28% of total refinery capacity as of 2023. IOCL is planning to expand and enhance its existing capacity by 327 kb/d by 2030. The 321 kb/d Panipat refinery operated by IOCL is gearing up for a significant 200 kb/d capacity boost by early 2027. This expansion is designed to enable the processing of heavier crude grades.**



## Indian refinery capacity, 2006-2030





**IOCL aims to commission the 60 kb/d expansion at its 140 kb/d Barauni refinery in 2025. The project involves the replacement of one of the plant's current 71 kb/d crude distillation unit (CDU) towers, with a new 131 kb/d unit. This project also includes the expansion of the hydrocracker, contributing to an increase in middle and light distillates output.**

**IOCL's Koyali refinery is planning to raise capacity by 60 kb/d by mid-2025, lifting crude processing capacity to 384 kb/d. The plant's FCC and diesel hydrotreater will also be expanded, ultimately boosting middle and light distillates output.**

**IOCL's Digboi oil refinery will undergo a minor expansion of 7 kb/d, taking its capacity from 13 kb/d to 20 kb/d by 2026.**

**Chennai Petroleum Corp. Ltd (CPCL), a subsidiary of IOCL, oversees the Nagapattinam refinery in the Cauvery Basin. The original 24 kb/d refinery, which was shut down for dismantling in 2019, is undergoing extensive refurbishment and expansion works, making up 18% of total additions for India out to 2030. The new refinery is set to have a capacity of 180 kb/d, with commissioning targeted for 2025.**

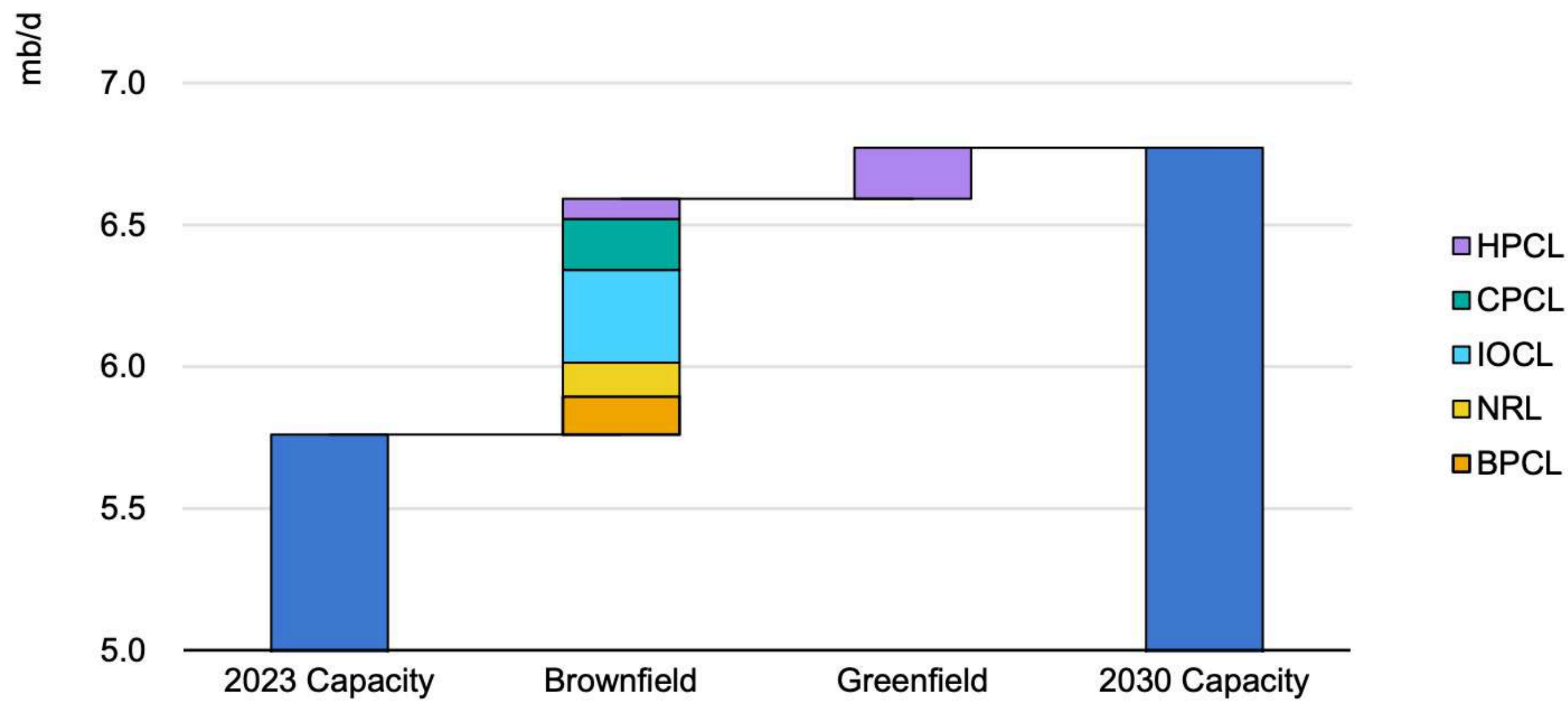
**Numaligarh Refinery Ltd (NRL), a division of Oil India Ltd, will increase the facility's capacity from 60 kb/d to 180 kb/d. The extensive project is expected to start up by the end of 2025 and will include significant upgrades in naphtha reforming, catalytic cracking, hydrocracking and hydrotreating capacities. The upgrades will lead to reduced LPG imports and increased middle distillate and polypropylene production.**



**Bharat Petroleum Corp. Ltd (BPCL) will see all three of its refineries add capacity, totalling 134 kb/d. The Bina manufacturing complex will include both a refinery expansion and a greenfield petrochemical project. The plans also include the addition of an ethylene cracker complex, utilising captive feedstocks such as naphtha, LPG and kerosene sourced from the refinery. The current refinery capacity at Bina stands at 170 kb/d, and the planned expansion will add 64 kb/d, bringing the total capacity to 234 kb/d upon commissioning in 2028. Other BPCL refineries will see some minor upgrades to existing facilities. The Kochi refinery is set to undergo a modest expansion of 30 kb/d in 2028, raising its capacity to 390 kb/d while in Mumbai a planned addition of 40 kb/d will push the overall capacity up from 300 kb/d to 340 kb/d by 2025.**



Indian refinery capacity growth, 2023-2030





**Reliance Industries currently operates the largest and most intricate refinery in India, which was one of the first full conversion refineries with significant coking and upgrading capacity. The Jamnagar refining complex has total capacity of 1.48 mb/d and can refine over 216 various types of crude. The refinery has no plans to expand on its current capacity, with future debottlenecking and upgrades dependent on the outlook for product markets.**

**Located in the Gulf of Kutch, Gujarat, Nayara Energy's Vadinar refinery stands as the second-largest refinery in the country. With a nameplate capacity of 410 kb/d, it can process a variety of complex crudes and is equipped to produce fuels that adhere to international standards such as Euro VI and BS VI grades. As of now, there are no confirmed plans for expansion at this refinery within the next five years.**

**Mangalore Refining & Petrochemicals (MRPL) operates the Mangaluru refinery, in the Karnataka state. The 363 kb/d plant has a versatile design with complex secondary processing units and a high flexibility to process crudes of various API, delivering a variety of quality products. The company has no plans to expand in the forecast horizon.**

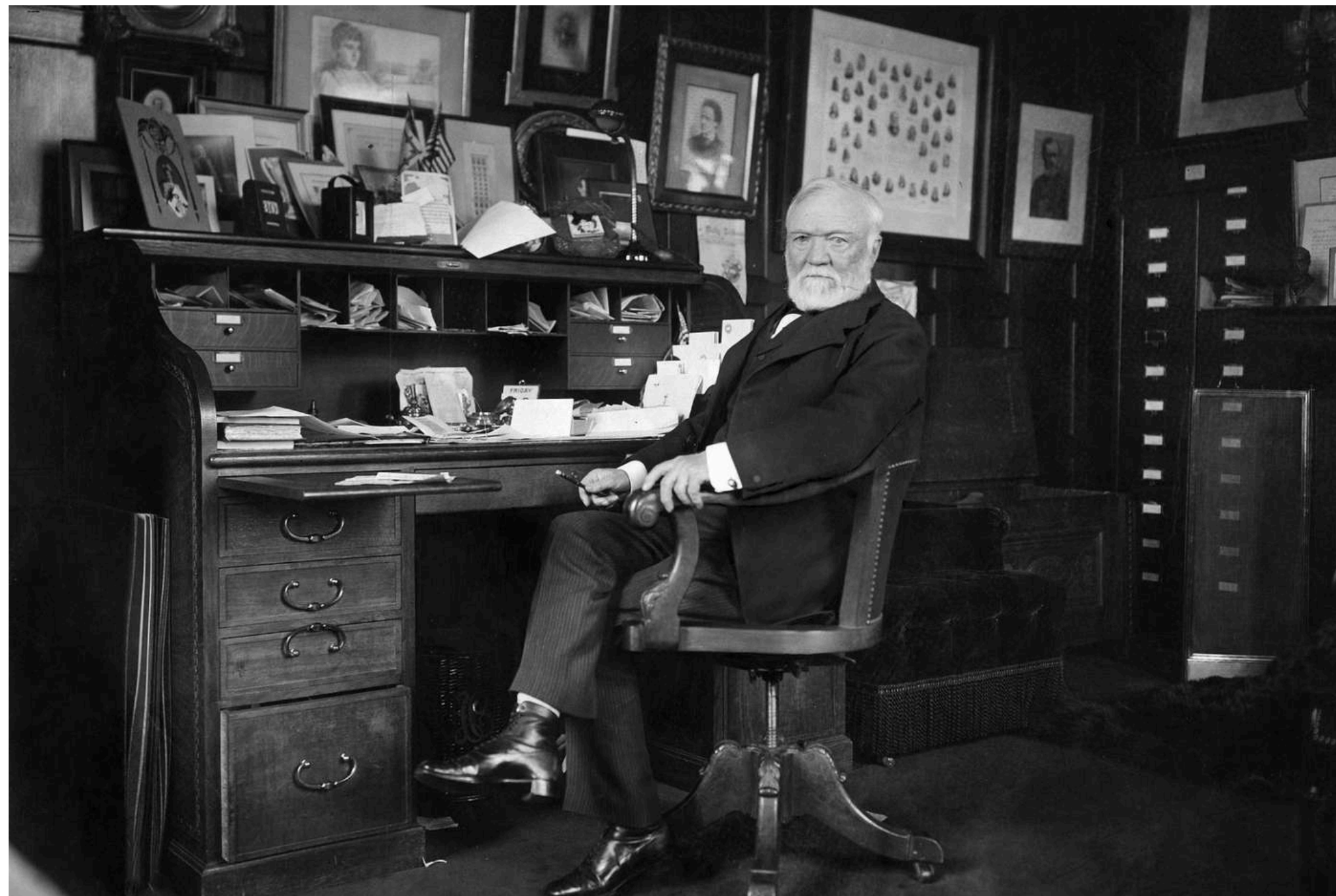
**ONGC runs the Tipitaka mini-refinery and condensate splitter in Hazirah, in the East Godavari district of Andhra Pradesh. The refinery produces and markets products such as LPG, kerosene, aromatic-rich naphtha and low-aromatic naphtha through fractionation of natural gas. Current capacity stands at 15 kb/d with no announced expansionary plans in the forecast.**



**In 2016, HPCL initiated significant expansion plans to address the disparity between regional production and demand. In response to this, the Mumbai refinery underwent a 50 kb/d expansion in 2021, boosting the total refinery capacity in the area to 210 kb/d. As the same time, HMEL (a joint venture between HPCL and ArcelorMittal), increased CDU capacity at its Bhatinda refinery from 220 kb/d to 260 kb/d to alleviate local product shortages.**

**HPCL's Visakhapatnam refinery, operating at around 200 kb/d since 2006, is set to undergo a sizeable and somewhat unique expansion project. The first instalment, of 70 kb/d, that took capacity to 270 kb/d was completed in 2023. The final phase is scheduled for 2024 and aims to further increase capacity to 340 kb/d, with a focus on enhancing coking and vacuum residue hydrocracking. The Barmer refinery and petrochemical complex, developed by a joint venture between HPCL Rajasthan Refinery Ltd (HRRL) and HPCL holding 74%, with the government of Rajasthan taking 26%, is a groundbreaking project for clean fuel production in the district. Once fully operational, likely by 2026, the refinery is expected to produce 180 kb/d. The facility will be the first of its kind in Rajasthan, producing BS-VI grade motor spirit, diesel fuel, as well as ethylene and propylene derivatives. These derivatives are intended for use as feedstock in various industries such as textiles, packaging and petroleum.**



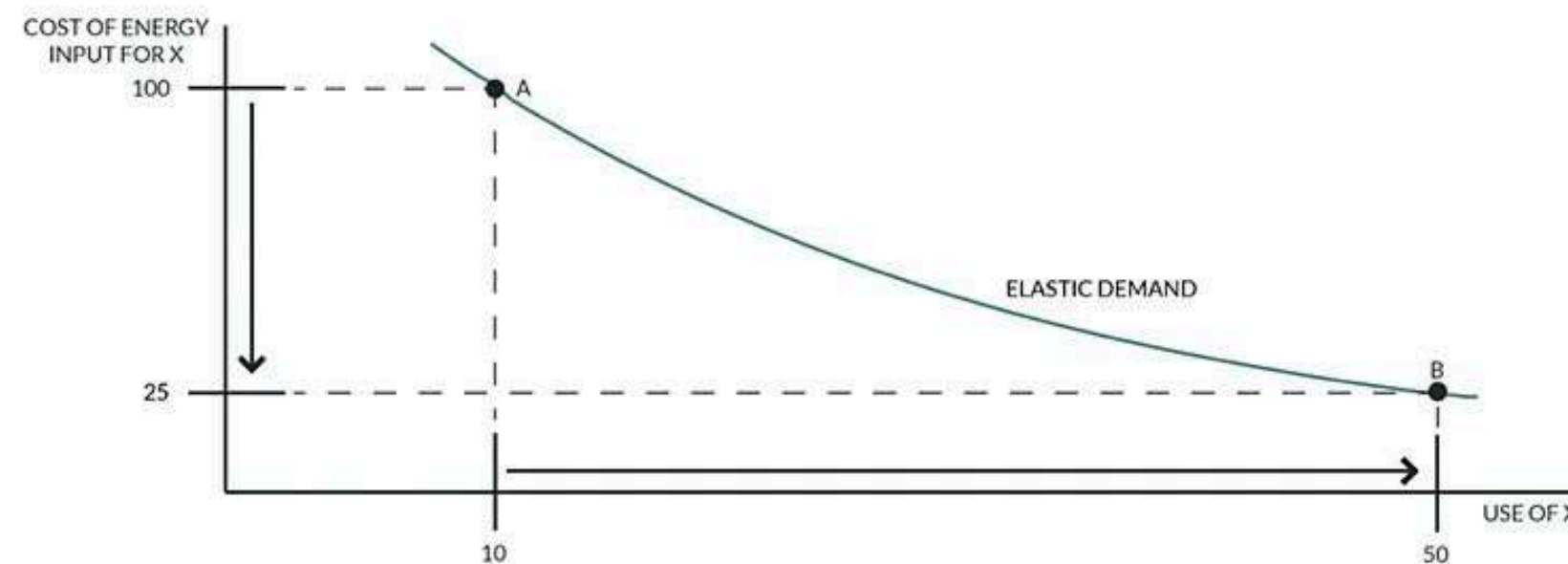


**The Law of the Surplus: “One great advantage which America will have in competing in the markets of the world is that her manufacturers will have the best home market. Upon this they can depend for a return upon capital, and the surplus product can be exported with advantage, even when the prices received for it do not more than cover actual cost, provided the exports be charged with their proportion of all expenses. The nation that has the best home market, especially if products are standardized, as ours are, can soon outsell the foreign producer.”**



# Jevons Paradox

In 1865, an English economist, William Stanley Jevons, detailed his eponymous Paradox in his book *The Coal Question*. Jevons noticed how improved steam engine efficiency actually led to much greater coal demand. At the time, economists were worried that England was running out of coal. Many argued that improved efficiency would temper demand and forestall a crisis. Jevon dismissed this logic, correctly concluding that improved efficiency would accelerate demand by promoting increased adoption. Jevon's Paradox was born: improved efficiency increases consumption. The mechanics were two-fold. First, better efficiency encourages more significant use. Secondly, as the same input unit generates more output, economic growth accelerates, increasing overall consumption.

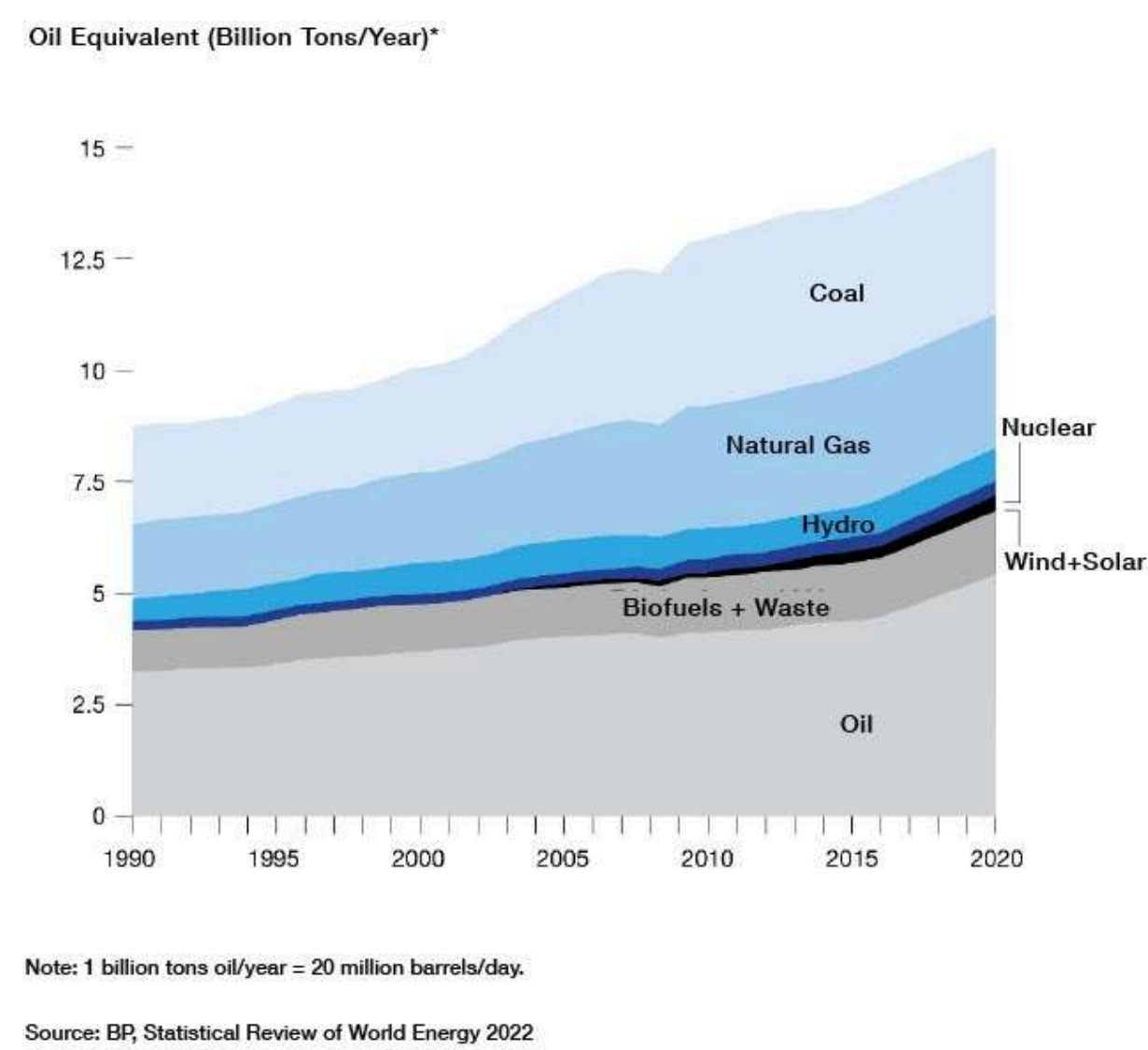


**JEVONS PARADOX = "BACKFIRE"**

EFFICIENCY IS COUNTERPRODUCTIVE  
TO REDUCING CONSUMPTION



**The shares of the world’s primary energy sources change very slowly. After at least \$5 trillion in spending over the past two decades, hydrocarbons still supply 84% of global energy, down just two percentage points. For context, burning wood still supplies more than five times the amount of global energy than do all the world’s solar panels. Meanwhile, the total demand for hydrocarbons has risen over those 20 years by an amount equal to six times the entire oil output of Saudi Arabia.**





# EROI

**Professor Hall developed the energy return on investment concept, or EROI, which measures how much input energy is required to generate a unit of usable power output – the key energetic measure of efficiency...**

**Despite being heralded as the future, wind and solar have terrible EROIs. Compared with coal or natural gas, sunlight and wind are not energy-dense. Compare the energy from a gas stove with a stiff breeze or a sunny afternoon; they are different orders of magnitude. Since renewable energy density is so low, their size must be enormous to generate the same output. A modern windmill stands 80 stories tall with rotor blades that are 600 feet in diameter. A 100 MW solar installation, enough to power 20,000 households, requires a staggering 139 million square feet of PV solar panels. Large size means copious raw materials, which consume enormous energy. As a result, the energy required to generate output is very high, and the EROI is low. A combined cycle natural gas plant enjoys an EROI of 30:1, compared with the best wind and solar at 10:1 and 5:1, respectively. Unfortunately, wind and solar are intermittent and must be “buffered” by either building redundant capacity or through grid-scale battery backup, reducing their overall EROI further to as low as 3-5:1. Based upon our framework, wind and solar could never replace conventional energy given their inferior EROI.**



**For most of history, society relied upon biofuels for energy: crops for food and fodder and wood for heat and construction materials. We estimated that such an energy economy had an EROI of 5:1. Given the low energy efficiency, economic growth was nearly impossible. It took sixteen centuries to double real per capita GDP, equating to an increase of 0.04% annually. The slow growth made sense when looking through the lens of energy. Energy consumption averaged 17 GJ per person yearly for most of human history. Given an EROI of 5:1, 3.5 GJ was spent generating energy. Food consumed 4 GJ per person annually, while other necessities, such as animal fodder and shelter, consumed 10 GJ per person. There was no surplus energy available, and with no surplus energy, economic growth proved impossible.**



**As London grew in the seventeenth century, it ran out of easily accessible wood and was forced to burn coal for heat. The improved energy efficiency was immediately apparent, and Britain's EROI jumped to 10:1. The energy needed to make energy dropped, allowing for material surplus energy for the first time in human history. Surplus energy allowed for economic growth; almost overnight, activity accelerated. After having taken sixteen centuries to double, real per capita GDP doubled again in 175 years, then 130 years, then 50 years, then 50 years again.**

**Coal gave way to oil and natural gas, each with ever-improving EROI. The result is today's modern world, in which developed economies consume 175 GJ of energy per capita annually – a ten-fold increase compared with the historical norm. Surplus energy, meanwhile, went from nil to nearly 150 GJ per capita annually – a seventy-fold improvement over the last 375 years.**

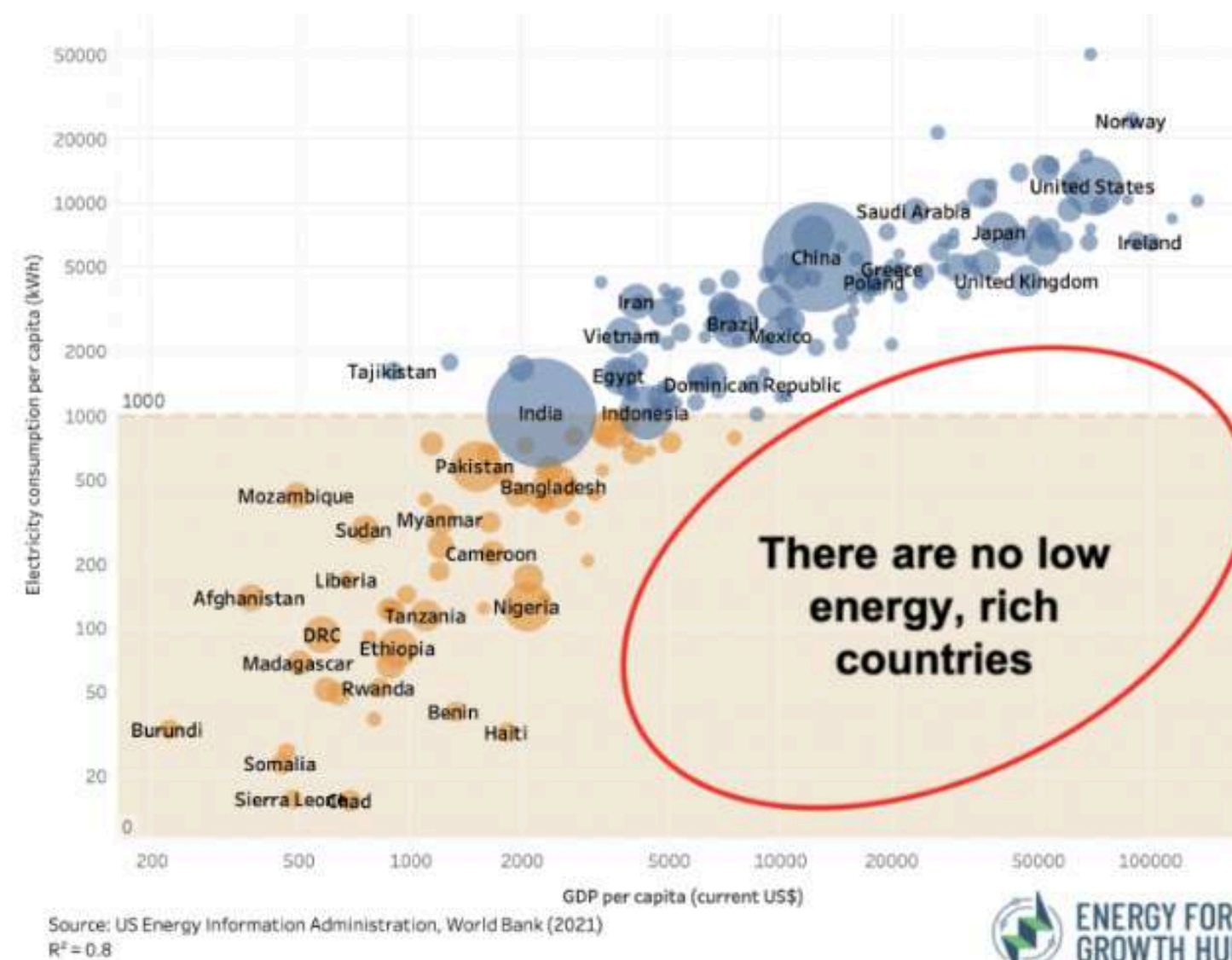
**Transitioning to wind and solar, with EROIs closer to biofuels than fossil fuels, would mean immediately lowering surplus energy by nearly 40% and returning to an energetic system incapable of delivering any real growth. This is not feasible.**



# Our civilization is built on energy

The amount of energy we can access, how densely we can store it, and how quickly we can deploy it are the closest things to measures of our ability to manipulate the physical world.

Energy creates a ceiling on what we can do. To cap our energy ambitions is to commit to permanent scarcity. Sustainability means perpetual scarcity – in our ability to explore, build, and create. It means a fixed pie, and the conflicts that inevitably erupt from it.





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